My Project

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Chapter 1

IMU - ICM-20948

1.1 Pin Connection List

1.1.1 ICM-20948 Pin Connections

VDD - Pin 1 (3.3V) GND - Pin 6 (GND) SDA - Pin 3 (I2C SDA, GPIO 2) SCL - Pin 5 (I2C SCL, GPIO 3) INT - Pin 13 (GPIO 27, data-ready interrupt) AD0 - GND (sets I2C address to 0x69)

1.2 I2C Configuration

I2C Bus : 1 I2C Address : 0x69 (AD0 tied to GND)

1.3 Sensor Settings

Accelerometer range : +/-2G Gyroscope range : +/-250 DPS Magnetometer mode : 100Hz continuous DLPF : enabled

1.4 Data-Ready Interrupt

The ICM-20948 asserts a rising edge on the INT pin when a new sample is ready. This is wired to GPIO 27 on the Raspberry Pi.

1.5 Hardware Assembly

1.6 Acknowledgements

The ICM-20948 IMU driver was adapted from the implementation written by the SnAIRbeats team (ENG 5220, University of Glasgow, 2025), available at: https://github.com/alepgr/SnAIRbeats/tree/main

Chapter 2

ERM Vibration Motor

A ERM (Eccentric Rotating Mass) vibration motor used to deliver haptic feedback proportional to swing intensity.

2.1 Pin Connection List

2.1.1 Motor Circuit Connections

Component	Connection
Motor +	5V (4x AA battery pack)
Motor -	NPN transistor collector
Transistor base	GPIO 18 via 220Ohm resistor
Transistor emitter	GND
Flyback diode	Across motor terminals (GND to +)

2.2 PWM Configuration

Parameter	Value
PWM GPIO	GPIO 18
PWM Frequency	30 Hz
Max Duty Cycle	80%
PWM Type	Software PWM

2.3 Motor Specifications

Parameter	Value
Operating Voltage	2V - 5V
Dimensions	10mm x 2.7mm
Current at 5V	100 mA
Current at 4V	80 mA
Current at 3V	60 mA
Current at 2V	40 mA
Weight	$<$ 1 gram

2.4 Driver Circuit

The motor is driven by an NPN transistor switch controlled by the Raspberry Pi GPIO 18 software PWM output:

- A **220Ohm resistor** on the transistor base limits the base current from GPIO 18.
- The **flyback diodes** across the motor terminals suppress the back-EMF spike when the motor is switched off.
- Power is supplied by a **4x AA battery pack (∼5V)** separate from

2.5 the Raspberry Pi supply to avoid voltage dips on the Pi rail.

2.6 Hardware Assembly

Chapter 3

ADS1115 Pin Connection List

ADC Connections	GPIO Pin	Physical Pin
VDD	3.3v	17
SDA	GPIO 2	3
SCL	GPIO 3	5
A0	Positive Piezo	N/A
ADDR	GND	6
GND	GND	6
ALRT	GPIO 26	37

3.1 Piezo Pin Connection

Piezo	Connections
Positive	A0
Negative	GND
Both	330Ω Resistor

3.2 ADS1115 Settings

- I2C bus: 1
- I2C address: 0x48
- Sampling rate: 128Hz
- Gain/full-scale range: +/-2.048V

3.3 Piezo Event Detector Settings

- Press threshold: 0.95V
- Release threshold: 0.70V

3.4 Hardware Assembly

3.5 Acknowledgements

The ADS1115rpi ADC was adapted from the implementation, rpi_ads1115, written by the Bernd Porr (University of Glasgow, 2025), available at: [(https://github.com/berndporr/rpi_ads1115)]

Chapter 4

README

4.1 Haptic Ping

Haptic Ping combines table tennis shadow practice with haptic feedback to simulate a training experience that more closely resembles real play. Shadow practice is a widely used technique where table tennis players practice their strokes and form, but in the absence of a coach it can be difficult for players to know whether their position and movements are correct.
Our prototype addresses this limitation by providing real-time haptic feedback, allowing players to practice effectively without the need for a coach or any additional table tennis equipment.
Haptic Ping integrates an inertial measurement unit (IMU) and a piezo sensor mounted on the bat to monitor both orientation and physical interaction. The piezo sensor detects correct grip and handling, which triggers an LED to confirm proper finger placement. Once the correct starting position is established, a second LED illuminates to indicate the system is ready for swing analysis. The IMU then tracks the bat's motion to analyse swing patterns in real time. When a correct swing is detected, the system provides immediate haptic feedback through an eccentric rotating motor (ERM).
A project from ENG 5220 Real-Time Embedded Programming University of Glasgow, 2026

4.1.1 Hardware Components

Component	Quantity	Link	Cost per unit
Raspberry Pi 5	1×	RS Online	∼£161.53
ICM-20948 IMU	1×	RS Online	∼£24.28
ERM Vibration Motor	1×	The Pi Hut	∼£1.70
ADS1115 ADC	1×	The Pi Hut	∼£14.40
Piezoelectric Sensor	1×	RS Online	∼£2.50
LEDs	2×	—	∼£0.45
Table Tennis Bat	1×	—	∼£70
AA Batteries	3×	—	∼£0.61

4.1.2 GPIO Pin Assignments

Signal	GPIO Pin
PWM (ERM motor)	GPIO 18
IMU LED	GPIO 22
Piezo LED	GPIO 25
ADS1115 DRDY interrupt	GPIO 26
IMU data-ready interrupt	GPIO 27
I2C SDA	GPIO 2
I2C SCL	GPIO 3

IMU I2C address: `0x69`, ADS1115 I2C address: `0x48`, I2C bus: `1`

4.1.3 Wiring Diagram

4.1.4 Hardware Assembly

- The IMU was mounted at the distal end of the bat using velcro (see below).
- The matrix board shown in the Wiring Diagram was affixed to one face of the bat using velcro.
- The piezolelectric sensor was attached to the bat at the area that ensures appropriate index finger placement for the user and therefore grip.
- Extended wiring ($\sim$1 metre for ease of use) was created by splicing female-to-female jumper cables with standard wire via soldering. Heat-shrink tubing was applied over each joint for insulation and mechanical reinforcement.
- The LED outputs were soldered to matrix boards with wire lengths tailored to suit the user's preferred placement and positioned on a secondary station in front of the user to ensure clear visibility.
- The battery holder wire lenght was also extended to $\sim$1 metre for ease of use.

4.1.5 <img src="images/IMU.jpeg" alt="IMU LOCATION"/>

4.1.6 How It Works

The system uses a gate logic sequence to ensure feedback is only triggered for a valid swing:

1. **Grip detection** — The piezo sensor on the bat is sampled at 128 Hz by the ADS1115 ADC. An EMA filter smooths the signal and a threshold of 0.95V triggers a press event, illuminating the piezo LED.
2. **Position confirmation** — The IMU data-ready interrupt (GPIO 27) wakes the reader thread for each sample. The position detector checks the acceleration vector against a reference orientation using a dot product. After 10 consecutive samples that meet the predetermined threshold for correct starting position, the system is considered to be in the correct starting position and the IMU LED illuminates.
3. **Swing analysis** — Once both grip and position are confirmed, the swing processor categorises the swing vibration magnitude through the magnitude of the IMU acceleration. Gravity bias is removed via a rolling calibration window. The net linear acceleration magnitude is classified into duty cycle levels:
   - $>= 25\ m/s^2$ → Highest Duty Cycle (long vibration burst)
   - $>= 15\ m/s^2$ → Medium Duty Cycle (double burst)
   - $>= 10\ m/s^2$ → Low Duty Cycle (short burst)
4. **Haptic feedback** — The ERM motor is driven by software PWM on GPIO 18 ramping up to max 80% duty cycle. The feedback pattern plays and the system resets for the next swing.

4.1.7 Realtime Design

The code is entirely event-driven with no polling loops:

- **IMU sampling** — driven by the data-ready GPIO interrupt via `epoll` + `eventfd`. The reader thread blocks on `epoll_wait` and wakes only when new data is available.
- **ADS1115 sampling** — driven by the ALERT/RDY GPIO edge event via `libgpiodcxx`. The worker thread blocks on `wait_edge_events`.
- **PWM generation** — runs in a dedicated thread using `timerfd` for precise on/off timing at 30 Hz. No `usleep` or busy-wait.
- **Haptic patterns** — played in a detached worker thread using `timerfd`-based waits for each ramp step.
- **Shutdown** — handled via `signalfd` blocking on `SIGINT`/`SIGHUP` rather than signal handlers.

4.1.8 Maximum Latency

4.1.8.1 IMU - ICM20948

The sampling frequency is calculated by

- **Accelerometer**: 1.125 kHz/(1+ sample_rate_div)
- **Gyrometer**: 1.1kHz/(1+ sample_rate_div) The maximum frequency for accelerometer is therefore 1.125 kHz and gyrometer is 1.1kHz. The maximum latency found for the accelerometer and gyrometer to work is at 18.75 Hz (sample_rate_div = 59).

4.1.8.2 Piezo - ADS1115

The available sample rate are 8Hz, 16Hz, 32Hz, 64Hz, 128Hz, 250Hz, 475Hz, and 860Hz. The maximum latency for the ADS1115 is at 860 Hz.
This can be adjusted in the main.cpp.

4.1.9 Prerequisites

Haptic Ping runs on Linux (Raspberry Pi OS) and is not compatible with Windows.
Prior to installing the required libraries, the system package list should be updated using:

```
sudo apt update
```

4.1.9.1 Enable I2C

```
sudo raspi-config
# Navigate to: Interface Options → I2C → Enable
```

4.1.9.2 Install dependencies

```
sudo apt update
sudo apt install -y libgpiod-dev libgpiod2 libi2c-dev i2c-tools libyaml-cpp-dev cmake build-essential
```

4.1.10 Libraries

4.1.10.1 libgpiod

- `libgpiod`
  Used for all GPIO control on the Raspberry Pi via the Linux character device interface. In this project it is used in three places: the IMU reader configures the data-ready line (GPIO 27) as a rising-edge input and uses `epoll` on the GPIO file descriptor to wake the reader thread; the ADS1115 driver configures the ALERT/↩ RDY line (GPIO 26) as a rising-edge input and blocks on `wait_edge_events`; and the LED controllers and PWM driver configure output lines to drive the LEDs and ERM motor.

4.1.10.2 PWM

- This library was developed by the project group.
  RPI_PWM is a software PWM library for the Raspberry Pi that drives GPIO 18 at 30 Hz using a dedicated worker thread and timerfd-based waits. It exposes a simple setDutyCycle() interface, clamps output to a maximum of 80% to reduce back-EMF spikes, and cleanly acquires and releases the GPIO line through libgpiod.

4.1.10.3 yaml-cpp

- `yaml-cpp`
  Used by the ICM-20948 driver settings class to load sensor configuration parameters — accelerometer scale, gyroscope scale, sample rate divider, and digital low-pass filter settings — from a YAML file. This allows sensor tuning without recompiling.

4.1.11 Compilation from Source

```
git clone <repo-url>
cd RTES_Table_Tennis

# Build main application
cmake -S src -B build
cmake --build build -j$(nproc)
```

4.1.12 Usage

```
sudo ./build/rtes_main
```

The system will:

1. Wait for correct grip detected by piezo sensor → **Piezo LED on**
2. Wait for correct bat starting position detected by IMU → **IMU LED on**
3. Analyse swing in real time
4. Vibrate ERM motor on correct swing detection

Press `Ctrl+C` or `SIGHUP` to stop cleanly.

A demo video is linked here (only accessible to UofG staff and students)

4.1.13 Running Tests

```
cmake -S src -B build -DBUILD_TESTS=ON
cmake --build build -j$(nproc)
ctest --test-dir build --output-on-failure
```

31 unit tests across 6 test suites covering:

Test Suite	What it tests
`PiezoEventDetector`	EMA filter, press/release thresholds, reset behaviour
`PositionDetector`	Stability counter, orientation detection, state transitions
`SwingCalibrator`	Bias averaging, recalibration trigger, callback firing
`SwingDetector`	Level classification, threshold boundaries, reset
`SwingFeedback`	PWM ramp patterns, reset callback, timeout guard
`SwingProcessor`	Gate logic, force/position interaction, magnitude output

4.1.14 Documentation

The documentation for this project can be found here:

4.1.15 Project Structure

```
RTES_Table_Tennis/
+-- src/
|   +-- main.cpp
|   +-- CMakeLists.txt
|   +-- libs/
|       +-- IMU/        # ICM-20948 driver and threaded reader
|       +-- IMU_math/   # Position detection, swing analysis, calibration
|       +-- LEDs/       # GPIO-backed LED controllers
|       +-- Motor/      # Software PWM and swing feedback patterns
|       +-- Piezo/      # ADS1115 ADC driver and event detector
+-- test/               # GTest unit tests
+-- images/
```

4.1.16 Authors

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4.1.17 Individual Contributions

Despina Charalambous: Motor code, soldering, organisation/clean-up
Najaree Janjerdsak: Latency test, IMU math code, soldering, organisation/clean-up
Natalia McCoy: IMU code, main.cpp, organisation/clean-up
Olivia Skinner: piezo code, circuit design, organisation/clean-up
Wiktoria Smolarek: piezo code, circuit design, organisation/clean-up

4.1.18 Media

- `Instagram`
- `TikTok`

4.1.19 License

This project is licensed under the GPL License — see the LICENSE file for details.

4.1.20 Acknowledgements

- `Google Test`
- University of Glasgow ENG 5220 teaching team

We would like to thank the lecturers for the course Real time Embedded Systems, Bernd Porr and Chongfeng Wei, for all their help. We would also like to expresss our gratitude to Thomas O'Hara for all his guidance and support.

This project used AI tools, including Claude (Anthropic), to assist with debugging, documentation, and refactoring. All suggestions were reviewed and validated by the development team, who retain full responsibility for the design decisions and final codebase.

Chapter 5

Namespace Index

5.1 Namespace List

Here is a list of all namespaces with brief descriptions:

icm20948
Settings types 21

Chapter 6

Hierarchical Index

6.1 Class Hierarchy

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Chapter 7

Class Index

7.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 9

Namespace Documentation

9.1 icm20948 Namespace Reference

Settings types.

Classes

- struct accel_settings
- struct gyro_settings

  Gyroscope runtime configuration.
- class ICM20948_I2C
- struct IMUSample
- struct magn_settings

  Magnetometer runtime configuration.
- struct settings

Typedefs

- typedef struct icm20948::accel_settings accel_settings
- typedef struct icm20948::gyro_settings gyro_settings

  Gyroscope runtime configuration.
- typedef struct icm20948::magn_settings magn_settings

  Magnetometer runtime configuration.
- typedef struct icm20948::settings settings

Enumerations

- enum accel_scale { ACCEL_2G = 0 , ACCEL_4G , ACCEL_8G , ACCEL_16G }
- enum accel_dlpf_config {
  ACCEL_DLPF_246HZ = 0 , ACCEL_DLPF_246HZ_2 , ACCEL_DLPF_111_4HZ , ACCEL_DLPF_50_4HZ ,
  ACCEL_DLPF_23_9HZ , ACCEL_DLPF_11_5HZ , ACCEL_DLPF_5_7HZ , ACCEL_DLPF_473HZ }

  Accelerometer digital low-pass filter config.
- enum gyro_scale { GYRO_250DPS = 0 , GYRO_500DPS , GYRO_1000DPS , GYRO_2000DPS }

  Gyroscope full-scale range.
- enum gyro_dlpf_config {
  GYRO_DLPF_196_6HZ = 0 , GYRO_DLPF_151_8HZ , GYRO_DLPF_119_5HZ , GYRO_DLPF_51_2HZ ,
  GYRO_DLPF_23_9HZ , GYRO_DLPF_11_6HZ , GYRO_DLPF_5_7HZ , GYRO_DLPF_361_4HZ }

  Gyroscope digital low-pass filter config.
- enum magn_mode {
  MAGN_SHUTDOWN = 0 , MAGN_SINGLE = 1 , MAGN_10HZ = 2 , MAGN_20HZ = 4 ,
  MAGN_50HZ = 6 , MAGN_100HZ = 8 , MAGN_SELF_TEST = 16 }

  Magnetometer operating mode.

Functions

- float accel_scale_factor (accel_scale scale)

  Convert accel range enum to raw-count scale factor.
- std::string accel_scale_to_str (accel_scale scale)
- std::string accel_dlpf_config_to_str (accel_dlpf_config config)

  Stringify accel DLPF enum.
- float gyro_scale_factor (gyro_scale scale)

  Convert gyro range enum to raw-count scale factor.
- std::string gyro_scale_to_str (gyro_scale scale)

  Stringify gyro range enum.
- std::string gyro_dlpf_config_to_str (gyro_dlpf_config config)

  Stringify gyro DLPF enum.
- std::string magn_mode_to_str (magn_mode mode)

  Stringify magnetometer mode enum.

9.1.1 Detailed Description

Settings types.

9.1.2 Typedef Documentation

9.1.2.1 accel_settings

`typedef struct icm20948::accel_settings icm20948::accel_settings`

Accelerometer runtime configuration. Grouping these parameters in one struct improves clarity and keeps the public interface cleaner than passing many separate constructor arguments.

9.1.2.2 gyro_settings

`typedef struct icm20948::gyro_settings icm20948::gyro_settings`

Gyroscope runtime configuration.

9.1.2.3 magn_settings

`typedef struct icm20948::magn_settings icm20948::magn_settings`

Magnetometer runtime configuration.

9.1.2.4 settings

`typedef struct icm20948::settings icm20948::settings`

Aggregate device configuration. This gives client code a single configuration object, which improves readability and makes the driver easier to construct and reconfigure.

9.1.3 Enumeration Type Documentation

9.1.3.1 accel_dlpf_config

`enum icm20948::accel_dlpf_config`

Accelerometer digital low-pass filter config.

Enumerator

ACCEL_DLPF_246HZ	
ACCEL_DLPF_246HZ↩ _2	
ACCEL_DLPF_111_4HZ	
ACCEL_DLPF_50_4HZ	

Enumerator

ACCEL_DLPF_23_9HZ	
ACCEL_DLPF_11_5HZ	
ACCEL_DLPF_5_7HZ	
ACCEL_DLPF_473HZ	

Definition at line 143 of file Icm20948driver.hpp.

9.1.3.2 accel_scale

`enum icm20948::accel_scale`

Accelerometer full-scale range. Using an enum keeps configuration values readable and avoids scattering raw register meanings through application code.

Enumerator

ACCEL_2G	
ACCEL_4G	
ACCEL_8G	
ACCEL_16G	

Definition at line 135 of file Icm20948driver.hpp.

9.1.3.3 gyro_dlpf_config

`enum icm20948::gyro_dlpf_config`

Gyroscope digital low-pass filter config.

Enumerator

GYRO_DLPF_196_6HZ	
GYRO_DLPF_151_8HZ	
GYRO_DLPF_119_5HZ	
GYRO_DLPF_51_2HZ	
GYRO_DLPF_23_9HZ	
GYRO_DLPF_11_6HZ	
GYRO_DLPF_5_7HZ	
GYRO_DLPF_361_4HZ	

Definition at line 180 of file Icm20948driver.hpp.

9.1.3.4 gyro_scale

`enum icm20948::gyro_scale`

Gyroscope full-scale range.

Enumerator

GYRO_250DPS	
GYRO_500DPS	
GYRO_1000DPS	
GYRO_2000DPS	

Definition at line 172 of file Icm20948driver.hpp.

9.1.3.5 magn_mode

```
enum icm20948::magn_mode
```

Magnetometer operating mode.

Enumerator

MAGN_SHUTDOWN	
MAGN_SINGLE	
MAGN_10HZ	
MAGN_20HZ	
MAGN_50HZ	
MAGN_100HZ	
MAGN_SELF_TEST	

Definition at line 207 of file Icm20948driver.hpp.

9.1.4 Function Documentation

9.1.4.1 accel_dlpf_config_to_str()

```
std::string icm20948::accel_dlpf_config_to_str (
            accel_dlpf_config config )
```

Stringify accel DLPF enum.
Definition at line 50 of file Icm20948driver.cpp.

9.1.4.2 accel_scale_factor()

```
float icm20948::accel_scale_factor (
            accel_scale scale )
```

Convert accel range enum to raw-count scale factor.
Helpers for converting config enums to scale factors / readable strings. These utility functions separate presentation and conversion logic from the main driver code, improving maintainability.
Definition at line 25 of file Icm20948driver.cpp.

9.1.4.3 accel_scale_to_str()

```
std::string icm20948::accel_scale_to_str (
            accel_scale scale )
```

Definition at line 39 of file Icm20948driver.cpp.

9.1.4.4 gyro_dlpf_config_to_str()

```
std::string icm20948::gyro_dlpf_config_to_str (
            gyro_dlpf_config config )
```

Stringify gyro DLPF enum.
Definition at line 89 of file Icm20948driver.cpp.

9.1.4.5 gyro_scale_factor()

```
float icm20948::gyro_scale_factor (
            gyro_scale scale )
```

Convert gyro range enum to raw-count scale factor.
Definition at line 65 of file Icm20948driver.cpp.

9.1.4.6 gyro_scale_to_str()

```
std::string icm20948::gyro_scale_to_str (
            gyro_scale scale )
```

Stringify gyro range enum.
Definition at line 78 of file Icm20948driver.cpp.

9.1.4.7 magn_mode_to_str()

```
std::string icm20948::magn_mode_to_str (
            magn_mode mode )
```

Stringify magnetometer mode enum.
Definition at line 104 of file Icm20948driver.cpp.

Chapter 10

Class Documentation

10.1 icm20948::accel_settings Struct Reference

```
#include <Icm20948driver.hpp>
```

Public Member Functions

- accel_settings (uint16_t sample_rate_div=0, accel_scale scale=ACCEL_2G, bool dlpf_enable=true, accel_dlpf_config dlpf_config=ACCEL_DLPF_246HZ)

Public Attributes

- uint16_t sample_rate_div
- accel_scale scale
- bool dlpf_enable
- accel_dlpf_config dlpf_config

10.1.1 Detailed Description

Accelerometer runtime configuration. Grouping these parameters in one struct improves clarity and keeps the public interface cleaner than passing many separate constructor arguments.
Definition at line 157 of file Icm20948driver.hpp.

10.1.2 Constructor & Destructor Documentation

10.1.2.1 accel_settings()

```
icm20948::accel_settings::accel_settings (
            uint16_t sample_rate_div = 0,
            accel_scale scale = ACCEL_2G,
            bool dlpf_enable = true,
            accel_dlpf_config dlpf_config = ACCEL_DLPF_246HZ )  [inline]
```

Definition at line 163 of file Icm20948driver.hpp.

10.1.3 Member Data Documentation

10.1.3.1 dlpf_config

```
accel_dlpf_config icm20948::accel_settings::dlpf_config
```

Definition at line 161 of file Icm20948driver.hpp.

10.1.3.2 dlpf_enable

```
bool icm20948::accel_settings::dlpf_enable
```

Definition at line 160 of file Icm20948driver.hpp.

10.1.3.3 sample_rate_div

`uint16_t icm20948::accel_settings::sample_rate_div`
Definition at line 158 of file Icm20948driver.hpp.

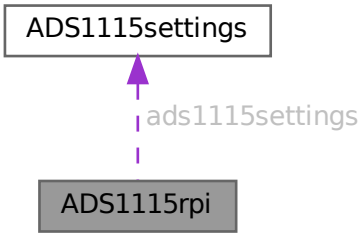
10.1.3.4 scale

`accel_scale icm20948::accel_settings::scale`
Definition at line 159 of file Icm20948driver.hpp.
The documentation for this struct was generated from the following file:

- src/libs/IMU/Include/Icm20948driver.hpp

10.2 ADS1115rpi Class Reference

Declares the ADS1115 Raspberry Pi interface, including callback support and the main functions for starting, stopping, configuring, and reading the device.
`#include <ads1115rpi.h>`
Collaboration diagram for ADS1115rpi:



Public Types

- using ADSCallbackInterface = std::function< void(float)>

Public Member Functions

- ~ADS1115rpi ()
- void registerCallback (ADSCallbackInterface ci)
- void setChannel (ADS1115settings::Input channel)

  Updates the input channel used by the ADS1115.
- void start (ADS1115settings settings=ADS1115settings())

  Opens I2C, configures the ADS1115, sets up the GPIO ready pin, and starts the worker thread.
- ADS1115settings getADS1115settings () const
- void stop ()

  Stops the worker thread and releases GPIO and I2C resources.
- void setRetryEnabled (bool enabled)

Private Member Functions

- void dataReady ()

  Reads the latest conversion result, converts it to a voltage, and sends it to the callback.
- void worker ()

Waits for ready interrupts and triggers a read when conversion data is available.

- void i2c_selectDevice ()

  Selects the ADS1115 device on the I2C bus.
- void i2c_writeWord (uint8_t reg, unsigned data)

  Writes a 16-bit value to the specified ADS1115 register.
- unsigned i2c_readWord (uint8_t reg)

  Reads a 16-bit value from the specified ADS1115 register.
- int i2c_readConversion ()

  Reads the latest conversion value using an atomic I2C transaction, with retries on failure.
- float fullScaleVoltage ()

  Returns the full-scale voltage based on the selected gain setting.

Private Attributes

- ADS1115settings ads1115settings
- const uint8_t reg_config = 1
- const uint8_t reg_lo_thres = 2
- const uint8_t reg_hi_thres = 3
- std::shared_ptr< gpiod::chip > chip
- std::shared_ptr< gpiod::line_request > request
- std::thread thr
- int fd_i2c = -1
- bool running = false
- ADSCallbackInterface adsCallbackInterface
- std::atomic< bool > retry_enabled_ {false}

Static Private Attributes

- static constexpr int64_t ISR_TIMEOUT_MS = 500

10.2.1 Detailed Description

Declares the ADS1115 Raspberry Pi interface, including callback support and the main functions for starting, stopping, configuring, and reading the device.
Definition at line 95 of file ads1115rpi.h.

10.2.2 Member Typedef Documentation

10.2.2.1 ADSCallbackInterface

```
using ADS1115rpi::ADSCallbackInterface = std::function<void(float)>
```

Definition at line 103 of file ads1115rpi.h.

10.2.3 Constructor & Destructor Documentation

10.2.3.1 ~ADS1115rpi()

```
ADS1115rpi::~ADS1115rpi ( ) [inline]
```

Definition at line 98 of file ads1115rpi.h.

10.2.4 Member Function Documentation

10.2.4.1 dataReady()

```
void ADS1115rpi::dataReady ( ) [private]
```

Reads the latest conversion result, converts it to a voltage, and sends it to the callback.
silent skip — retries already logged inside i2c_readConversion
Definition at line 127 of file ads1115rpi.cpp.

10.2.4.2 fullScaleVoltage()

```
float ADS1115rpi::fullScaleVoltage ( ) [inline], [private]
```

Returns the full-scale voltage based on the selected gain setting.
Definition at line 138 of file ads1115rpi.h.

10.2.4.3 getADS1115settings()

```
ADS1115settings ADS1115rpi::getADS1115settings ( ) const [inline]
```

Definition at line 112 of file ads1115rpi.h.

10.2.4.4 i2c_readConversion()

```
int ADS1115rpi::i2c_readConversion ( ) [private]
```

Reads the latest conversion value using an atomic I2C transaction, with retries on failure.
Definition at line 250 of file ads1115rpi.cpp.

10.2.4.5 i2c_readWord()

```
unsigned ADS1115rpi::i2c_readWord (
            uint8_t reg ) [private]
```

Reads a 16-bit value from the specified ADS1115 register.
Definition at line 228 of file ads1115rpi.cpp.

10.2.4.6 i2c_selectDevice()

```
void ADS1115rpi::i2c_selectDevice ( ) [private]
```

Selects the ADS1115 device on the I2C bus.
Definition at line 203 of file ads1115rpi.cpp.

10.2.4.7 i2c_writeWord()

```
void ADS1115rpi::i2c_writeWord (
            uint8_t reg,
            unsigned data ) [private]
```

Writes a 16-bit value to the specified ADS1115 register.
Definition at line 211 of file ads1115rpi.cpp.

10.2.4.8 registerCallback()

```
void ADS1115rpi::registerCallback (
            ADSCallbackInterface ci ) [inline]
```

Definition at line 105 of file ads1115rpi.h.

10.2.4.9 setChannel()

```
void ADS1115rpi::setChannel (
            ADS1115settings::Input channel )
```

Updates the input channel used by the ADS1115.
Definition at line 117 of file ads1115rpi.cpp.

10.2.4.10 setRetryEnabled()

```
void ADS1115rpi::setRetryEnabled (
            bool enabled ) [inline]
```

Enable retries on I2C read failure. Safe to call from any thread. Default: disabled. Enable only when highest duty cycle is active.
Definition at line 120 of file ads1115rpi.h.

10.2.4.11 start()

```
void ADS1115rpi::start (
            ADS1115settings settings = ADS1115settings() )
```

Opens I2C, configures the ADS1115, sets up the GPIO ready pin, and starts the worker thread.
Definition at line 37 of file ads1115rpi.cpp.

10.2.4.12 stop()

```
void ADS1115rpi::stop ( )
```

Stops the worker thread and releases GPIO and I2C resources.
Definition at line 176 of file ads1115rpi.cpp.

10.2.4.13 worker()

```
void ADS1115rpi::worker ( )  [private]
```

Waits for ready interrupts and triggers a read when conversion data is available.
Definition at line 143 of file ads1115rpi.cpp.

10.2.5 Member Data Documentation

10.2.5.1 ads1115settings

```
ADS1115settings ADS1115rpi::ads1115settings  [private]
```

Definition at line 123 of file ads1115rpi.h.

10.2.5.2 adsCallbackInterface

```
ADSCallbackInterface ADS1115rpi::adsCallbackInterface  [private]
```

Definition at line 160 of file ads1115rpi.h.

10.2.5.3 chip

```
std::shared_ptr<gpiod::chip> ADS1115rpi::chip  [private]
```

Definition at line 155 of file ads1115rpi.h.

10.2.5.4 fd_i2c

```
int ADS1115rpi::fd_i2c = -1  [private]
```

Definition at line 158 of file ads1115rpi.h.

10.2.5.5 ISR_TIMEOUT_MS

```
constexpr int64_t ADS1115rpi::ISR_TIMEOUT_MS = 500  [static], [constexpr], [private]
```

Definition at line 163 of file ads1115rpi.h.

10.2.5.6 reg_config

```
const uint8_t ADS1115rpi::reg_config = 1  [private]
```

Definition at line 133 of file ads1115rpi.h.

10.2.5.7 reg_hi_thres

```
const uint8_t ADS1115rpi::reg_hi_thres = 3  [private]
```

Definition at line 135 of file ads1115rpi.h.

10.2.5.8 reg_lo_thres

```
const uint8_t ADS1115rpi::reg_lo_thres = 2  [private]
```

Definition at line 134 of file ads1115rpi.h.

10.2.5.9 request

`std::shared_ptr<gpiod::line_request> ADS1115rpi::request [private]`
Definition at line 156 of file ads1115rpi.h.

10.2.5.10 retry_enabled_

`std::atomic<bool> ADS1115rpi::retry_enabled_ {false} [private]`
Definition at line 161 of file ads1115rpi.h.

10.2.5.11 running

`bool ADS1115rpi::running = false [private]`
Definition at line 159 of file ads1115rpi.h.

10.2.5.12 thr

`std::thread ADS1115rpi::thr [private]`
Definition at line 157 of file ads1115rpi.h.
The documentation for this class was generated from the following files:

- src/libs/Piezo/Include/ads1115rpi.h
- src/libs/Piezo/ads1115rpi.cpp

10.3 ADS1115settings Struct Reference

Stores the ADS1115 configuration values, including I2C settings, sampling rate, gain, input channel, and GPIO settings.
`#include <ads1115rpi.h>`

Public Types

- enum SamplingRates {
  FS8HZ = 0 , FS16HZ = 1 , FS32HZ = 2 , FS64HZ = 3 ,
  FS128HZ = 4 , FS250HZ = 5 , FS475HZ = 6 , FS860HZ = 7 }
- enum PGA { FSR2_048 = 2 , FSR1_024 = 3 , FSR0_512 = 4 , FSR0_256 = 5 }
- enum Input { AIN0 = 0 , AIN1 = 1 , AIN2 = 2 , AIN3 = 3 }

Public Member Functions

- unsigned getSamplingRate () const

  Returns the sampling rate value that matches the selected enum setting.

Public Attributes

- int i2c_bus = 1
- uint8_t address = DEFAULT_ADS1115_ADDRESS
- SamplingRates samplingRate = FS128HZ
- PGA pgaGain = FSR2_048
- Input channel = AIN0
- int drdy_chip = 0
- int drdy_gpio = DEFAULT_ALERT_RDY_TO_GPIO

Static Public Attributes

- static constexpr uint8_t DEFAULT_ADS1115_ADDRESS = 0x48
- static constexpr int DEFAULT_ALERT_RDY_TO_GPIO = 26

10.3.1 Detailed Description

Stores the ADS1115 configuration values, including I2C settings, sampling rate, gain, input channel, and GPIO settings.
Definition at line 37 of file ads1115rpi.h.

10.3.2 Member Enumeration Documentation

10.3.2.1 Input

`enum ADS1115settings::Input`

Enumerator

AIN0	
AIN1	
AIN2	
AIN3	

Definition at line 75 of file ads1115rpi.h.

10.3.2.2 PGA

`enum ADS1115settings::PGA`

Enumerator

FSR2_048	
FSR1_024	
FSR0_512	
FSR0_256	

Definition at line 65 of file ads1115rpi.h.

10.3.2.3 SamplingRates

`enum ADS1115settings::SamplingRates`

Enumerator

FS8HZ	
FS16HZ	
FS32HZ	
FS64HZ	
FS128HZ	
FS250HZ	
FS475HZ	
FS860HZ	

Definition at line 44 of file ads1115rpi.h.

10.3.3 Member Function Documentation

10.3.3.1 getSamplingRate()

`unsigned ADS1115settings::getSamplingRate ( ) const [inline]`
Returns the sampling rate value that matches the selected enum setting.
Definition at line 56 of file ads1115rpi.h.

10.3.4 Member Data Documentation

10.3.4.1 address

```
uint8_t ADS1115settings::address = DEFAULT_ADS1115_ADDRESS
```

Definition at line 42 of file ads1115rpi.h.

10.3.4.2 channel

```
Input ADS1115settings::channel = AIN0
```

Definition at line 83 of file ads1115rpi.h.

10.3.4.3 DEFAULT_ADS1115_ADDRESS

```
constexpr uint8_t ADS1115settings::DEFAULT_ADS1115_ADDRESS = 0x48  [static], [constexpr]
```

Definition at line 41 of file ads1115rpi.h.

10.3.4.4 DEFAULT_ALERT_RDY_TO_GPIO

```
constexpr int ADS1115settings::DEFAULT_ALERT_RDY_TO_GPIO = 26  [static], [constexpr]
```

Definition at line 87 of file ads1115rpi.h.

10.3.4.5 drdy_chip

```
int ADS1115settings::drdy_chip = 0
```

Definition at line 85 of file ads1115rpi.h.

10.3.4.6 drdy_gpio

```
int ADS1115settings::drdy_gpio = DEFAULT_ALERT_RDY_TO_GPIO
```

Definition at line 88 of file ads1115rpi.h.

10.3.4.7 i2c_bus

```
int ADS1115settings::i2c_bus = 1
```

Definition at line 39 of file ads1115rpi.h.

10.3.4.8 pgaGain

```
PGA ADS1115settings::pgaGain = FSR2_048
```

Definition at line 73 of file ads1115rpi.h.

10.3.4.9 samplingRate

```
SamplingRates ADS1115settings::samplingRate = FS128HZ
```

Definition at line 63 of file ads1115rpi.h.

The documentation for this struct was generated from the following file:

- src/libs/Piezo/Include/ads1115rpi.h

10.4 PositionDetector::Config Struct Reference

```
#include <positiondetector.hpp>
```

Public Attributes

- float refX = 9.88f
- float refY = -0.87f
- float refZ = 0.10f
- float minG = 7.0f

- float maxG = 12.5f
- float minDot = 0.90f
- float refHeadingDeg = 25.0f
- float headingToleranceDeg = 180.0f
- int stabilitySamples = 10

10.4.1 Detailed Description

Definition at line 31 of file positiondetector.hpp.

10.4.2 Member Data Documentation

10.4.2.1 headingToleranceDeg

```
float PositionDetector::Config::headingToleranceDeg = 180.0f
```

Definition at line 41 of file positiondetector.hpp.

10.4.2.2 maxG

```
float PositionDetector::Config::maxG = 12.5f
```

Definition at line 37 of file positiondetector.hpp.

10.4.2.3 minDot

```
float PositionDetector::Config::minDot = 0.90f
```

Definition at line 38 of file positiondetector.hpp.

10.4.2.4 minG

```
float PositionDetector::Config::minG = 7.0f
```

Definition at line 36 of file positiondetector.hpp.

10.4.2.5 refHeadingDeg

```
float PositionDetector::Config::refHeadingDeg = 25.0f
```

Definition at line 40 of file positiondetector.hpp.

10.4.2.6 refX

```
float PositionDetector::Config::refX = 9.88f
```

Definition at line 32 of file positiondetector.hpp.

10.4.2.7 refY

```
float PositionDetector::Config::refY = -0.87f
```

Definition at line 33 of file positiondetector.hpp.

10.4.2.8 refZ

```
float PositionDetector::Config::refZ = 0.10f
```

Definition at line 34 of file positiondetector.hpp.

10.4.2.9 stabilitySamples

```
int PositionDetector::Config::stabilitySamples = 10
```

Definition at line 43 of file positiondetector.hpp.

The documentation for this struct was generated from the following file:

- src/libs/IMU_math/Include/positiondetector.hpp

10.5 icm20948::gyro_settings Struct Reference

Gyroscope runtime configuration.
`#include <Icm20948driver.hpp>`

Public Member Functions

- gyro_settings (uint8_t sample_rate_div=0, gyro_scale scale=GYRO_250DPS, bool dlpf_enable=true, gyro_dlpf_config dlpf_config=GYRO_DLPF_196_6HZ)

Public Attributes

- uint8_t sample_rate_div
- gyro_scale scale
- bool dlpf_enable
- gyro_dlpf_config dlpf_config

10.5.1 Detailed Description

Gyroscope runtime configuration.
Definition at line 192 of file Icm20948driver.hpp.

10.5.2 Constructor & Destructor Documentation

10.5.2.1 gyro_settings()

```
icm20948::gyro_settings::gyro_settings (
            uint8_t sample_rate_div = 0,
            gyro_scale scale = GYRO_250DPS,
            bool dlpf_enable = true,
            gyro_dlpf_config dlpf_config = GYRO_DLPF_196_6HZ )  [inline]
```

Definition at line 198 of file Icm20948driver.hpp.

10.5.3 Member Data Documentation

10.5.3.1 dlpf_config

```
gyro_dlpf_config icm20948::gyro_settings::dlpf_config
```

Definition at line 196 of file Icm20948driver.hpp.

10.5.3.2 dlpf_enable

```
bool icm20948::gyro_settings::dlpf_enable
```

Definition at line 195 of file Icm20948driver.hpp.

10.5.3.3 sample_rate_div

```
uint8_t icm20948::gyro_settings::sample_rate_div
```

Definition at line 193 of file Icm20948driver.hpp.

10.5.3.4 scale

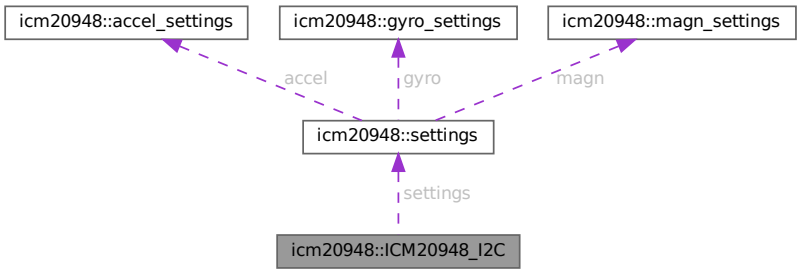
```
gyro_scale icm20948::gyro_settings::scale
```

Definition at line 194 of file Icm20948driver.hpp.
The documentation for this struct was generated from the following file:

- src/libs/IMU/Include/Icm20948driver.hpp

10.6 icm20948::ICM20948_I2C Class Reference

`#include <Icm20948driver.hpp>`
Collaboration diagram for icm20948::ICM20948_I2C:



Public Member Functions

- ICM20948_I2C (unsigned i2c_bus, unsigned i2c_address=ICM20948_I2C_ADDR, icm20948::settings settings=icm20948::settings())

  Open the I2C device and initialise cached driver state.
- ~ICM20948_I2C ()
- bool init ()
- bool reset ()
- bool wake ()
- bool set_settings ()
- bool read_accel_gyro ()
- bool read_magn ()
- bool read_sample (IMUSample &sample)
- bool check_DRDY_INT ()

Public Attributes

- float accel [3]
- float gyro [3]
- float magn [3]
- icm20948::settings settings

Private Member Functions

- bool _enable_data_ready_interrupt ()
- bool _set_bank (uint8_t bank)
- bool _set_accel_sample_rate_div ()
- bool _set_accel_range_dlpf ()
- bool _set_gyro_sample_rate_div ()
- bool _set_gyro_range_dlpf ()
- bool _magnetometer_init ()
- bool _magnetometer_enable ()
- bool _magnetometer_set_mode ()
- bool _magnetometer_configured ()
- bool _magnetometer_set_readout ()
- bool _chip_i2c_master_reset ()
- bool _write_byte (uint8_t bank, uint8_t reg, uint8_t byte)

- bool _read_byte (uint8_t bank, uint8_t reg, uint8_t &byte)
- bool _write_bit (uint8_t bank, uint8_t reg, uint8_t bit_pos, bool bit)
- bool _read_bit (uint8_t bank, uint8_t reg, uint8_t bit_pos, bool &bit)
- bool _read_block_bytes (uint8_t bank, uint8_t start_reg, uint8_t *bytes, int length)
- bool _write_mag_byte (uint8_t mag_reg, uint8_t byte)
- bool _read_mag_byte (uint8_t mag_reg, uint8_t &byte)

Private Attributes

- int _i2c_fd
- unsigned _i2c_bus
- unsigned _i2c_address
- uint8_t _current_bank
- float _accel_scale_factor
- float _gyro_scale_factor
- float _magn_scale_factor

10.6.1 Detailed Description

ICM20948_I2C encapsulates the device driver for the IMU over Linux I2C.
SOLID principles: S - This class focuses on IMU communication, configuration, and conversion of raw register data into decoded sensor values. O - Client code interacts through configuration objects and high-level methods, allowing internal implementation details to evolve without forcing changes on users of the class.
Definition at line 274 of file Icm20948driver.hpp.

10.6.2 Constructor & Destructor Documentation

10.6.2.1 ICM20948_I2C()

```
icm20948::ICM20948_I2C::ICM20948_I2C (
            unsigned i2c_bus,
            unsigned i2c_address = ICM20948_I2C_ADDR,
            icm20948::settings settings = icm20948::settings() )
```

Open the I2C device and initialise cached driver state.
Set a kernel-side I2C timeout.
Bind this fd to the IMU slave address.
Zero cached sensor outputs.
Definition at line 163 of file Icm20948driver.cpp.

10.6.2.2 ~ICM20948_I2C()

```
icm20948::ICM20948_I2C::~ICM20948_I2C ( )
```

Definition at line 200 of file Icm20948driver.cpp.

10.6.3 Member Function Documentation

10.6.3.1 _chip_i2c_master_reset()

```
bool icm20948::ICM20948_I2C::_chip_i2c_master_reset ( )  [private]
```

Reset the chip's internal I2C master controller.
Definition at line 495 of file Icm20948driver.cpp.

10.6.3.2 _enable_data_ready_interrupt()

```
bool icm20948::ICM20948_I2C::_enable_data_ready_interrupt ( )  [private]
```

Device setup helpers. These methods isolate configuration subtasks, which keeps the public interface small and the implementation more maintainable. Enable raw-data-ready interrupt.
Definition at line 276 of file Icm20948driver.cpp.

10.6.3.3 _magnetometer_configured()

```
bool icm20948::ICM20948_I2C::_magnetometer_configured ( ) [private]
```

Probe the magnetometer and reset the internal master on failure.
Definition at line 467 of file Icm20948driver.cpp.

10.6.3.4 _magnetometer_enable()

```
bool icm20948::ICM20948_I2C::_magnetometer_enable ( ) [private]
```

Enable the chip's internal I2C master path for magnetometer access.
Definition at line 439 of file Icm20948driver.cpp.

10.6.3.5 _magnetometer_init()

```
bool icm20948::ICM20948_I2C::_magnetometer_init ( ) [private]
```

Magnetometer setup via the ICM-20948 internal I2C master. This complexity is hidden from callers so they can treat the driver as a single IMU abstraction. Bring up the internal AK09916 through the chip's I2C master.
Definition at line 424 of file Icm20948driver.cpp.

10.6.3.6 _magnetometer_set_mode()

```
bool icm20948::ICM20948_I2C::_magnetometer_set_mode ( ) [private]
```

Switch through shutdown before applying target mode.
Definition at line 457 of file Icm20948driver.cpp.

10.6.3.7 _magnetometer_set_readout()

```
bool icm20948::ICM20948_I2C::_magnetometer_set_readout ( ) [private]
```

Configure SLV0 to continuously fetch magnetometer data into EXT_SLV_SENS_DATA.
Definition at line 480 of file Icm20948driver.cpp.

10.6.3.8 _read_bit()

```
bool icm20948::ICM20948_I2C::_read_bit (
            uint8_t bank,
            uint8_t reg,
            uint8_t bit_pos,
            bool & bit ) [private]
```

Extract one bit from a register.
Definition at line 550 of file Icm20948driver.cpp.

10.6.3.9 _read_block_bytes()

```
bool icm20948::ICM20948_I2C::_read_block_bytes (
            uint8_t bank,
            uint8_t start_reg,
            uint8_t * bytes,
            int length ) [private]
```

Sequential block read starting at start_reg.
Definition at line 560 of file Icm20948driver.cpp.

10.6.3.10 _read_byte()

```
bool icm20948::ICM20948_I2C::_read_byte (
            uint8_t bank,
            uint8_t reg,
            uint8_t & byte ) [private]
```

Definition at line 511 of file Icm20948driver.cpp.

10.6.3.11 _read_mag_byte()

```
bool icm20948::ICM20948_I2C::_read_mag_byte (
            uint8_t mag_reg,
            uint8_t & byte )  [private]
```

Route a read from the external magnetometer via SLV4.
Poll for completion/ACK.
Definition at line 612 of file Icm20948driver.cpp.

10.6.3.12 _set_accel_range_dlpf()

```
bool icm20948::ICM20948_I2C::_set_accel_range_dlpf ( )  [private]
```

Pack accel DLPF enable, range, and cutoff into CONFIG_1.
Definition at line 389 of file Icm20948driver.cpp.

10.6.3.13 _set_accel_sample_rate_div()

```
bool icm20948::ICM20948_I2C::_set_accel_sample_rate_div ( )  [private]
```

Sample-rate divider is split across MSB/LSB registers.
Definition at line 376 of file Icm20948driver.cpp.

10.6.3.14 _set_bank()

```
bool icm20948::ICM20948_I2C::_set_bank (
            uint8_t bank )  [private]
```

Skip the write if the requested bank is already selected.
Definition at line 351 of file Icm20948driver.cpp.

10.6.3.15 _set_gyro_range_dlpf()

```
bool icm20948::ICM20948_I2C::_set_gyro_range_dlpf ( )  [private]
```

Pack gyro DLPF enable, range, and cutoff into CONFIG_1.
Definition at line 410 of file Icm20948driver.cpp.

10.6.3.16 _set_gyro_sample_rate_div()

```
bool icm20948::ICM20948_I2C::_set_gyro_sample_rate_div ( )  [private]
```

Definition at line 403 of file Icm20948driver.cpp.

10.6.3.17 _write_bit()

```
bool icm20948::ICM20948_I2C::_write_bit (
            uint8_t bank,
            uint8_t reg,
            uint8_t bit_pos,
            bool bit )  [private]
```

Read-modify-write a single register bit.
Definition at line 535 of file Icm20948driver.cpp.

10.6.3.18 _write_byte()

```
bool icm20948::ICM20948_I2C::_write_byte (
            uint8_t bank,
            uint8_t reg,
            uint8_t byte )  [private]
```

Low-level register access helpers. These centralise raw byte/bit operations so direct register handling is not scattered across the driver implementation. Select bank, then write one register byte.
Definition at line 502 of file Icm20948driver.cpp.

10.6.3.19 _write_mag_byte()

```
bool icm20948::ICM20948_I2C::_write_mag_byte (
            uint8_t mag_reg,
            uint8_t byte )  [private]
```

Route a write to the external magnetometer via SLV4.
Poll for completion/ACK.
Definition at line 586 of file Icm20948driver.cpp.

10.6.3.20 check_DRDY_INT()

```
bool icm20948::ICM20948_I2C::check_DRDY_INT ( )
```

Definition at line 640 of file Icm20948driver.cpp.

10.6.3.21 init()

```
bool icm20948::ICM20948_I2C::init ( )
```

Public API for device lifecycle and sensor access. Higher-level code interacts through these methods rather than directly manipulating device registers. Basic bring-up: identify device, reset, wake, configure, enable DRDY.
Definition at line 208 of file Icm20948driver.cpp.

10.6.3.22 read_accel_gyro()

```
bool icm20948::ICM20948_I2C::read_accel_gyro ( )
```

Read accel and gyro output block in one transaction.
Convert raw counts to SI units.
Definition at line 283 of file Icm20948driver.cpp.

10.6.3.23 read_magn()

```
bool icm20948::ICM20948_I2C::read_magn ( )
```

Read magnetometer data staged through the IMU external-sensor window.
Convert raw magnetometer counts using fixed scale factor.
Definition at line 311 of file Icm20948driver.cpp.

10.6.3.24 read_sample()

```
bool icm20948::ICM20948_I2C::read_sample (
            IMUSample & sample )
```

Accel/gyro are required for a valid sample.
Magnetometer intentionally disabled for stability testing. if (!read_magn()) magn[0] = magn[1] = magn[2] = 0.0f;
Definition at line 332 of file Icm20948driver.cpp.

10.6.3.25 reset()

```
bool icm20948::ICM20948_I2C::reset ( )
```

Trigger device reset via PWR_MGMT_1.
Poll until the reset bit clears.
Definition at line 235 of file Icm20948driver.cpp.

10.6.3.26 set_settings()

```
bool icm20948::ICM20948_I2C::set_settings ( )
```

Push cached accel/gyro settings to the device.
Definition at line 265 of file Icm20948driver.cpp.

10.6.3.27 wake()

```
bool icm20948::ICM20948_I2C::wake ( )
```

Clear sleep bit.

Definition at line 256 of file Icm20948driver.cpp.

10.6.4 Member Data Documentation

10.6.4.1 _accel_scale_factor

`float icm20948::ICM20948_I2C::_accel_scale_factor [private]`
Cached conversion factors from raw counts to physical units. Keeping these values private prevents duplication of conversion logic.
Definition at line 319 of file Icm20948driver.hpp.

10.6.4.2 _current_bank

`uint8_t icm20948::ICM20948_I2C::_current_bank [private]`
Cached register bank to avoid redundant bank switches. This is a performance and consistency optimisation internal to the driver.
Definition at line 315 of file Icm20948driver.hpp.

10.6.4.3 _gyro_scale_factor

`float icm20948::ICM20948_I2C::_gyro_scale_factor [private]`
Definition at line 320 of file Icm20948driver.hpp.

10.6.4.4 _i2c_address

`unsigned icm20948::ICM20948_I2C::_i2c_address [private]`
Definition at line 311 of file Icm20948driver.hpp.

10.6.4.5 _i2c_bus

`unsigned icm20948::ICM20948_I2C::_i2c_bus [private]`
Definition at line 310 of file Icm20948driver.hpp.

10.6.4.6 _i2c_fd

`int icm20948::ICM20948_I2C::_i2c_fd [private]`
Linux I2C file descriptor and device identity. These implementation details are kept private so the transport layer remains hidden from the rest of the application.
Definition at line 309 of file Icm20948driver.hpp.

10.6.4.7 _magn_scale_factor

`float icm20948::ICM20948_I2C::_magn_scale_factor [private]`
Definition at line 321 of file Icm20948driver.hpp.

10.6.4.8 accel

`float icm20948::ICM20948_I2C::accel[3]`
Most recent decoded sensor values. These cached arrays provide direct access to the latest successful reading while keeping raw register handling internal to the class.
Definition at line 280 of file Icm20948driver.hpp.

10.6.4.9 gyro

`float icm20948::ICM20948_I2C::gyro[3]`
Definition at line 281 of file Icm20948driver.hpp.

10.6.4.10 magn

`float icm20948::ICM20948_I2C::magn[3]`
Definition at line 282 of file Icm20948driver.hpp.

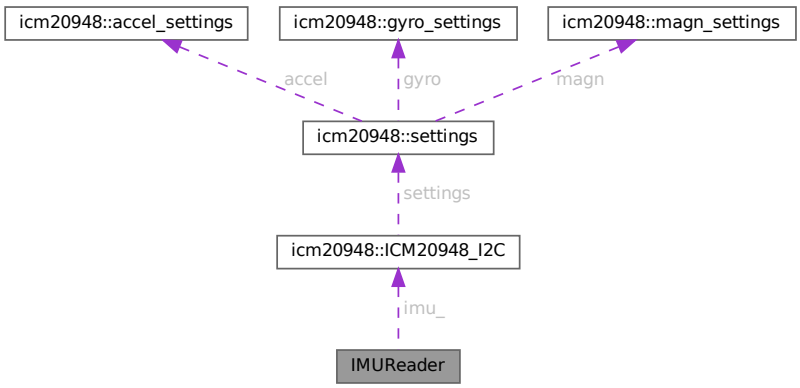
10.6.4.11 settings

`icm20948::settings icm20948::ICM20948_I2C::settings`
Active driver settings. Storing settings here keeps configuration tied to the device instance.
Definition at line 286 of file Icm20948driver.hpp.
The documentation for this class was generated from the following files:

- src/libs/IMU/Include/Icm20948driver.hpp
- src/libs/IMU/Icm20948driver.cpp

10.7 IMUReader Class Reference

`#include <imureader.hpp>`
Collaboration diagram for IMUReader:



Public Types

- using SampleCallback = std::function< void(float ax, float ay, float az, float gx, float gy, float gz, float mx, float my)>

Public Member Functions

- IMUReader (unsigned i2c_bus, unsigned i2c_address=ICM20948_I2C_ADDR, const char *gpiochip="/dev/gpiochip0", unsigned rdy_line=27, icm20948::settings settings=icm20948::settings())
- ~IMUReader ()
- bool init ()

  Initialise the underlying IMU device before starting the worker thread.
- void setCallback (SampleCallback cb)
- bool start ()

  Start the background reader thread and GPIO-driven event loop.
- void stop ()

  Stop the worker thread and release runtime resources cleanly.
- icm20948::ICM20948_I2C & getIMU ()

Private Member Functions

- void worker ()

Private Attributes

- icm20948::ICM20948_I2C imu_
- SampleCallback callback_
- std::thread worker_thread_
- std::atomic_bool running_
- const char * gpiochip_path_
- unsigned rdy_line_
- gpiod_chip * chip_
- gpiod_line_request * request_
- int gpio_fd_
- int stop_fd_

10.7.1 Detailed Description

IMUReader provides a threaded, event-driven wrapper around the low-level ICM-20948 driver.
SOLID principles: S - This class is responsible for coordinating asynchronous IMU sample acquisition from the data-ready GPIO and forwarding decoded samples to client code. O - New behaviours can be attached through the callback interface without changing the reader logic itself. D - The low-level register and I2C interaction remain inside the IMU driver, while this class focuses on threading and sample dispatch.
Threaded IMU reader driven by the data-ready GPIO.
Definition at line 25 of file imureader.hpp.

10.7.2 Member Typedef Documentation

10.7.2.1 SampleCallback

```
using IMUReader::SampleCallback = std::function<void(float ax, float ay, float az, float gx,
float gy, float gz, float mx, float my)>
```

Callback invoked for each decoded IMU sample. Using a callback keeps the class open for extension: logging, filtering, fusion, motion detection, or UI updates can be added without modifying the reader implementation.
Definition at line 32 of file imureader.hpp.

10.7.3 Constructor & Destructor Documentation

10.7.3.1 IMUReader()

```
IMUReader::IMUReader (
            unsigned i2c_bus,
            unsigned i2c_address = ICM20948_I2C_ADDR,
            const char * gpiochip = "/dev/gpiochip0",
            unsigned rdy_line = 27,
            icm20948::settings settings = icm20948::settings() )
```

Construct the reader with its hardware configuration and runtime settings. The IMU configuration is injected from outside so the reader can be reused with different buses, addresses, and sensor settings.
Definition at line 9 of file imureader.cpp.

10.7.3.2 ~IMUReader()

```
IMUReader::~IMUReader ( )
```

Destructor ensures the threaded reader can clean up safely through the class interface. Ensure the worker thread and GPIO resources are cleaned up.
Definition at line 25 of file imureader.cpp.

10.7.4 Member Function Documentation

10.7.4.1 getIMU()

```
icm20948::ICM20948_I2C & IMUReader::getIMU ( )
```

Definition at line 37 of file imureader.cpp.

10.7.4.2 init()

```
bool IMUReader::init ( )
```

Initialise the underlying IMU device before starting the worker thread.
Initialise the underlying IMU driver.
Definition at line 31 of file imureader.cpp.

10.7.4.3 setCallback()

```
void IMUReader::setCallback (
            SampleCallback cb )
```

Register client code to receive new IMU samples. This keeps sample processing outside the reader itself, improving flexibility and separation of concerns. Register the callback for decoded IMU samples.
Definition at line 42 of file imureader.cpp.

10.7.4.4 start()

```
bool IMUReader::start ( )
```

Start the background reader thread and GPIO-driven event loop.
Ignore duplicate starts.
Open the GPIO chip exposing the DRDY line.
Configure the DRDY GPIO as a rising-edge input.
Tag the line request for easier debugging.
Request ownership of the DRDY line.
Get the GPIO event fd and create an eventfd for thread shutdown.
Launch the background event loop.
Definition at line 48 of file imureader.cpp.

10.7.4.5 stop()

```
void IMUReader::stop ( )
```

Stop the worker thread and release runtime resources cleanly.
Signal the worker loop to exit.
Wake epoll_wait() so the worker can notice shutdown.
Wait for the worker thread to finish before releasing resources.
Release GPIO request and chip handles.
Close the shutdown event fd.
Definition at line 92 of file imureader.cpp.

10.7.4.6 worker()

```
void IMUReader::worker ( )  [private]
```

Background worker loop waiting for DRDY events and dispatching samples. This keeps blocking/event-loop behaviour internal to the class so client code does not need to manage low-level synchronisation details. epoll waits on both the GPIO event fd and the shutdown event fd.
Single-slot buffer is enough since events are handled one at a time.
hold the shared I2C bus mutex for the full read_sample() call so the ADS1115 thread cannot interleave its own ioctl(I2C_SLAVE) + read between our register writes.
Optional DRDY status check after the sample read.
Forward decoded sample to the client callback
Definition at line 123 of file imureader.cpp.

10.7.5 Member Data Documentation

10.7.5.1 callback_

```
SampleCallback IMUReader::callback_  [private]
```

Client callback for newly acquired samples. Stored privately so callback ownership and invocation remain controlled by the reader.
Definition at line 77 of file imureader.hpp.

10.7.5.2 chip_

`gpiod_chip* IMUReader::chip_ [private]`
libgpiod handles for the DRDY line. These low-level resources are kept private to preserve encapsulation.
Definition at line 96 of file imureader.hpp.

10.7.5.3 gpio_fd_

`int IMUReader::gpio_fd_ [private]`
File descriptors used by the event loop. Internal event-loop state is hidden so callers interact only through the public start/stop interface.
Definition at line 102 of file imureader.hpp.

10.7.5.4 gpiochip_path_

`const char* IMUReader::gpiochip_path_ [private]`
GPIO chip path and DRDY line number. These hardware configuration details are stored with the reader instance so callers only configure them once during construction.
Definition at line 91 of file imureader.hpp.

10.7.5.5 imu_

`icm20948::ICM20948_I2C IMUReader::imu_ [private]`
Low-level IMU driver instance. Hardware-specific register access is delegated to the driver so this class can focus on threaded sample acquisition and dispatch.
Definition at line 72 of file imureader.hpp.

10.7.5.6 rdy_line_

`unsigned IMUReader::rdy_line_ [private]`
Definition at line 92 of file imureader.hpp.

10.7.5.7 request_

`gpiod_line_request* IMUReader::request_ [private]`
Definition at line 97 of file imureader.hpp.

10.7.5.8 running_

`std::atomic_bool IMUReader::running_ [private]`
Run flag shared with the worker thread. Atomic state allows safe coordination between start/stop and the worker.
Definition at line 86 of file imureader.hpp.

10.7.5.9 stop_fd_

`int IMUReader::stop_fd_ [private]`
Definition at line 103 of file imureader.hpp.

10.7.5.10 worker_thread_

`std::thread IMUReader::worker_thread_ [private]`
Background reader thread. Thread lifetime is encapsulated inside the class rather than exposed to higher-level application code.
Definition at line 82 of file imureader.hpp.
The documentation for this class was generated from the following files:

- src/libs/IMU/Include/imureader.hpp
- src/libs/IMU/imureader.cpp

10.8 icm20948::IMUSample Struct Reference

```
#include <Icm20948driver.hpp>
```

Public Attributes

- float ax
- float ay
- float az
- float gx
- float gy
- float gz
- float mx
- float my
- float mz

10.8.1 Detailed Description

One fully decoded IMU sample. This struct provides a clean data-transfer object so callers can work with physical units rather than raw register bytes.
Definition at line 258 of file Icm20948driver.hpp.

10.8.2 Member Data Documentation

10.8.2.1 ax

```
float icm20948::IMUSample::ax
```
Definition at line 259 of file Icm20948driver.hpp.

10.8.2.2 ay

```
float icm20948::IMUSample::ay
```
Definition at line 259 of file Icm20948driver.hpp.

10.8.2.3 az

```
float icm20948::IMUSample::az
```
Definition at line 259 of file Icm20948driver.hpp.

10.8.2.4 gx

```
float icm20948::IMUSample::gx
```
Definition at line 260 of file Icm20948driver.hpp.

10.8.2.5 gy

```
float icm20948::IMUSample::gy
```
Definition at line 260 of file Icm20948driver.hpp.

10.8.2.6 gz

```
float icm20948::IMUSample::gz
```
Definition at line 260 of file Icm20948driver.hpp.

10.8.2.7 mx

```
float icm20948::IMUSample::mx
```
Definition at line 261 of file Icm20948driver.hpp.

10.8.2.8 my

`float icm20948::IMUSample::my`
Definition at line 261 of file Icm20948driver.hpp.

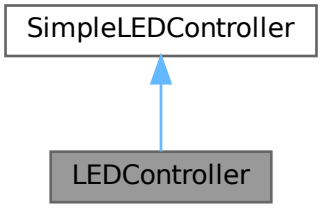
10.8.2.9 mz

`float icm20948::IMUSample::mz`
Definition at line 261 of file Icm20948driver.hpp.
The documentation for this struct was generated from the following file:

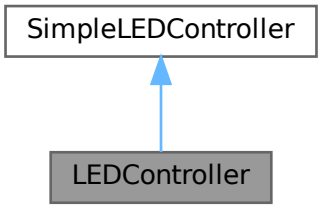
- src/libs/IMU/Include/Icm20948driver.hpp

10.9 LEDController Class Reference

Declares the LED controller class and its main functions for setting, switching, flashing, and updating the LED output.
`#include <led_controller.h>`
Inheritance diagram for LEDController:



Collaboration diagram for LEDController:



Public Member Functions

- LEDController (LEDControllerSettings settings=LEDControllerSettings())

  Initialises the LED controller, clears the LED state, and updates the output.
- ~LEDController () override

  Turns all LED output off when the controller is destroyed.

- void set (bool on) override

  Sets the green LED state directly and updates the output.
- void greenOn ()

  Turns the green LED on and clears any flash timing.
- void greenOff ()

  Turns the green LED off and clears any flash timing.
- void allOff ()

  Turns all LED output off and clears any flash timing.
- void flashGreen (int flashMs)

  Turns the green LED on for a set time and stores when it should switch off.
- void service ()

  Checks whether the flash time has ended and switches the LED off if needed.

Public Member Functions inherited from SimpleLEDController

- SimpleLEDController (int pinNo, int chipNo=0)

  Initialises the LED controller by setting up the GPIO chip, output line, and line request for the selected pin.
- virtual ~SimpleLEDController ()

  Releases the GPIO resources and makes sure the LED is turned off before closing.

Private Member Functions

- void updateOutputs ()

  Declares the internal update function and private members used to track the green LED state and flash timing.

Private Attributes

- bool greenActive_ = false
- std::chrono::steady_clock::time_point greenOffTime_ {}

Additional Inherited Members

Protected Member Functions inherited from SimpleLEDController

- SimpleLEDController ()=default

Protected Attributes inherited from SimpleLEDController

- int pin = -1
- int chipNumber = -1
- std::shared_ptr< gpiod::chip > chip

  resource lifetime is managed safely without manual memory handling.
- std::shared_ptr< gpiod::line_request > request

10.9.1 Detailed Description

Declares the LED controller class and its main functions for setting, switching, flashing, and updating the LED output.
Definition at line 29 of file led_controller.h.

10.9.2 Constructor & Destructor Documentation

10.9.2.1 LEDController()

```
LEDController::LEDController (
            LEDControllerSettings settings = LEDControllerSettings() )
```

Initialises the LED controller, clears the LED state, and updates the output.
Definition at line 24 of file led_controller.cpp.

10.9.2.2 ~LEDController()

```
LEDController::~LEDController ( ) [override]
```

Turns all LED output off when the controller is destroyed.
Definition at line 33 of file led_controller.cpp.

10.9.3 Member Function Documentation

10.9.3.1 allOff()

```
void LEDController::allOff ( )
```

Turns all LED output off and clears any flash timing.
Definition at line 65 of file led_controller.cpp.

10.9.3.2 flashGreen()

```
void LEDController::flashGreen (
            int flashMs ) [virtual]
```

Turns the green LED on for a set time and stores when it should switch off.
Reimplemented from SimpleLEDController.
Definition at line 72 of file led_controller.cpp.

10.9.3.3 greenOff()

```
void LEDController::greenOff ( )
```

Turns the green LED off and clears any flash timing.
Definition at line 57 of file led_controller.cpp.

10.9.3.4 greenOn()

```
void LEDController::greenOn ( )
```

Turns the green LED on and clears any flash timing.
Definition at line 49 of file led_controller.cpp.

10.9.3.5 service()

```
void LEDController::service ( ) [virtual]
```

Checks whether the flash time has ended and switches the LED off if needed.
Reimplemented from SimpleLEDController.
Definition at line 80 of file led_controller.cpp.

10.9.3.6 set()

```
void LEDController::set (
            bool on ) [override], [virtual]
```

Sets the green LED state directly and updates the output.
Reimplemented from SimpleLEDController.
Definition at line 39 of file led_controller.cpp.

10.9.3.7 updateOutputs()

```
void LEDController::updateOutputs ( ) [private]
```

Declares the internal update function and private members used to track the green LED state and flash timing.
Updates the physical LED output to match the current stored state.
Definition at line 92 of file led_controller.cpp.

10.9.4 Member Data Documentation

10.9.4.1 greenActive_

```
bool LEDController::greenActive_ = false [private]
```

Definition at line 48 of file led_controller.h.

10.9.4.2 greenOffTime_

```
std::chrono::steady_clock::time_point LEDController::greenOffTime_ {}  [private]
```

Definition at line 49 of file led_controller.h.
The documentation for this class was generated from the following files:

- src/libs/LEDs/Include/led_controller.h
- src/libs/LEDs/led_controller.cpp

10.10 LEDControllerSettings Struct Reference

Declares the LED controller settings, including the chip number and green LED GPIO pin.
`#include <led_controller.h>`

Public Attributes

- int chipNumber = 0
- unsigned int greenGpio = 25

10.10.1 Detailed Description

Declares the LED controller settings, including the chip number and green LED GPIO pin.
This header declares the LED controller settings and class. It manages green LED output, supports timed flashing, and exposes a small LED-facing interface through the inherited controller.
SOLID principles: S - This class is responsible for LED-specific behaviour only. O - Client code can request LED actions through the public interface without modifying the internal output logic. L - The concrete LEDController builds on the simpler base controller so it can be substituted wherever that LED abstraction is expected. I - Client code uses a small LED-facing interface rather than dealing with GPIO details or unrelated hardware operations. D - Higher-level code can depend on the simpler SimpleLEDController abstraction rather than this concrete GPIO-backed implementation.
Definition at line 23 of file led_controller.h.

10.10.2 Member Data Documentation

10.10.2.1 chipNumber

```
int LEDControllerSettings::chipNumber = 0
```

Definition at line 24 of file led_controller.h.

10.10.2.2 greenGpio

```
unsigned int LEDControllerSettings::greenGpio = 25
```

Definition at line 25 of file led_controller.h.
The documentation for this struct was generated from the following file:

- src/libs/LEDs/Include/led_controller.h

10.11 icm20948::magn_settings Struct Reference

Magnetometer runtime configuration.
`#include <Icm20948driver.hpp>`

Public Member Functions

- magn_settings (magn_mode mode=MAGN_100HZ)

Public Attributes

- magn_mode mode

10.11.1 Detailed Description

Magnetometer runtime configuration.
Definition at line 218 of file Icm20948driver.hpp.

10.11.2 Constructor & Destructor Documentation

10.11.2.1 magn_settings()

```
icm20948::magn_settings::magn_settings (
            magn_mode mode = MAGN_100HZ )  [inline], [explicit]
```

Definition at line 220 of file Icm20948driver.hpp.

10.11.3 Member Data Documentation

10.11.3.1 mode

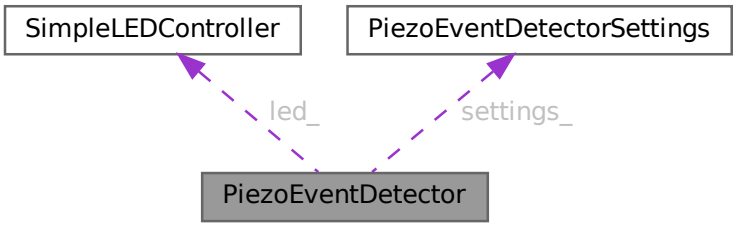
```
magn_mode icm20948::magn_settings::mode
```

Definition at line 219 of file Icm20948driver.hpp.
The documentation for this struct was generated from the following file:

- src/libs/IMU/Include/Icm20948driver.hpp

10.12 PiezoEventDetector Class Reference

Declares the detector class and its public interface.
`#include <event_detector.h>`
Collaboration diagram for PiezoEventDetector:



Public Types

- using ForceCallback = std::function< void(bool)>

  Defines the callback type used for press state changes.

Public Member Functions

- PiezoEventDetector (SimpleLEDController &ledRef, PiezoEventDetectorSettings settings)

  Creates the detector with an LED reference and settings.
- void setForceCallback (ForceCallback cb)

  Sets the callback used when the press state changes.

- void processSample (float v)

  Sets the callback function for state changes.
- void reset ()

  Processes a new sensor sample.

Private Attributes

- SimpleLEDController & led_

  Resets the detector state.
- PiezoEventDetectorSettings settings_

  Reference to the LED controller.
- ForceCallback forceCallback_

  Stores the detector settings.
- bool pressed_ = false

  Stores the callback function.
- bool filter_init_ = false
- float filtered_v_ = 0.0f

10.12.1 Detailed Description

Declares the detector class and its public interface.
Definition at line 30 of file event_detector.h.

10.12.2 Member Typedef Documentation

10.12.2.1 ForceCallback

```
using PiezoEventDetector::ForceCallback = std::function<void(bool)>
```

Defines the callback type used for press state changes.
Definition at line 34 of file event_detector.h.

10.12.3 Constructor & Destructor Documentation

10.12.3.1 PiezoEventDetector()

```
PiezoEventDetector::PiezoEventDetector (
            SimpleLEDController & ledRef,
            PiezoEventDetectorSettings settings )
```

Creates the detector with an LED reference and settings.
Stores the LED reference and detector settings.
This class processes piezo sensor samples to detect press and release events. It smooths the input signal, compares it against press/release thresholds, updates the LED state, and triggers a callback when the state changes.
Definition at line 11 of file event_detector.cpp.

10.12.4 Member Function Documentation

10.12.4.1 processSample()

```
void PiezoEventDetector::processSample (
            float v )
```

Sets the callback function for state changes.
Filters the sample, checks thresholds, updates state, and triggers outputs.
Definition at line 33 of file event_detector.cpp.

10.12.4.2 reset()

```
void PiezoEventDetector::reset ( )
```

Processes a new sensor sample.
Resets the detector state and turns the LED off.
Definition at line 24 of file event_detector.cpp.

10.12.4.3 setForceCallback()

```
void PiezoEventDetector::setForceCallback (
            ForceCallback cb )
```

Sets the callback used when the press state changes.
Definition at line 18 of file event_detector.cpp.

10.12.5 Member Data Documentation

10.12.5.1 filter_init_

```
bool PiezoEventDetector::filter_init_ = false [private]
```

Definition at line 51 of file event_detector.h.

10.12.5.2 filtered_v_

```
float PiezoEventDetector::filtered_v_ = 0.0f [private]
```

Definition at line 52 of file event_detector.h.

10.12.5.3 forceCallback_

```
ForceCallback PiezoEventDetector::forceCallback_ [private]
```

Stores the detector settings.
Definition at line 47 of file event_detector.h.

10.12.5.4 led_

```
SimpleLEDController& PiezoEventDetector::led_ [private]
```

Resets the detector state.
Definition at line 45 of file event_detector.h.

10.12.5.5 pressed_

```
bool PiezoEventDetector::pressed_ = false [private]
```

Stores the callback function.
Tracks whether the detector is currently pressed, whether the filter has been initialised and stores the current filtered sample value.
Definition at line 50 of file event_detector.h.

10.12.5.6 settings_

```
PiezoEventDetectorSettings PiezoEventDetector::settings_ [private]
```

Reference to the LED controller.
Definition at line 46 of file event_detector.h.
The documentation for this class was generated from the following files:

- src/libs/Piezo/Include/event_detector.h
- src/libs/Piezo/event_detector.cpp

10.13 PiezoEventDetectorSettings Struct Reference

Stores the threshold and debug settings for the detector.

```
#include <event_detector.h>
```

Public Attributes

- float pressThreshold = 0.95f
- float releaseThreshold = 0.70f
- bool enableDebugPrints = false

10.13.1 Detailed Description

Stores the threshold and debug settings for the detector.
This header defines the piezo event detector and its settings. The detector is responsible for interpreting sensor samples from the ADC, tracking press/release state, and exposing higher-level event behaviour through a callback while also providing direct LED feedback.
SOLID principle: S - The main responsibility of this class is to interpret piezo samples using filtering and threshold logic. O - New behaviours can be attached through the callback without changing the detector logic itself. L/D - The detector depends on a simpler LED abstraction and a callback rather than on broader application logic, helping to reduce coupling.
Definition at line 22 of file event_detector.h.

10.13.2 Member Data Documentation

10.13.2.1 enableDebugPrints

```
bool PiezoEventDetectorSettings::enableDebugPrints = false
```
Definition at line 26 of file event_detector.h.

10.13.2.2 pressThreshold

```
float PiezoEventDetectorSettings::pressThreshold = 0.95f
```
Definition at line 24 of file event_detector.h.

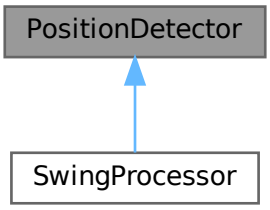
10.13.2.3 releaseThreshold

```
float PiezoEventDetectorSettings::releaseThreshold = 0.70f
```
Definition at line 25 of file event_detector.h.
The documentation for this struct was generated from the following file:

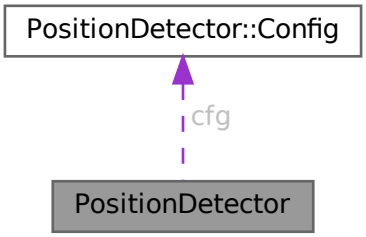
- src/libs/Piezo/Include/event_detector.h

10.14 PositionDetector Class Reference

Declares the position detector class, including its callback type, configuration settings, and main functions for processing samples and resetting state.
`#include <positiondetector.hpp>`
Inheritance diagram for PositionDetector:



Collaboration diagram for PositionDetector:



Classes

- struct Config

Public Types

- using StateCallback = std::function< void(bool upright)>

Public Member Functions

- PositionDetector ()

  Default constructor delegates to config-based constructor.
- PositionDetector (const Config &c)
- virtual ~PositionDetector ()=default
- virtual void onSample (float ax, float ay, float az, float mx, float my)
- void setCallback (StateCallback cb)
- void resetState ()

Static Private Member Functions

- static float wrapAngleDeg (float a)

  Declares the helper function and private members used to store configuration, reference values, stability count, state, and callback data.

Private Attributes

- Config cfg
- float nRefX
- float nRefY
- float nRefZ
- int stableCount

  normalised ref vector, computed from cfg
- bool confirmed
- StateCallback callback

10.14.1 Detailed Description

Declares the position detector class, including its callback type, configuration settings, and main functions for processing samples and resetting state.
This header declares the position detector class. It stores the configuration for detecting an upright position, processes sensor samples, tracks stability, and uses a callback when the required position has been confirmed.
SOLID principles: S - This class is responsible for analysing incoming motion/orientation samples and deciding whether the required upright position has been stably detected. O - Client code can extend what happens after detection by attaching a callback, without modifying the detector logic itself. L - The virtual destructor and virtual onSample() allow derived detector types to replace the base class safely, provided they preserve the same detection contract. I - The public API is intentionally small, exposing only the operations needed by client code: process samples, register a callback, and reset state.
Definition at line 27 of file positiondetector.hpp.

10.14.2 Member Typedef Documentation

10.14.2.1 StateCallback

```
using PositionDetector::StateCallback = std::function<void(bool upright)>
```
Definition at line 29 of file positiondetector.hpp.

10.14.3 Constructor & Destructor Documentation

10.14.3.1 PositionDetector() [1/2]

```
PositionDetector::PositionDetector ( )
```
Default constructor delegates to config-based constructor.
Definition at line 4 of file positiondetector.cpp.

10.14.3.2 PositionDetector() [2/2]

```
PositionDetector::PositionDetector (
            const Config & c )  [explicit]
```
Precompute normalised reference gravity vector.
Definition at line 7 of file positiondetector.cpp.

10.14.3.3 ~PositionDetector()

```
virtual PositionDetector::~PositionDetector ( )  [virtual], [default]
```

10.14.4 Member Function Documentation

10.14.4.1 onSample()

```
void PositionDetector::onSample (
            float ax,
            float ay,
            float az,
            float mx,
            float my )  [virtual]
```
Compute acceleration magnitude.
Compute heading from magnetometer.
Check heading against reference (with wrap-around).
Check if acceleration magnitude is within expected bounds.
Normalise acceleration vector.
Compare with reference orientation using dot product.
Position valid only if both checks pass.
Count consecutive valid samples.
Reset on failure.
Confirm only after enough consecutive valid samples.

Emit callback only on state change.
Reimplemented in SwingProcessor.
Definition at line 32 of file positiondetector.cpp.

10.14.4.2 resetState()

```
void PositionDetector::resetState ( )
```

Clears stability count and confirmed state so the callback can re-fire after a cycle reset while the device stays at the same position. Reset stability tracking and confirmation state.
Definition at line 25 of file positiondetector.cpp.

10.14.4.3 setCallback()

```
void PositionDetector::setCallback (
            StateCallback cb )  [inline]
```

Definition at line 51 of file positiondetector.hpp.

10.14.4.4 wrapAngleDeg()

```
float PositionDetector::wrapAngleDeg (
            float a )  [static], [private]
```

Declares the helper function and private members used to store configuration, reference values, stability count, state, and callback data.
Wrap angle into [-180, 180] range.
Definition at line 17 of file positiondetector.cpp.

10.14.5 Member Data Documentation

10.14.5.1 callback

```
StateCallback PositionDetector::callback  [private]
```

Definition at line 64 of file positiondetector.hpp.

10.14.5.2 cfg

```
Config PositionDetector::cfg  [private]
```

Definition at line 60 of file positiondetector.hpp.

10.14.5.3 confirmed

```
bool PositionDetector::confirmed  [private]
```

Definition at line 63 of file positiondetector.hpp.

10.14.5.4 nRefX

```
float PositionDetector::nRefX  [private]
```

Definition at line 61 of file positiondetector.hpp.

10.14.5.5 nRefY

```
float PositionDetector::nRefY  [private]
```

Definition at line 61 of file positiondetector.hpp.

10.14.5.6 nRefZ

```
float PositionDetector::nRefZ  [private]
```

Definition at line 61 of file positiondetector.hpp.

10.14.5.7 stableCount

`int PositionDetector::stableCount [private]`
normalised ref vector, computed from cfg
Definition at line 62 of file positiondetector.hpp.
The documentation for this class was generated from the following files:

- src/libs/IMU_math/Include/positiondetector.hpp
- src/libs/IMU_math/positiondetector.cpp

10.15 RPI_PWM Class Reference

`#include <rpi_pwm.h>`

Public Member Functions

- RPI_PWM ()
- bool start ()
- void stop ()

  Stops the PWM, turns the output off, and releases the GPIO resources.
- void setDutyCycle (float percent)

  Updates the PWM duty cycle and keeps it within a valid range.
- float getDutyCycle () const

  Returns the current duty cycle value.
- ~RPI_PWM ()

  Stops the PWM automatically when the object is destroyed.

Private Member Functions

- void waitUs (int tfd, int us)

  Waits for the given number of microseconds using a timer file descriptor.
- void loop ()

  Runs the PWM loop by switching the GPIO on and off based on the duty cycle.

Static Private Member Functions

- static itimerspec usToTimerspec (long us)

  Converts a time value in microseconds into a timer specification.

Private Attributes

- std::atomic< float > duty_
- std::atomic< bool > running_
- std::thread thread_
- gpiod_chip * chip_
- gpiod_line_request * request_

10.15.1 Detailed Description

Definition at line 32 of file rpi_pwm.h.

10.15.2 Constructor & Destructor Documentation

10.15.2.1 RPI_PWM()

`RPI_PWM::RPI_PWM ( ) [inline]`
Initialises the PWM class to a safe default state with no active output or GPIO resources in use.
Definition at line 38 of file rpi_pwm.h.

10.15.2.2 ~RPI_PWM()

```
RPI_PWM::~RPI_PWM ( ) [inline]
```

Stops the PWM automatically when the object is destroyed.
Definition at line 123 of file rpi_pwm.h.

10.15.3 Member Function Documentation

10.15.3.1 getDutyCycle()

```
float RPI_PWM::getDutyCycle ( ) const [inline]
```

Returns the current duty cycle value.
Definition at line 120 of file rpi_pwm.h.

10.15.3.2 loop()

```
void RPI_PWM::loop ( ) [inline], [private]
```

Runs the PWM loop by switching the GPIO on and off based on the duty cycle.
Work out the full PWM period in microseconds.
Create a timer used for accurate PWM delays.
Keep generating PWM while the class is marked as running.
Calculate the on and off times for this PWM cycle.
Keep the output fully on for a 100% duty cycle.
Keep the output fully off for a 0% duty cycle.
Otherwise, switch on for part of the period and off for the rest.
Close the timer once the loop finishes.
Definition at line 157 of file rpi_pwm.h.

10.15.3.3 setDutyCycle()

```
void RPI_PWM::setDutyCycle (
            float percent ) [inline]
```

Updates the PWM duty cycle and keeps it within a valid range.
Clamp the value so it stays between 0% and 100%.
Store the new duty cycle value.
Definition at line 110 of file rpi_pwm.h.

10.15.3.4 start()

```
bool RPI_PWM::start ( ) [inline]
```

Sets up the GPIO output and starts the PWM worker thread. Returns true if the PWM starts successfully. Open the GPIO chip used for the PWM output pin.
Create and configure the GPIO line as an output, starting inactive.
Apply the line settings to the chosen PWM GPIO line.
Create the request configuration and name this GPIO user.
Request control of the GPIO line.
Free the temporary configuration objects after the request is made.
If the line request fails, close the chip and report failure.
Mark the PWM engine as active and launch the worker thread.
Print a message to confirm that PWM has started.
Definition at line 44 of file rpi_pwm.h.

10.15.3.5 stop()

```
void RPI_PWM::stop ( ) [inline]
```

Stops the PWM, turns the output off, and releases the GPIO resources.
Mark the PWM as no longer running.
Wait for the worker thread to finish if it is active.
Set the GPIO output low, then release the line request.
Close the GPIO chip and clear the stored pointer.

Definition at line 87 of file rpi_pwm.h.

10.15.3.6 usToTimerspec()

```
static itimerspec RPI_PWM::usToTimerspec (
            long us )  [inline], [static], [private]
```

Converts a time value in microseconds into a timer specification.
Store the whole seconds part of the delay.
Store the remaining time in nanoseconds.
Disable repeating so the timer only runs once.
Definition at line 127 of file rpi_pwm.h.

10.15.3.7 waitUs()

```
void RPI_PWM::waitUs (
            int tfd,
            int us )  [inline], [private]
```

Waits for the given number of microseconds using a timer file descriptor.
Convert the delay into a timer format.
Start the timer with the requested delay.
Read from the timer so the function blocks until the timer expires.
Definition at line 144 of file rpi_pwm.h.

10.15.4 Member Data Documentation

10.15.4.1 chip_

```
gpiod_chip* RPI_PWM::chip_  [private]
```

Definition at line 200 of file rpi_pwm.h.

10.15.4.2 duty_

```
std::atomic<float> RPI_PWM::duty_  [private]
```

Definition at line 197 of file rpi_pwm.h.

10.15.4.3 request_

```
gpiod_line_request* RPI_PWM::request_  [private]
```

Definition at line 201 of file rpi_pwm.h.

10.15.4.4 running_

```
std::atomic<bool> RPI_PWM::running_  [private]
```

Definition at line 198 of file rpi_pwm.h.

10.15.4.5 thread_

```
std::thread RPI_PWM::thread_  [private]
```

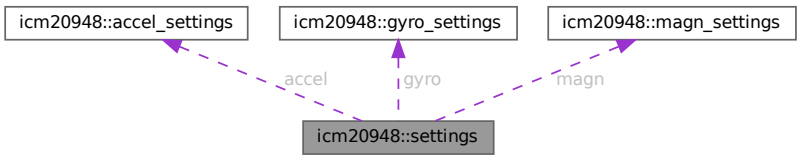
Definition at line 199 of file rpi_pwm.h.
The documentation for this class was generated from the following file:

- src/libs/Motor/Include/rpi_pwm.h

10.16 icm20948::settings Struct Reference

```
#include <Icm20948driver.hpp>
```

Collaboration diagram for icm20948::settings:



Public Member Functions

- settings (accel_settings accel=accel_settings(), gyro_settings gyro=gyro_settings(), magn_settings magn=magn_settings())
- settings (YAML::Node config_file_node)

  Parse driver settings from YAML.

Public Attributes

- accel_settings accel
- gyro_settings gyro
- magn_settings magn

10.16.1 Detailed Description

Aggregate device configuration. This gives client code a single configuration object, which improves readability and makes the driver easier to construct and reconfigure.
Definition at line 226 of file Icm20948driver.hpp.

10.16.2 Constructor & Destructor Documentation

10.16.2.1 settings() [1/2]

```
icm20948::settings::settings (
            accel_settings accel = accel_settings(),
            gyro_settings gyro = gyro_settings(),
            magn_settings magn = magn_settings() )  [inline]
```

Definition at line 231 of file Icm20948driver.hpp.

10.16.2.2 settings() [2/2]

```
icm20948::settings::settings (
            YAML::Node config_file_node )  [explicit]
```

Parse driver settings from YAML.
Construct settings from a YAML config node. This supports configuration-driven setup without hard-coding all parameters in application code. Magnetometer YAML config intentionally ignored for stability testing.
Definition at line 118 of file Icm20948driver.cpp.

10.16.3 Member Data Documentation

10.16.3.1 accel

```
accel_settings icm20948::settings::accel
```

Definition at line 227 of file Icm20948driver.hpp.

10.16.3.2 gyro

`gyro_settings icm20948::settings::gyro`
Definition at line 228 of file Icm20948driver.hpp.

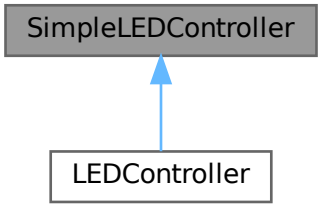
10.16.3.3 magn

`magn_settings icm20948::settings::magn`
Definition at line 229 of file Icm20948driver.hpp.
The documentation for this struct was generated from the following files:

- src/libs/IMU/Include/Icm20948driver.hpp
- src/libs/IMU/Icm20948driver.cpp

10.17 SimpleLEDController Class Reference

`#include <ledcallback.hpp>`
Inheritance diagram for SimpleLEDController:



Public Member Functions

- SimpleLEDController (int pinNo, int chipNo=0)

  Initialises the LED controller by setting up the GPIO chip, output line, and line request for the selected pin.
- virtual ~SimpleLEDController ()

  Releases the GPIO resources and makes sure the LED is turned off before closing.
- virtual void set (bool on)

  Sets the LED output state to on or off.
- virtual void flashGreen (int)
- virtual void service ()

Protected Member Functions

- SimpleLEDController ()=default

Protected Attributes

- int pin = -1
- int chipNumber = -1
- std::shared_ptr< gpiod::chip > chip

  resource lifetime is managed safely without manual memory handling.
- std::shared_ptr< gpiod::line_request > request

10.17.1 Detailed Description

This header declares a simple LED controller base class. This is to control the LED associated with the IMU swing position.
SOLID principles: S - This class is responsible only for wrapping the basic GPIO-backed LED control mechanism and exposing a small interface for derived LED classes. O - Derived classes can extend behaviour such as flashing or timed service updates without changing the base class interface. L - Client code can use this base class interface and substitute any derived LED controller that implements the same contract. I/D - Other parts of the system can depend on this small LED abstraction rather than on a larger concrete GPIO implementation. SimpleLEDController acts as a narrow abstraction for LED behaviour. Higher-level logic can depend on this interface rather than depending directly on a concrete GPIO-specific LED implementation.
Definition at line 23 of file ledcallback.hpp.

10.17.2 Constructor & Destructor Documentation

10.17.2.1 SimpleLEDController() [1/2]

```
SimpleLEDController::SimpleLEDController (
            int pinNo,
            int chipNo = 0 )
```

Initialises the LED controller by setting up the GPIO chip, output line, and line request for the selected pin.
Construct the controller with the GPIO pin number and optional chip number. The dependency on the physical GPIO configuration is contained here so that client code does not need to manage those hardware details itself.
This file implements the simple LED controller class. It sets up the GPIO line for LED output, releases the GPIO resources when finished, and allows the LED state to be switched on or off.
Definition at line 12 of file ledcallback.cpp.

10.17.2.2 ~SimpleLEDController()

```
SimpleLEDController::~SimpleLEDController ( ) [virtual]
```

Releases the GPIO resources and makes sure the LED is turned off before closing.
Virtual destructor ensures correct cleanup through the base-class interface, supporting safe substitution of derived LED controller types.
Definition at line 38 of file ledcallback.cpp.

10.17.2.3 SimpleLEDController() [2/2]

```
SimpleLEDController::SimpleLEDController ( ) [protected], [default]
```

Protected default constructor allows derived classes controlled access without exposing incomplete construction to general client code.

10.17.3 Member Function Documentation

10.17.3.1 flashGreen()

```
virtual void SimpleLEDController::flashGreen (
            int ) [inline], [virtual]
```

Optional extension point for derived classes that support timed flashing. Default implementation does nothing so derived classes are not forced to implement behaviour they do not need.
Reimplemented in LEDController.
Definition at line 42 of file ledcallback.hpp.

10.17.3.2 service()

```
virtual void SimpleLEDController::service ( ) [inline], [virtual]
```

Optional extension point for derived classes that need periodic servicing, such as time-based LED state updates.
Reimplemented in LEDController.
Definition at line 46 of file ledcallback.hpp.

10.17.3.3 set()

```
void SimpleLEDController::set (
            bool on ) [virtual]
```

Sets the LED output state to on or off.
Minimal public LED interface: callers express intent (on/off) while the hardware-specific details remain encapsulated inside the implementation.
Reimplemented in LEDController.
Definition at line 55 of file ledcallback.cpp.

10.17.4 Member Data Documentation

10.17.4.1 chip

```
std::shared_ptr<gpiod::chip> SimpleLEDController::chip [protected]
```

resource lifetime is managed safely without manual memory handling.
Shared ownership is used for the GPIO chip/request resources so that the
Definition at line 60 of file ledcallback.hpp.

10.17.4.2 chipNumber

```
int SimpleLEDController::chipNumber = -1 [protected]
```

Definition at line 56 of file ledcallback.hpp.

10.17.4.3 pin

```
int SimpleLEDController::pin = -1 [protected]
```

Internal GPIO configuration state is protected so it is available to derived implementations, while still hidden from external users.
Definition at line 55 of file ledcallback.hpp.

10.17.4.4 request

```
std::shared_ptr<gpiod::line_request> SimpleLEDController::request [protected]
```

Definition at line 61 of file ledcallback.hpp.
The documentation for this class was generated from the following files:

- src/libs/LEDs/Include/ledcallback.hpp
- src/libs/LEDs/ledcallback.cpp

10.18 SwingCalibrator Class Reference

```
#include <SwingCalibrator.hpp>
```

Public Types

- using DoneCallback = std::function< void(float bx, float by, float bz, bool initial)>

Public Member Functions

- SwingCalibrator (int initial_samples=200, int recal_samples=50)

  Start in initial-calibration mode with empty accumulators and zero bias.
- void setCallback (DoneCallback cb)

  Register the callback fired when a calibration pass completes.
- bool feed (float ax, float ay, float az)
- void triggerRecal ()
- bool isReady () const
- float biasX () const
- float biasY () const
- float biasZ () const

Private Types

- enum class State { INITIAL , READY , RECAL }

Private Member Functions

- void accumulate (float ax, float ay, float az, int target, bool initial)

  Accumulate one sample and finalise when target samples are reached.

Private Attributes

- int initial_samples_
- int recal_samples_
- State state_
- int count_
- float sum_x_

  Running sums for the current calibration window.
- float sum_y_
- float sum_z_
- float bias_x_

  Latest completed bias estimate.
- float bias_y_
- float bias_z_
- DoneCallback callback_

10.18.1 Detailed Description

SOLID Principles: S - This file has exactly one function - to remove the gravitational bias from the acceleration value of each axis. I - The accumulate() is kept private as it is only used for moving average calibration, other caller doesn't need it. D - The std::function, which is an abstraction, allows the SwingCalibrator to do its job without needing to know the caller or the results. Manages gravity-bias calibration for the swing detector.
Usage:

1. Construct and set a callback.
2. Call feed() on every IMU sample.
   - While calibrating, feed() returns true (caller should skip measurement).
   - Once done, feed() returns false and the bias is ready via biasX/Y/Z().
3. Call triggerRecal() whenever the position detector confirms the start position. The next RECAL_SAMPLES samples will be used to refresh the bias.

Definition at line 24 of file SwingCalibrator.hpp.

10.18.2 Member Typedef Documentation

10.18.2.1 DoneCallback

```
using SwingCalibrator::DoneCallback = std::function<void(float bx, float by, float bz, bool
initial)>
```

Fired when a calibration phase completes. bx/by/bz – new gravity bias (m/s^2) initial – true for the first calibration, false for recalibrations
Definition at line 30 of file SwingCalibrator.hpp.

10.18.3 Member Enumeration Documentation

10.18.3.1 State

```
enum class SwingCalibrator::State  [strong], [private]
```

Enumerator

INITIAL	
READY	
RECAL	

Definition at line 52 of file SwingCalibrator.hpp.

10.18.4 Constructor & Destructor Documentation

10.18.4.1 SwingCalibrator()

```
SwingCalibrator::SwingCalibrator (
            int initial_samples = 200,
            int recal_samples = 50 )  [explicit]
```

Start in initial-calibration mode with empty accumulators and zero bias.
Definition at line 4 of file SwingCalibrator.cpp.

10.18.5 Member Function Documentation

10.18.5.1 accumulate()

```
void SwingCalibrator::accumulate (
            float ax,
            float ay,
            float az,
            int target,
            bool initial )  [private]
```

Accumulate one sample and finalise when target samples are reached.
Update running sums for the current calibration window.
Finalise once the target sample count is reached.
Reset state for normal operation.
Notify caller that a new bias is available.
Definition at line 58 of file SwingCalibrator.cpp.

10.18.5.2 biasX()

```
float SwingCalibrator::biasX ( ) const  [inline]
```

Definition at line 47 of file SwingCalibrator.hpp.

10.18.5.3 biasY()

```
float SwingCalibrator::biasY ( ) const  [inline]
```

Definition at line 48 of file SwingCalibrator.hpp.

10.18.5.4 biasZ()

```
float SwingCalibrator::biasZ ( ) const  [inline]
```

Definition at line 49 of file SwingCalibrator.hpp.

10.18.5.5 feed()

```
bool SwingCalibrator::feed (
            float ax,
            float ay,
            float az )
```

Feed one raw accel sample (m/s^2). Returns true $\rightarrow$ sample was consumed for calibration; skip measurement output. Returns false $\rightarrow$ calibration is idle; use biasX/Y/Z() to correct the sample. Consume samples for initial calibration.

Consume samples for recalibration.
Calibration idle; caller can use the current bias.
Definition at line 20 of file SwingCalibrator.cpp.

10.18.5.6 isReady()

```
bool SwingCalibrator::isReady ( ) const
```
Definition at line 53 of file SwingCalibrator.cpp.

10.18.5.7 setCallback()

```
void SwingCalibrator::setCallback (
            DoneCallback cb )
```
Register the callback fired when a calibration pass completes.
Register completion callback.
Definition at line 14 of file SwingCalibrator.cpp.

10.18.5.8 triggerRecal()

```
void SwingCalibrator::triggerRecal ( )
```
Request a recalibration burst (e.g. triggered by the position detector). Ignored if a calibration phase is already in progress. Ignore if a calibration pass is already in progress.
already calibrating
Reset accumulators and enter recalibration state.
Definition at line 41 of file SwingCalibrator.cpp.

10.18.6 Member Data Documentation

10.18.6.1 bias_x_

```
float SwingCalibrator::bias_x_ [private]
```
Latest completed bias estimate.
Definition at line 63 of file SwingCalibrator.hpp.

10.18.6.2 bias_y_

```
float SwingCalibrator::bias_y_ [private]
```
Definition at line 63 of file SwingCalibrator.hpp.

10.18.6.3 bias_z_

```
float SwingCalibrator::bias_z_ [private]
```
Definition at line 63 of file SwingCalibrator.hpp.

10.18.6.4 callback_

```
DoneCallback SwingCalibrator::callback_ [private]
```
Definition at line 65 of file SwingCalibrator.hpp.

10.18.6.5 count_

```
int SwingCalibrator::count_ [private]
```
Definition at line 57 of file SwingCalibrator.hpp.

10.18.6.6 initial_samples_

```
int SwingCalibrator::initial_samples_ [private]
```
Definition at line 54 of file SwingCalibrator.hpp.

10.18.6.7 recal_samples_

`int SwingCalibrator::recal_samples_ [private]`
Definition at line 55 of file SwingCalibrator.hpp.

10.18.6.8 state_

`State SwingCalibrator::state_ [private]`
Definition at line 56 of file SwingCalibrator.hpp.

10.18.6.9 sum_x_

`float SwingCalibrator::sum_x_ [private]`
Running sums for the current calibration window.
Definition at line 60 of file SwingCalibrator.hpp.

10.18.6.10 sum_y_

`float SwingCalibrator::sum_y_ [private]`
Definition at line 60 of file SwingCalibrator.hpp.

10.18.6.11 sum_z_

`float SwingCalibrator::sum_z_ [private]`
Definition at line 60 of file SwingCalibrator.hpp.
The documentation for this class was generated from the following files:

- src/libs/IMU_math/Include/SwingCalibrator.hpp
- src/libs/IMU_math/SwingCalibrator.cpp

10.19 SwingDetector Class Reference

`#include <SwingDetector.hpp>`

Public Types

- using EventCallback = std::function< void(const char *level)>

Public Member Functions

- SwingDetector ()

  Start with no previously emitted level.
- void setCallback (EventCallback cb)

  Register the callback fired on level changes.
- void detect (float accel_mag)

  Feed one accel magnitude sample (m/s^2). Fires callback on level change.
- void reset ()

  Reset last known level so the next detect() always fires a callback.

Private Types

- enum class Level { NONE , LOW , MEDIUM , HIGH }

Static Private Member Functions

- static Level classify (float accel_mag)

  Convert accel magnitude to a duty-cycle band.
- static const char * label (Level level)

  Convert enum value to callback label.

Private Attributes

- Level last_level_

  Last emitted band.
- EventCallback callback_

10.19.1 Detailed Description

SOLID principles: S - SwingDetector categorised the swing speed intro 3 levels, and creates a callback when the level changes. This actually does 2 things but it is very small so we decided to wrap them up with each other. I - The calculation of the level classification in made private as other caller doesn't need to know. D - The std::function is an abstraction which allows for any inputs to the function. Classifies linear acceleration magnitude into duty cycle levels. Fires the registered callback only when the level changes.
Thresholds (m/s^2): a >= 25 → "Highest Duty Cycle" 15 <= a < 25 → "Medium Duty Cycle" 5 <= a < 15 → "Low Duty Cycle" a < 5 → "No Duty Cycle"
Definition at line 21 of file SwingDetector.hpp.

10.19.2 Member Typedef Documentation

10.19.2.1 EventCallback

```
using SwingDetector::EventCallback = std::function<void(const char* level)>
```
Definition at line 23 of file SwingDetector.hpp.

10.19.3 Member Enumeration Documentation

10.19.3.1 Level

```
enum class SwingDetector::Level  [strong], [private]
```

Enumerator

NONE	
LOW	
MEDIUM	
HIGH	

Definition at line 36 of file SwingDetector.hpp.

10.19.4 Constructor & Destructor Documentation

10.19.4.1 SwingDetector()

```
SwingDetector::SwingDetector ( )
```
Start with no previously emitted level.
Definition at line 4 of file SwingDetector.cpp.

10.19.5 Member Function Documentation

10.19.5.1 classify()

```
SwingDetector::Level SwingDetector::classify (
            float accel_mag )  [static], [private]
```
Convert accel magnitude to a duty-cycle band.
Map acceleration magnitude to a discrete level.
Definition at line 32 of file SwingDetector.cpp.

10.19.5.2 detect()

```
void SwingDetector::detect (
```

```
            float accel_mag )
```

Feed one accel magnitude sample (m/s^2). Fires callback on level change.
Classify the current sample into a duty-cycle band.
Only emit when the band changes.
Definition at line 20 of file SwingDetector.cpp.

10.19.5.3 label()

```
const char * SwingDetector::label (
            Level level )  [static], [private]
```

Convert enum value to callback label.
Convert enum level to the external string label.
Definition at line 41 of file SwingDetector.cpp.

10.19.5.4 reset()

```
void SwingDetector::reset ( )
```

Reset last known level so the next detect() always fires a callback.
Clear cached state so the next detect() call always emits.
Definition at line 15 of file SwingDetector.cpp.

10.19.5.5 setCallback()

```
void SwingDetector::setCallback (
            EventCallback cb )
```

Register the callback fired on level changes.
Definition at line 9 of file SwingDetector.cpp.

10.19.6 Member Data Documentation

10.19.6.1 callback_

```
EventCallback SwingDetector::callback_  [private]
```

Definition at line 47 of file SwingDetector.hpp.

10.19.6.2 last_level_

```
Level SwingDetector::last_level_  [private]
```

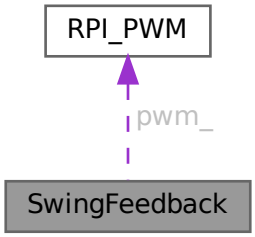
Last emitted band.
Definition at line 45 of file SwingDetector.hpp.
The documentation for this class was generated from the following files:

- src/libs/IMU_math/Include/SwingDetector.hpp
- src/libs/IMU_math/SwingDetector.cpp

10.20 SwingFeedback Class Reference

Drives PWM feedback from detected swing levels.
`#include <swingfeedback.h>`

Collaboration diagram for SwingFeedback:



Public Types

- using ResetCallback = std::function< void() >

  Fired when the feedback cycle times out and needs a reset.

Public Member Functions

- SwingFeedback (RPI_PWM &pwm)

  Initialises the feedback object and sets the PWM output to off.
- void setResetCallback (ResetCallback cb)

  Stores the callback used when the feedback state is reset.
- void checkTimeout ()

  Checks whether the feedback has been active for too long and forces it off if needed.
- void onLevel (const char *level)

  Receives a feedback level, stores the highest pending level, and starts the worker if needed.
- void forceOff ()

  Forces the PWM output off, clears the busy state, and triggers the reset callback.

Private Member Functions

- void worker ()

  Internal worker for applying queued feedback updates.

Private Attributes

- RPI_PWM & pwm_

  PWM output owned externally.
- ResetCallback reset_cb_

  Optional callback for timeout/reset handling.
- std::atomic< bool > busy_ {false}

  True while a feedback cycle is active.
- std::atomic< int > pending_level_ {0}

  Pending duty-cycle level to apply.
- std::atomic< int64_t > busy_since_ms_ {0}

Static Private Attributes

- static constexpr int64_t TIMEOUT_MS = 3000

10.20.1 Detailed Description

Drives PWM feedback from detected swing levels.
Definition at line 17 of file swingfeedback.h.

10.20.2 Member Typedef Documentation

10.20.2.1 ResetCallback

```
using SwingFeedback::ResetCallback = std::function<void()>
```

Fired when the feedback cycle times out and needs a reset.
Definition at line 20 of file swingfeedback.h.

10.20.3 Constructor & Destructor Documentation

10.20.3.1 SwingFeedback()

```
SwingFeedback::SwingFeedback (
            RPI_PWM & pwm )  [explicit]
```

Initialises the feedback object and sets the PWM output to off.
Definition at line 102 of file swingfeedback.cpp.

10.20.4 Member Function Documentation

10.20.4.1 checkTimeout()

```
void SwingFeedback::checkTimeout ( )
```

Checks whether the feedback has been active for too long and forces it off if needed.
Definition at line 114 of file swingfeedback.cpp.

10.20.4.2 forceOff()

```
void SwingFeedback::forceOff ( )
```

Forces the PWM output off, clears the busy state, and triggers the reset callback.
Definition at line 172 of file swingfeedback.cpp.

10.20.4.3 onLevel()

```
void SwingFeedback::onLevel (
            const char * level )
```

Receives a feedback level, stores the highest pending level, and starts the worker if needed.
Level dropped to zero — swing has ended. Fire the worker now so it plays the actual peak that was accumulated, not just the first edge.
Accumulate the highest level reached during the swing. Do NOT spawn the worker yet — wait for the swing to end (rank == 0 above).
Definition at line 149 of file swingfeedback.cpp.

10.20.4.4 setResetCallback()

```
void SwingFeedback::setResetCallback (
            ResetCallback cb )
```

Stores the callback used when the feedback state is reset.
Definition at line 109 of file swingfeedback.cpp.

10.20.4.5 worker()

```
void SwingFeedback::worker ( )  [private]
```

Internal worker for applying queued feedback updates.
Runs the feedback output, updates the busy state, and calls the reset callback when finished. Clear any level that accumulated during playback so it doesn't contaminate the next swing.
Definition at line 125 of file swingfeedback.cpp.

10.20.5 Member Data Documentation

10.20.5.1 busy_

`std::atomic<bool> SwingFeedback::busy_ {false} [private]`
True while a feedback cycle is active.
Definition at line 40 of file swingfeedback.h.

10.20.5.2 busy_since_ms_

`std::atomic<int64_t> SwingFeedback::busy_since_ms_ {0} [private]`
Start time of the active feedback cycle in ms since epoch. Zero means idle.
Definition at line 47 of file swingfeedback.h.

10.20.5.3 pending_level_

`std::atomic<int> SwingFeedback::pending_level_ {0} [private]`
Pending duty-cycle level to apply.
Definition at line 43 of file swingfeedback.h.

10.20.5.4 pwm_

`RPI_PWM& SwingFeedback::pwm_ [private]`
PWM output owned externally.
Definition at line 34 of file swingfeedback.h.

10.20.5.5 reset_cb_

`ResetCallback SwingFeedback::reset_cb_ [private]`
Optional callback for timeout/reset handling.
Definition at line 37 of file swingfeedback.h.

10.20.5.6 TIMEOUT_MS

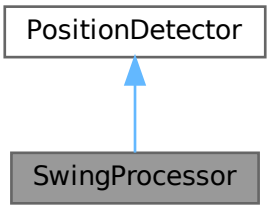
`constexpr int64_t SwingFeedback::TIMEOUT_MS = 3000 [static], [constexpr], [private]`
Definition at line 49 of file swingfeedback.h.
The documentation for this class was generated from the following files:

- src/libs/Motor/Include/swingfeedback.h
- src/libs/Motor/swingfeedback.cpp

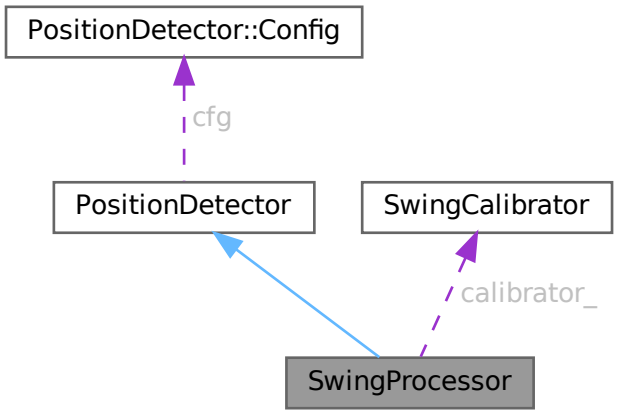
10.21 SwingProcessor Class Reference

raised to reject motor EMI noise floor
`#include <SwingProcessor.hpp>`

Inheritance diagram for SwingProcessor:



Collaboration diagram for SwingProcessor:



Public Types

- using MagnitudeCallback = std::function< void(float accel_mag)>
- using PositionStateCallback = std::function< void(bool upright)>

Public Types inherited from PositionDetector

- using StateCallback = std::function< void(bool upright)>

Public Member Functions

- SwingProcessor (float accel_deadband=ACCEL_DEADBAND, int calib_samples=CALIB_SAMPLES, int recal_samples=RECAL_SAMPLES)

  Initialise the deadband and embedded gravity-bias calibrator.
- void setMagnitudeCallback (MagnitudeCallback cb)

  Register callback for filtered linear-acceleration magnitude.
- void setPositionCallback (PositionStateCallback cb)

  Register callback for upright/not-upright state changes.

- void onForceReady (bool correctForce)
- bool isActive () const
- void onPositionReady (bool upright)
- void reset ()
- void onSample (float ax, float ay, float az, float mx, float my) override
- void operator() (float ax, float ay, float az, float gx, float gy, float gz, float mx, float my)

Public Member Functions inherited from PositionDetector

- PositionDetector ()

  Default constructor delegates to config-based constructor.
- PositionDetector (const Config &c)
- virtual ~PositionDetector ()=default
- void setCallback (StateCallback cb)
- void resetState ()

Private Attributes

- float accel_deadband_

  Declares the private members used to store thresholds, calibration, callbacks, activity state, and reset timing.
- SwingCalibrator calibrator_
- MagnitudeCallback mag_callback_
- PositionStateCallback pos_callback_
- bool isArmed_ = false
- bool isActive_ = false

  position confirmed, ready for piezo
- bool hasLeftStart_ = false

  gate open, motor feedback running
- std::chrono::steady_clock::time_point reset_time_ {}

Static Private Attributes

- static constexpr int COOLDOWN_MS = 500

  paddle has left the starting position since activation

10.21.1 Detailed Description

raised to reject motor EMI noise floor
Gate logic:

1. When the correct start position is held, isArmed_ becomes true and the bias is recalibrated. The system must still be in this valid position when the piezo is triggered.

2. When the piezo is pressed while armed, isActive_ becomes true and the motor feedback starts. Once active, any further piezo events are ignored to avoid false triggers caused by motor vibration.

3. When the SwingFeedback timer ends, reset() is called to clear both state flags. Declares the swing processor class, including its callback types and main functions for handling force, position, reset, and sensor sample processing.

Definition at line 31 of file SwingProcessor.hpp.

10.21.2 Member Typedef Documentation

10.21.2.1 MagnitudeCallback

```
using SwingProcessor::MagnitudeCallback = std::function<void(float accel_mag)>
```

Definition at line 33 of file SwingProcessor.hpp.

10.21.2.2 PositionStateCallback

```
using SwingProcessor::PositionStateCallback = std::function<void(bool upright)>
```

Definition at line 34 of file SwingProcessor.hpp.

10.21.3 Constructor & Destructor Documentation

10.21.3.1 SwingProcessor()

```
SwingProcessor::SwingProcessor (
            float accel_deadband = ACCEL_DEADBAND,
            int calib_samples = CALIB_SAMPLES,
            int recal_samples = RECAL_SAMPLES )  [explicit]
```

Initialise the deadband and embedded gravity-bias calibrator.
Default no-op calibration callback.
Forward position events to any external callback, then update swing state.
Definition at line 6 of file SwingProcessor.cpp.

10.21.4 Member Function Documentation

10.21.4.1 isActive()

```
bool SwingProcessor::isActive ( ) const  [inline]
```

Definition at line 44 of file SwingProcessor.hpp.

10.21.4.2 onForceReady()

```
void SwingProcessor::onForceReady (
            bool correctForce )
```

Drop output to zero if force is lost mid-swing.
Ignore force events unless the system is armed from the start position.
Ignore duplicate triggers while already active.
Force confirmation starts the swing.
must leave start position before magnitude is measured
Definition at line 54 of file SwingProcessor.cpp.

10.21.4.3 onPositionReady()

```
void SwingProcessor::onPositionReady (
            bool upright )
```

Ignore position events during the post-reset cooldown.
Reaching the start position arms the swing and refreshes bias.
Left the start position before force confirmation; cancel arming.
Left position before piezo — disarm
Left position after piezo — swing has begun
If already active, leaving the start position is expected.
Definition at line 32 of file SwingProcessor.cpp.

10.21.4.4 onSample()

```
void SwingProcessor::onSample (
            float ax,
            float ay,
            float az,
            float mx,
            float my )  [override], [virtual]
```

Feed the sample into the position detector first.
Only process magnitude while a swing is active.
don't measure until bat leaves start position
Hold output until the current calibration window completes.

Remove gravity bias to estimate linear acceleration.
Collapse 3D linear acceleration into a scalar magnitude.
Clamp small values to zero to suppress noise.
Reimplemented from PositionDetector.
Definition at line 85 of file SwingProcessor.cpp.

10.21.4.5 operator()()

```
void SwingProcessor::operator() (
            float ax,
            float ay,
            float az,
            float gx,
            float gy,
            float gz,
            float mx,
            float my )
```

Adapter for callback-style IMU interfaces.
Definition at line 113 of file SwingProcessor.cpp.

10.21.4.6 reset()

```
void SwingProcessor::reset ( )
```

Clear swing state and restart cooldown.
Definition at line 76 of file SwingProcessor.cpp.

10.21.4.7 setMagnitudeCallback()

```
void SwingProcessor::setMagnitudeCallback (
            MagnitudeCallback cb )
```

Register callback for filtered linear-acceleration magnitude.
Definition at line 20 of file SwingProcessor.cpp.

10.21.4.8 setPositionCallback()

```
void SwingProcessor::setPositionCallback (
            PositionStateCallback cb )
```

Register callback for upright/not-upright state changes.
Definition at line 26 of file SwingProcessor.cpp.

10.21.5 Member Data Documentation

10.21.5.1 accel_deadband_

```
float SwingProcessor::accel_deadband_ [private]
```

Declares the private members used to store thresholds, calibration, callbacks, activity state, and reset timing.
Definition at line 57 of file SwingProcessor.hpp.

10.21.5.2 calibrator_

```
SwingCalibrator SwingProcessor::calibrator_ [private]
```

Definition at line 58 of file SwingProcessor.hpp.

10.21.5.3 COOLDOWN_MS

```
constexpr int SwingProcessor::COOLDOWN_MS = 500 [static], [constexpr], [private]
```

paddle has left the starting position since activation
Definition at line 67 of file SwingProcessor.hpp.

10.21.5.4 hasLeftStart_

`bool SwingProcessor::hasLeftStart_ = false [private]`
gate open, motor feedback running
Definition at line 64 of file SwingProcessor.hpp.

10.21.5.5 isActive_

`bool SwingProcessor::isActive_ = false [private]`
position confirmed, ready for piezo
Definition at line 63 of file SwingProcessor.hpp.

10.21.5.6 isArmed_

`bool SwingProcessor::isArmed_ = false [private]`
Definition at line 62 of file SwingProcessor.hpp.

10.21.5.7 mag_callback_

`MagnitudeCallback SwingProcessor::mag_callback_ [private]`
Definition at line 59 of file SwingProcessor.hpp.

10.21.5.8 pos_callback_

`PositionStateCallback SwingProcessor::pos_callback_ [private]`
Definition at line 60 of file SwingProcessor.hpp.

10.21.5.9 reset_time_

`std::chrono::steady_clock::time_point SwingProcessor::reset_time_ {} [private]`
Definition at line 68 of file SwingProcessor.hpp.
The documentation for this class was generated from the following files:

- src/libs/IMU_math/Include/SwingProcessor.hpp
- src/libs/IMU_math/SwingProcessor.cpp

Chapter 11

File Documentation

11.1 src/CMakeFiles/3.31.6/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- #define __has_include(x) 0
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X) #X
- #define STRINGIFY(X) STRINGIFY_HELPER(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define CXX_STD_98 199711L
- #define CXX_STD_11 201103L
- #define CXX_STD_14 201402L
- #define CXX_STD_17 201703L
- #define CXX_STD_20 202002L
- #define CXX_STD_23 202302L
- #define CXX_STD __cplusplus

Functions

- int main (int argc, char *argv[ ])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

11.1.1 Macro Definition Documentation

11.1.1.1 __has_include

```
#define __has_include(
            x ) 0
```

Definition at line 11 of file CMakeCXXCompilerId.cpp.

11.1.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 724 of file CMakeCXXCompilerId.cpp.

11.1.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 427 of file CMakeCXXCompilerId.cpp.

11.1.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

Definition at line 861 of file CMakeCXXCompilerId.cpp.

11.1.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

Definition at line 810 of file CMakeCXXCompilerId.cpp.

11.1.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

Definition at line 811 of file CMakeCXXCompilerId.cpp.

11.1.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

Definition at line 812 of file CMakeCXXCompilerId.cpp.

11.1.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

Definition at line 813 of file CMakeCXXCompilerId.cpp.

11.1.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

Definition at line 814 of file CMakeCXXCompilerId.cpp.

11.1.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

Definition at line 809 of file CMakeCXXCompilerId.cpp.

11.1.1.11 DEC

```
#define DEC(
              n )
```

Value:

```
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))
```

Definition at line 728 of file CMakeCXXCompilerId.cpp.

11.1.1.12 HEX

```
#define HEX(
            n )
```

Value:

```
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))
```

Definition at line 739 of file CMakeCXXCompilerId.cpp.

11.1.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 558 of file CMakeCXXCompilerId.cpp.

11.1.1.14 STRINGIFY

```
#define STRINGIFY(
            X ) STRINGIFY_HELPER(X)
```

Definition at line 448 of file CMakeCXXCompilerId.cpp.

11.1.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X ) #X
```

Definition at line 447 of file CMakeCXXCompilerId.cpp.

11.1.2 Function Documentation

11.1.2.1 main()

```
int main (
            int argc,
            char * argv[] )
```

Definition at line 894 of file CMakeCXXCompilerId.cpp.

11.1.3 Variable Documentation

11.1.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 805 of file CMakeCXXCompilerId.cpp.

11.1.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 434 of file CMakeCXXCompilerId.cpp.

11.1.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
= "INFO" ":" "extensions_default["



  "OFF"

"]"
```

Definition at line 882 of file CMakeCXXCompilerId.cpp.

11.1.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
= "INFO" ":" "standard_default["
```

```
  "98"
"]"
```

Definition at line 864 of file CMakeCXXCompilerId.cpp.

11.1.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 804 of file CMakeCXXCompilerId.cpp.

11.2 CMakeCXXCompilerId.cpp

Go to the documentation of this file.

```
00001 /* This source file must have a .cpp extension so that all C++ compilers
00002    recognize the extension without flags.  Borland does not know .cxx for
00003    example.  */
00004 #ifndef __cplusplus
00005 # error "A C compiler has been selected for C++."
00006 #endif
00007
00008 #if !defined(__has_include)
00009 /* If the compiler does not have __has_include, pretend the answer is
00010    always no.  */
00011 #  define __has_include(x) 0
00012 #endif
00013
00014
00015 /* Version number components: V=Version, R=Revision, P=Patch
00016    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00017
00018 #if defined(__INTEL_COMPILER) || defined(__ICC)
00019 # define COMPILER_ID "Intel"
00020 # if defined(_MSC_VER)
00021 #  define SIMULATE_ID "MSVC"
00022 # endif
00023 # if defined(__GNUC__)
00024 #  define SIMULATE_ID "GNU"
00025 # endif
00026   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00027      except that a few beta releases use the old format with V=2021.  */
00028 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00029 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00030 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00031 #  if defined(__INTEL_COMPILER_UPDATE)
00032 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00033 #  else
00034 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00035 #  endif
00036 # else
00037 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
00038 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
00039    /* The third version component from --version is an update index,
00040       but no macro is provided for it.  */
00041 #  define COMPILER_VERSION_PATCH DEC(0)
00042 # endif
00043 # if defined(__INTEL_COMPILER_BUILD_DATE)
00044    /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
00045 #  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
00046 # endif
```

```
# if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
  /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
  /* __WATCOMC__ = VVRP + 1100 */
```

```
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_CC)
# define COMPILER_ID "SunPro"
# if __SUNPRO_CC >= 0x5100
   /* __SUNPRO_CC = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# endif

#elif defined(__HP_aCC)
# define COMPILER_ID "HP"
  /* __HP_aCC = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_aCC/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_aCC/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_aCC     % 100)

#elif defined(__DECCXX)
# define COMPILER_ID "Compaq"
  /* __DECCXX_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECCXX_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECCXX_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECCXX_VER         % 10000)

#elif defined(__IBMCPP__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ >= 800
# define COMPILER_ID "XL"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

```

```
#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
```

```
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__) || defined(__GNUG__)
# define COMPILER_ID "GNU"
# if defined(__GNUC__)
#  define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# else
#  define COMPILER_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

```

```
#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
    defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
    defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif


/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"
```

```

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
```

```
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"
```

```

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                   \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)                   \
  ('0' + ((n)>>28 & 0xF)), \
```

```
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define CXX_STD_98 199711L
#define CXX_STD_11 201103L
#define CXX_STD_14 201402L
#define CXX_STD_17 201703L
#define CXX_STD_20 202002L
#define CXX_STD_23 202302L

#if defined(__INTEL_COMPILER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > CXX_STD_17
#    define CXX_STD _MSVC_LANG
#  elif _MSVC_LANG == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14 && __cplusplus > CXX_STD_17
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14
#    define CXX_STD CXX_STD_17
#  elif defined(__INTEL_CXX11_MODE__) && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
```

```
#  elif defined(__INTEL_CXX11_MODE__)
#    define CXX_STD CXX_STD_11
#  else
#    define CXX_STD CXX_STD_98
#  endif
#elif defined(_MSC_VER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > __cplusplus
#    define CXX_STD _MSVC_LANG
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__NVCOMPILER)
#  if __cplusplus == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__INTEL_COMPILER) || defined(__PGI)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_namespace_attributes)
#    define CXX_STD CXX_STD_17
#  elif __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif (defined(__IBMCPP__) || defined(__ibmxl__)) && defined(__linux__)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif __cplusplus == 1 && defined(__GXX_EXPERIMENTAL_CXX0X__)
#  define CXX_STD CXX_STD_11
#else
#  define CXX_STD __cplusplus
#endif

const char* info_language_standard_default = "INFO" ":" "standard_default["
#if CXX_STD > CXX_STD_23
  "26"
#elif CXX_STD > CXX_STD_20
  "23"
#elif CXX_STD > CXX_STD_17
  "20"
#elif CXX_STD > CXX_STD_14
  "17"
#elif CXX_STD > CXX_STD_11
  "14"
#elif CXX_STD >= CXX_STD_11
  "11"
#else
  "98"
#endif
"]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||           \
     defined(__TI_COMPILER_VERSION__)) &&                                     \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

int main(int argc, char* argv[])
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#ifdef COMPILER_VERSION_INTERNAL
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
```

```
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
```

11.3 src/CMakeFiles/rtes_main.dir/link.d File Reference

11.4 link.d

Go to the documentation of this file.

```
rtes_main: \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/Scrt1.o \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/crti.o \
  /usr/lib/gcc/aarch64-linux-gnu/14/crtbeginS.o \
  CMakeFiles/rtes_main.dir/main.cpp.o \
  libs/IMU/libimu_lib.a \
  libs/IMU_math/libimu_math_lib.a \
  libs/LEDs/libleds_lib.a \
  libs/Motor/libmotor_lib.a \
  libs/Piezo/libpiezo_lib.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libyaml-cpp.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libpthread.a \
  libs/IMU/libimu_lib.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libgpiod.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libyaml-cpp.so \
  libs/LEDs/libleds_lib.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libpthread.a \
  /usr/lib/aarch64-linux-gnu/libgpiodcxx.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/libstdc++.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libm.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libm.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libm.so \
  /lib/aarch64-linux-gnu/libm.so.6 \
  /lib/aarch64-linux-gnu/libmvec.so.1 \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libgcc_s.so.1 \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libc.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libc.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libc.so \
  /lib/aarch64-linux-gnu/libc.so.6 \
  /usr/lib/aarch64-linux-gnu/libc_nonshared.a \
  /lib/ld-linux-aarch64.so.1 \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libgcc_s.so.1 \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a \
  /usr/lib/gcc/aarch64-linux-gnu/14/crtendS.o \
  /usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/crtn.o \
  /lib/ld-linux-aarch64.so.1

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/Scrt1.o:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/crti.o:

/usr/lib/gcc/aarch64-linux-gnu/14/crtbeginS.o:

CMakeFiles/rtes_main.dir/main.cpp.o:

libs/IMU/libimu_lib.a:

libs/IMU_math/libimu_math_lib.a:

libs/LEDs/libleds_lib.a:

libs/Motor/libmotor_lib.a:

libs/Piezo/libpiezo_lib.a:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libyaml-cpp.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libpthread.a:

libs/IMU/libimu_lib.a:

```

```
/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libgpiod.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libyaml-cpp.so:

libs/LEDs/libleds_lib.a:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libpthread.a:

/usr/lib/aarch64-linux-gnu/libgpiodcxx.so:

/usr/lib/gcc/aarch64-linux-gnu/14/libstdc++.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libm.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libm.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libm.so:

/lib/aarch64-linux-gnu/libm.so.6:

/lib/aarch64-linux-gnu/libmvec.so.1:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libgcc_s.so.1:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libc.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libc.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libc.so:

/lib/aarch64-linux-gnu/libc.so.6:

/usr/lib/aarch64-linux-gnu/libc_nonshared.a:

/lib/ld-linux-aarch64.so.1:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc_s.so:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/libgcc_s.so.1:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a:

/usr/lib/gcc/aarch64-linux-gnu/14/libgcc.a:

/usr/lib/gcc/aarch64-linux-gnu/14/crtendS.o:

/usr/lib/gcc/aarch64-linux-gnu/14/../../../aarch64-linux-gnu/crtn.o:

/lib/ld-linux-aarch64.so.1:
```

11.5 src/CMakeFiles/rtes_main.dir/main.cpp.o.d File Reference

11.6 main.cpp.o.d

Go to the documentation of this file.

```
CMakeFiles/rtes_main.dir/main.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/main.cpp /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/positiondetector.hpp \
 /usr/include/c++/14/cmath /usr/include/c++/14/bits/requires_hosted.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
```

```
/usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
/usr/include/c++/14/pstl/pstl_config.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
/usr/include/c++/14/bits/version.h /usr/include/c++/14/ext/type_traits.h \
/usr/include/math.h \
/usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
/usr/include/aarch64-linux-gnu/bits/types.h \
/usr/include/aarch64-linux-gnu/bits/typesizes.h \
/usr/include/aarch64-linux-gnu/bits/time64.h \
/usr/include/aarch64-linux-gnu/bits/math-vector.h \
/usr/include/aarch64-linux-gnu/bits/libm-simd-decl-stubs.h \
/usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/include/aarch64-linux-gnu/bits/flt-eval-method.h \
/usr/include/aarch64-linux-gnu/bits/fp-logb.h \
/usr/include/aarch64-linux-gnu/bits/fp-fast.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-macros.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-helper-functions.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-narrow.h \
/usr/include/aarch64-linux-gnu/bits/iscanonical.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/stdlib.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
/usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
/usr/include/aarch64-linux-gnu/sys/types.h \
/usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
/usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
/usr/include/aarch64-linux-gnu/bits/types/time_t.h \
/usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
/usr/include/aarch64-linux-gnu/bits/stdint-intn.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/endian.h \
/usr/include/aarch64-linux-gnu/bits/endianness.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/specfun.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/type_traits \
/usr/include/c++/14/bits/move.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/compare /usr/include/c++/14/concepts \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/stl_iterator.h /usr/include/c++/14/new \
/usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/limits /usr/include/c++/14/tr1/gamma.tcc \
/usr/include/c++/14/tr1/special_function_util.h \
/usr/include/c++/14/tr1/bessel_function.tcc \
/usr/include/c++/14/tr1/beta_function.tcc \
/usr/include/c++/14/tr1/ell_integral.tcc \
/usr/include/c++/14/tr1/exp_integral.tcc \
/usr/include/c++/14/tr1/hypergeometric.tcc \
/usr/include/c++/14/tr1/legendre_function.tcc \
/usr/include/c++/14/tr1/modified_bessel_func.tcc \
/usr/include/c++/14/tr1/poly_hermite.tcc \
/usr/include/c++/14/tr1/poly_laguerre.tcc \
/usr/include/c++/14/tr1/riemann_zeta.tcc /usr/include/c++/14/functional \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
```

```
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h /usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
/usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h \
/usr/include/c++/14/bits/memoryfwd.h \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/erase_if.h \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array \
/home/group5/RTES_Table_Tennis/src/libs/IMU/Include/imureader.hpp \
/usr/include/c++/14/atomic /usr/include/c++/14/bits/atomic_base.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
/usr/include/aarch64-linux-gnu/bits/wchar.h \
/usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
/usr/include/aarch64-linux-gnu/bits/stdint-least.h \
/usr/include/c++/14/bits/atomic_lockfree_defines.h \
/usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/cstdint \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h /usr/include/linux/types.h \
/usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
/usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
/usr/include/aarch64-linux-gnu/bits/cpu-set.h /usr/include/time.h \
/usr/include/aarch64-linux-gnu/bits/time.h \
/usr/include/aarch64-linux-gnu/bits/timex.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/c++/14/cerrno /usr/include/errno.h \
/usr/include/aarch64-linux-gnu/bits/errno.h /usr/include/linux/errno.h \
/usr/lib/linux/uapi/arm64/asm/errno.h /usr/include/asm-generic/errno.h \
/usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/std_mutex.h /usr/include/c++/14/thread \
/usr/include/c++/14/stop_token /usr/include/c++/14/bits/std_thread.h \
/usr/include/c++/14/iosfwd /usr/include/c++/14/bits/stringfwd.h \
/usr/include/c++/14/bits/postypes.h /usr/include/c++/14/cwchar \
/usr/include/wchar.h /usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
```

```
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/bits/unique_ptr.h /usr/include/c++/14/ostream \
/usr/include/c++/14/ios /usr/include/c++/14/exception \
/usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/basic_string.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/c++/14/cstdio /usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc \
/usr/include/c++/14/bits/ostream.tcc /usr/include/c++/14/semaphore \
/usr/include/c++/14/bits/semaphore_base.h \
/usr/include/c++/14/bits/chrono.h /usr/include/c++/14/ratio \
/usr/include/c++/14/ctime /usr/include/c++/14/bits/parse_numbers.h \
/usr/include/c++/14/bits/atomic_timed_wait.h \
/usr/include/c++/14/bits/this_thread_sleep.h \
/usr/include/aarch64-linux-gnu/sys/time.h /usr/include/semaphore.h \
/usr/include/aarch64-linux-gnu/bits/semaphore.h \
/home/group5/RTES_Table_Tennis/src/libs/IMU/Include/Icm20948driver.hpp \
/usr/include/linux/i2c-dev.h /usr/include/linux/i2c.h \
/usr/include/yaml-cpp/yaml.h /usr/include/yaml-cpp/parser.h \
/usr/include/c++/14/memory /usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/align.h /usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/ext/concurrence.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/yaml-cpp/dll.h \
/usr/include/yaml-cpp/emitter.h /usr/include/c++/14/sstream \
/usr/include/c++/14/istream /usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/yaml-cpp/binary.h \
/usr/include/yaml-cpp/emitterdef.h /usr/include/yaml-cpp/emittermanip.h \
/usr/include/yaml-cpp/null.h /usr/include/yaml-cpp/ostream_wrapper.h \
/usr/include/yaml-cpp/emitterstyle.h /usr/include/yaml-cpp/stlemitter.h \
/usr/include/c++/14/list /usr/include/c++/14/bits/stl_list.h \
/usr/include/c++/14/bits/list.tcc /usr/include/c++/14/set \
/usr/include/c++/14/bits/stl_tree.h /usr/include/c++/14/bits/stl_set.h \
/usr/include/c++/14/bits/stl_multiset.h /usr/include/c++/14/map \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/yaml-cpp/exceptions.h /usr/include/yaml-cpp/mark.h \
/usr/include/yaml-cpp/noexcept.h /usr/include/yaml-cpp/traits.h \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/yaml-cpp/node/node.h \
/usr/include/yaml-cpp/node/detail/iterator_fwd.h \
/usr/include/yaml-cpp/node/ptr.h /usr/include/yaml-cpp/node/type.h \
/usr/include/yaml-cpp/node/impl.h \
/usr/include/yaml-cpp/node/detail/memory.h \
/usr/include/yaml-cpp/node/detail/node.h \
```

```
/usr/include/yaml-cpp/node/detail/node_ref.h \
/usr/include/yaml-cpp/node/detail/node_data.h \
/usr/include/yaml-cpp/node/detail/node_iterator.h \
/usr/include/c++/14/iterator /usr/include/c++/14/bits/stream_iterator.h \
/usr/include/yaml-cpp/node/iterator.h \
/usr/include/yaml-cpp/node/detail/iterator.h \
/usr/include/yaml-cpp/node/convert.h /usr/include/c++/14/valarray \
/usr/include/c++/14/algorithm /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/ranges_algo.h \
/usr/include/c++/14/pstl/glue_algorithm_defs.h \
/usr/include/c++/14/bits/valarray_array.h \
/usr/include/c++/14/bits/valarray_array.tcc \
/usr/include/c++/14/bits/valarray_before.h \
/usr/include/c++/14/bits/slice_array.h \
/usr/include/c++/14/bits/valarray_after.h \
/usr/include/c++/14/bits/gslice.h \
/usr/include/c++/14/bits/gslice_array.h \
/usr/include/c++/14/bits/mask_array.h \
/usr/include/c++/14/bits/indirect_array.h \
/usr/include/yaml-cpp/node/detail/impl.h \
/usr/include/yaml-cpp/node/parse.h /usr/include/yaml-cpp/node/emit.h \
/usr/include/gpiod.h /usr/lib/gcc/aarch64-linux-gnu/14/include/stdbool.h \
/usr/include/c++/14/future /usr/include/c++/14/mutex \
/usr/include/c++/14/bits/unique_lock.h \
/usr/include/c++/14/condition_variable \
/usr/include/c++/14/bits/atomic_futex.h \
/home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingProcessor.hpp \
/home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingCalibrator.hpp \
/usr/include/c++/14/chrono /usr/include/c++/14/bits/chrono_io.h \
/usr/include/c++/14/iomanip /usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/charconv /usr/include/c++/14/optional \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h \
/home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingDetector.hpp \
/home/group5/RTES_Table_Tennis/src/libs/Motor/Include/swingfeedback.h \
/home/group5/RTES_Table_Tennis/src/libs/Motor/Include/rpi_pwm.h \
/usr/include/c++/14/iostream \
/usr/include/aarch64-linux-gnu/sys/timerfd.h \
/usr/include/aarch64-linux-gnu/bits/timerfd.h \
/usr/include/c++/14/csignal /usr/include/signal.h \
/usr/include/aarch64-linux-gnu/bits/signum-generic.h \
/usr/include/aarch64-linux-gnu/bits/signum-arch.h \
/usr/include/aarch64-linux-gnu/bits/types/sig_atomic_t.h \
/usr/include/aarch64-linux-gnu/bits/types/siginfo_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigval_t.h \
/usr/include/aarch64-linux-gnu/bits/siginfo-arch.h \
/usr/include/aarch64-linux-gnu/bits/siginfo-consts.h \
/usr/include/aarch64-linux-gnu/bits/siginfo-consts-arch.h \
/usr/include/aarch64-linux-gnu/bits/types/sigval_t.h \
/usr/include/aarch64-linux-gnu/bits/types/sigevent_t.h \
/usr/include/aarch64-linux-gnu/bits/sigevent-consts.h \
/usr/include/aarch64-linux-gnu/bits/sigaction.h \
/usr/include/aarch64-linux-gnu/bits/sigcontext.h \
/usr/lib/linux/uapi/arm64/asm/sigcontext.h \
/usr/lib/linux/uapi/arm64/asm/sve_context.h \
/usr/include/aarch64-linux-gnu/bits/types/stack_t.h \
/usr/include/aarch64-linux-gnu/sys/ucontext.h \
/usr/include/aarch64-linux-gnu/sys/procfs.h \
/usr/include/aarch64-linux-gnu/sys/user.h \
/usr/include/aarch64-linux-gnu/bits/procfs.h \
/usr/include/aarch64-linux-gnu/bits/procfs-id.h \
/usr/include/aarch64-linux-gnu/bits/procfs-prregset.h \
/usr/include/aarch64-linux-gnu/bits/procfs-extra.h \
/usr/include/aarch64-linux-gnu/bits/sigstack.h \
/usr/include/aarch64-linux-gnu/bits/sigstksz.h \
/usr/include/aarch64-linux-gnu/bits/ss_flags.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sigstack.h \
/usr/include/aarch64-linux-gnu/bits/sigthread.h \
/usr/include/aarch64-linux-gnu/bits/signal_ext.h \
/usr/include/aarch64-linux-gnu/sys/signalfd.h \
/usr/include/aarch64-linux-gnu/bits/signalfd.h \
/home/group5/RTES_Table_Tennis/src/libs/Piezo/Include/ads1115rpi.h \
/usr/include/c++/14/stdlib.h /usr/include/assert.h \
/usr/include/gpiod.hpp /usr/include/gpiodcxx/chip.hpp \
/usr/include/c++/14/filesystem /usr/include/c++/14/bits/fs_fwd.h \
/usr/include/c++/14/bits/fs_path.h /usr/include/c++/14/codecvt \
```

```
/usr/include/c++/14/bits/fs_dir.h /usr/include/c++/14/bits/fs_ops.h \
/usr/include/gpiodcxx/line.hpp /usr/include/gpiodcxx/chip-info.hpp \
/usr/include/gpiodcxx/edge-event.hpp /usr/include/gpiodcxx/timestamp.hpp \
/usr/include/gpiodcxx/edge-event-buffer.hpp \
/usr/include/gpiodcxx/exception.hpp /usr/include/gpiodcxx/info-event.hpp \
/usr/include/gpiodcxx/line.hpp /usr/include/gpiodcxx/line-config.hpp \
/usr/include/gpiodcxx/line-info.hpp \
/usr/include/gpiodcxx/line-request.hpp /usr/include/gpiodcxx/misc.hpp \
/usr/include/gpiodcxx/line-settings.hpp \
/usr/include/gpiodcxx/request-builder.hpp \
/usr/include/gpiodcxx/request-config.hpp \
/home/group5/RTES_Table_Tennis/src/libs/Piezo/Include/event_detector.h \
/home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/led_controller.h \
/home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/ledcallback.hpp \
/home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/ledcallback.hpp
```

11.7 src/libs/IMU/CMakeFiles/imu_lib.dir/Icm20948driver.cpp.o.d File Reference

11.8 Icm20948driver.cpp.o.d

Go to the documentation of this file.

```
libs/IMU/CMakeFiles/imu_lib.dir/Icm20948driver.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/IMU/Icm20948driver.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU/Include/Icm20948driver.hpp \
 /usr/include/c++/14/cstdint \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
 /usr/include/aarch64-linux-gnu/bits/time64.h \
 /usr/include/aarch64-linux-gnu/bits/wchar.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-least.h \
 /usr/include/c++/14/string /usr/include/c++/14/bits/requires_hosted.h \
 /usr/include/c++/14/bits/stringfwd.h \
 /usr/include/c++/14/bits/memoryfwd.h \
 /usr/include/c++/14/bits/char_traits.h \
 /usr/include/c++/14/bits/postypes.h /usr/include/c++/14/cwchar \
 /usr/include/wchar.h /usr/include/aarch64-linux-gnu/bits/floatn.h \
 /usr/include/aarch64-linux-gnu/bits/floatn-common.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
 /usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
 /usr/include/aarch64-linux-gnu/bits/types/FILE.h \
 /usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
 /usr/include/c++/14/type_traits /usr/include/c++/14/bits/version.h \
 /usr/include/c++/14/compare /usr/include/c++/14/concepts \
 /usr/include/c++/14/bits/stl_construct.h /usr/include/c++/14/new \
 /usr/include/c++/14/bits/exception.h /usr/include/c++/14/bits/move.h \
 /usr/include/c++/14/bits/stl_iterator_base_types.h \
 /usr/include/c++/14/bits/iterator_concepts.h \
 /usr/include/c++/14/bits/ptr_traits.h \
 /usr/include/c++/14/bits/ranges_cmp.h \
 /usr/include/c++/14/bits/stl_iterator_base_funcs.h \
 /usr/include/c++/14/bits/concept_check.h \
 /usr/include/c++/14/debug/assertions.h \
 /usr/include/c++/14/bits/allocator.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
 /usr/include/c++/14/bits/new_allocator.h \
 /usr/include/c++/14/bits/functexcept.h \
 /usr/include/c++/14/bits/exception_defines.h \
 /usr/include/c++/14/bits/cpp_type_traits.h \
 /usr/include/c++/14/bits/localefwd.h \
```

```
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/iosfwd \
/usr/include/c++/14/cctype /usr/include/ctype.h \
/usr/include/aarch64-linux-gnu/bits/endian.h \
/usr/include/aarch64-linux-gnu/bits/endianness.h \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/ext/type_traits.h \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/bits/refwrap.h /usr/include/c++/14/bits/invoke.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/basic_string.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/stdlib.h /usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/sys/types.h \
/usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
/usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
/usr/include/aarch64-linux-gnu/bits/types/time_t.h \
/usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
/usr/include/endian.h /usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/c++/14/cstdio \
/usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/cerrno /usr/include/errno.h \
/usr/include/aarch64-linux-gnu/bits/errno.h /usr/include/linux/errno.h \
/usr/lib/linux/uapi/arm64/asm/errno.h /usr/include/asm-generic/errno.h \
/usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/uses_allocator_args.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/ranges_util.h /usr/include/linux/i2c-dev.h \
/usr/include/linux/types.h /usr/lib/linux/uapi/arm64/asm/types.h \
/usr/include/asm-generic/types.h /usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h /usr/include/linux/i2c.h \
/usr/include/yaml-cpp/yaml.h /usr/include/yaml-cpp/parser.h \
/usr/include/c++/14/ios /usr/include/c++/14/exception \
/usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/typeinfo /usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/ios_base.h /usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
```

```
/usr/include/aarch64-linux-gnu/bits/cpu-set.h /usr/include/time.h \
/usr/include/aarch64-linux-gnu/bits/time.h \
/usr/include/aarch64-linux-gnu/bits/timex.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc /usr/include/c++/14/memory \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/align.h /usr/include/c++/14/bits/unique_ptr.h \
/usr/include/c++/14/ostream /usr/include/c++/14/bits/ostream.tcc \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/concurrence.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/bits/atomic_base.h \
/usr/include/c++/14/bits/atomic_lockfree_defines.h \
/usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/std_mutex.h \
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/yaml-cpp/dll.h \
/usr/include/yaml-cpp/emitter.h /usr/include/c++/14/cmath \
/usr/include/math.h /usr/include/aarch64-linux-gnu/bits/math-vector.h \
/usr/include/aarch64-linux-gnu/bits/libm-simd-decl-stubs.h \
/usr/include/aarch64-linux-gnu/bits/flt-eval-method.h \
/usr/include/aarch64-linux-gnu/bits/fp-logb.h \
/usr/include/aarch64-linux-gnu/bits/fp-fast.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-macros.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-helper-functions.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-narrow.h \
/usr/include/aarch64-linux-gnu/bits/iscanonical.h \
/usr/include/c++/14/bits/specfun.h /usr/include/c++/14/limits \
/usr/include/c++/14/tr1/gamma.tcc \
/usr/include/c++/14/tr1/special_function_util.h \
/usr/include/c++/14/tr1/bessel_function.tcc \
/usr/include/c++/14/tr1/beta_function.tcc \
/usr/include/c++/14/tr1/ell_integral.tcc \
/usr/include/c++/14/tr1/exp_integral.tcc \
/usr/include/c++/14/tr1/hypergeometric.tcc \
/usr/include/c++/14/tr1/legendre_function.tcc \
/usr/include/c++/14/tr1/modified_bessel_func.tcc \
/usr/include/c++/14/tr1/poly_hermite.tcc \
/usr/include/c++/14/tr1/poly_laguerre.tcc \
/usr/include/c++/14/tr1/riemann_zeta.tcc /usr/include/c++/14/sstream \
```

```
/usr/include/c++/14/istream /usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/yaml-cpp/binary.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/yaml-cpp/emitterdef.h \
/usr/include/yaml-cpp/emittermanip.h /usr/include/yaml-cpp/null.h \
/usr/include/yaml-cpp/ostream_wrapper.h \
/usr/include/yaml-cpp/emitterstyle.h /usr/include/yaml-cpp/stlemitter.h \
/usr/include/c++/14/list /usr/include/c++/14/bits/stl_list.h \
/usr/include/c++/14/bits/list.tcc /usr/include/c++/14/set \
/usr/include/c++/14/bits/stl_tree.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/stl_set.h \
/usr/include/c++/14/bits/stl_multiset.h \
/usr/include/c++/14/bits/erase_if.h /usr/include/c++/14/map \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/yaml-cpp/exceptions.h /usr/include/yaml-cpp/mark.h \
/usr/include/yaml-cpp/noexcept.h /usr/include/yaml-cpp/traits.h \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/yaml-cpp/node/node.h \
/usr/include/yaml-cpp/node/detail/iterator_fwd.h \
/usr/include/yaml-cpp/node/ptr.h /usr/include/yaml-cpp/node/type.h \
/usr/include/yaml-cpp/node/impl.h \
/usr/include/yaml-cpp/node/detail/memory.h \
/usr/include/yaml-cpp/node/detail/node.h \
/usr/include/yaml-cpp/node/detail/node_ref.h \
/usr/include/yaml-cpp/node/detail/node_data.h \
/usr/include/yaml-cpp/node/detail/node_iterator.h \
/usr/include/c++/14/iterator /usr/include/c++/14/bits/stream_iterator.h \
/usr/include/yaml-cpp/node/iterator.h \
/usr/include/yaml-cpp/node/detail/iterator.h /usr/include/c++/14/atomic \
/usr/include/yaml-cpp/node/convert.h /usr/include/c++/14/array \
/usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/valarray /usr/include/c++/14/algorithm \
/usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/ranges_algo.h \
/usr/include/c++/14/pstl/glue_algorithm_defs.h \
/usr/include/c++/14/bits/valarray_array.h \
/usr/include/c++/14/bits/valarray_array.tcc \
/usr/include/c++/14/bits/valarray_before.h \
/usr/include/c++/14/bits/slice_array.h \
/usr/include/c++/14/bits/valarray_after.h \
/usr/include/c++/14/bits/gslice.h \
/usr/include/c++/14/bits/gslice_array.h \
/usr/include/c++/14/bits/mask_array.h \
/usr/include/c++/14/bits/indirect_array.h \
/usr/include/yaml-cpp/node/detail/impl.h \
/usr/include/yaml-cpp/node/parse.h /usr/include/yaml-cpp/node/emit.h \
/usr/include/c++/14/chrono /usr/include/c++/14/bits/chrono.h \
/usr/include/c++/14/ratio /usr/include/c++/14/ctime \
/usr/include/c++/14/bits/parse_numbers.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/charconv /usr/include/c++/14/optional \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h /usr/include/c++/14/iostream \
/usr/include/c++/14/thread /usr/include/c++/14/stop_token \
/usr/include/c++/14/bits/std_thread.h /usr/include/c++/14/semaphore \
/usr/include/c++/14/bits/semaphore_base.h \
/usr/include/c++/14/bits/atomic_timed_wait.h \
/usr/include/c++/14/bits/this_thread_sleep.h \
/usr/include/aarch64-linux-gnu/sys/time.h /usr/include/semaphore.h \
/usr/include/aarch64-linux-gnu/bits/semaphore.h /usr/include/fcntl.h \
/usr/include/aarch64-linux-gnu/bits/fcntl.h \
/usr/include/aarch64-linux-gnu/bits/fcntl-linux.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_iovec.h \
/usr/include/linux/falloc.h /usr/include/aarch64-linux-gnu/bits/stat.h \
/usr/include/aarch64-linux-gnu/bits/struct_stat.h \
/usr/include/aarch64-linux-gnu/sys/ioctl.h \
/usr/include/aarch64-linux-gnu/bits/ioctls.h \
/usr/lib/linux/uapi/arm64/asm/ioctls.h /usr/include/asm-generic/ioctls.h \
```

```
/usr/include/linux/ioctl.h /usr/lib/linux/uapi/arm64/asm/ioctl.h \
/usr/include/asm-generic/ioctl.h \
/usr/include/aarch64-linux-gnu/bits/ioctl-types.h \
/usr/include/aarch64-linux-gnu/sys/ttydefaults.h \
/usr/include/c++/14/cstring /usr/include/string.h /usr/include/strings.h
```

11.9 src/libs/IMU/CMakeFiles/imu_lib.dir/imureader.cpp.o.d File Reference

11.10 imureader.cpp.o.d

Go to the documentation of this file.

```
libs/IMU/CMakeFiles/imu_lib.dir/imureader.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/IMU/imureader.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU/Include/imureader.hpp \
 /usr/include/c++/14/atomic /usr/include/c++/14/bits/version.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h \
 /usr/include/c++/14/bits/atomic_base.h /usr/include/c++/14/new \
 /usr/include/c++/14/bits/exception.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
 /usr/include/aarch64-linux-gnu/bits/time64.h \
 /usr/include/aarch64-linux-gnu/bits/wchar.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-least.h \
 /usr/include/c++/14/bits/atomic_lockfree_defines.h \
 /usr/include/c++/14/bits/move.h /usr/include/c++/14/type_traits \
 /usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/cstdint \
 /usr/include/c++/14/bits/functional_hash.h \
 /usr/include/c++/14/bits/hash_bytes.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
 /usr/include/pthread.h /usr/include/sched.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/include/aarch64-linux-gnu/bits/types/time_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
 /usr/include/aarch64-linux-gnu/bits/endian.h \
 /usr/include/aarch64-linux-gnu/bits/endianness.h \
 /usr/include/aarch64-linux-gnu/bits/sched.h \
 /usr/include/linux/sched/types.h /usr/include/linux/types.h \
 /usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
 /usr/include/asm-generic/int-ll64.h \
 /usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
 /usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
 /usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
 /usr/include/asm-generic/posix_types.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
 /usr/include/aarch64-linux-gnu/bits/cpu-set.h /usr/include/time.h \
 /usr/include/aarch64-linux-gnu/bits/time.h \
 /usr/include/aarch64-linux-gnu/bits/timex.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
 /usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
 /usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
 /usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
 /usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
 /usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
 /usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
 /usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
 /usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
 /usr/include/aarch64-linux-gnu/bits/setjmp.h \
 /usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
```

```
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
/usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/cerrno \
/usr/include/errno.h /usr/include/aarch64-linux-gnu/bits/errno.h \
/usr/include/linux/errno.h /usr/lib/linux/uapi/arm64/asm/errno.h \
/usr/include/asm-generic/errno.h /usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/std_mutex.h /usr/include/c++/14/functional \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/compare /usr/include/c++/14/concepts \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
/usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/requires_hosted.h \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h \
/usr/include/c++/14/bits/memoryfwd.h \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/erase_if.h \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array \
/usr/include/c++/14/thread /usr/include/c++/14/stop_token \
/usr/include/c++/14/bits/std_thread.h /usr/include/c++/14/iosfwd \
/usr/include/c++/14/bits/stringfwd.h /usr/include/c++/14/bits/postypes.h \
/usr/include/c++/14/cwchar /usr/include/wchar.h \
/usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
```

```
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/bits/unique_ptr.h /usr/include/c++/14/ostream \
/usr/include/c++/14/ios /usr/include/c++/14/exception \
/usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/basic_string.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/stdlib.h /usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/sys/types.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/c++/14/cstdio \
/usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc \
/usr/include/c++/14/bits/ostream.tcc /usr/include/c++/14/semaphore \
/usr/include/c++/14/bits/semaphore_base.h \
/usr/include/c++/14/bits/chrono.h /usr/include/c++/14/ratio \
/usr/include/c++/14/limits /usr/include/c++/14/ctime \
/usr/include/c++/14/bits/parse_numbers.h \
/usr/include/c++/14/bits/atomic_timed_wait.h \
/usr/include/c++/14/bits/this_thread_sleep.h \
/usr/include/aarch64-linux-gnu/sys/time.h /usr/include/semaphore.h \
/usr/include/aarch64-linux-gnu/bits/semaphore.h \
/home/group5/RTES_Table_Tennis/src/libs/IMU/Include/Icm20948driver.hpp \
/usr/include/linux/i2c-dev.h /usr/include/linux/i2c.h \
/usr/include/yaml-cpp/yaml.h /usr/include/yaml-cpp/parser.h \
/usr/include/c++/14/memory /usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/align.h /usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/ext/concurrence.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/yaml-cpp/dll.h \
/usr/include/yaml-cpp/emitter.h /usr/include/c++/14/cmath \
/usr/include/math.h /usr/include/aarch64-linux-gnu/bits/math-vector.h \
/usr/include/aarch64-linux-gnu/bits/libm-simd-decl-stubs.h \
/usr/include/aarch64-linux-gnu/bits/flt-eval-method.h \
/usr/include/aarch64-linux-gnu/bits/fp-logb.h \
/usr/include/aarch64-linux-gnu/bits/fp-fast.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-macros.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-helper-functions.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-narrow.h \
/usr/include/aarch64-linux-gnu/bits/iscanonical.h \
```

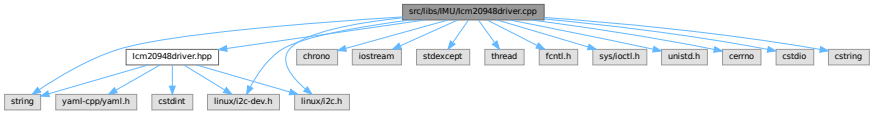
```
/usr/include/c++/14/bits/specfun.h /usr/include/c++/14/tr1/gamma.tcc \
/usr/include/c++/14/tr1/special_function_util.h \
/usr/include/c++/14/tr1/bessel_function.tcc \
/usr/include/c++/14/tr1/beta_function.tcc \
/usr/include/c++/14/tr1/ell_integral.tcc \
/usr/include/c++/14/tr1/exp_integral.tcc \
/usr/include/c++/14/tr1/hypergeometric.tcc \
/usr/include/c++/14/tr1/legendre_function.tcc \
/usr/include/c++/14/tr1/modified_bessel_func.tcc \
/usr/include/c++/14/tr1/poly_hermite.tcc \
/usr/include/c++/14/tr1/poly_laguerre.tcc \
/usr/include/c++/14/tr1/riemann_zeta.tcc /usr/include/c++/14/sstream \
/usr/include/c++/14/istream /usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/yaml-cpp/binary.h \
/usr/include/yaml-cpp/emitterdef.h /usr/include/yaml-cpp/emittermanip.h \
/usr/include/yaml-cpp/null.h /usr/include/yaml-cpp/ostream_wrapper.h \
/usr/include/yaml-cpp/emitterstyle.h /usr/include/yaml-cpp/stlemitter.h \
/usr/include/c++/14/list /usr/include/c++/14/bits/stl_list.h \
/usr/include/c++/14/bits/list.tcc /usr/include/c++/14/set \
/usr/include/c++/14/bits/stl_tree.h /usr/include/c++/14/bits/stl_set.h \
/usr/include/c++/14/bits/stl_multiset.h /usr/include/c++/14/map \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/yaml-cpp/exceptions.h /usr/include/yaml-cpp/mark.h \
/usr/include/yaml-cpp/noexcept.h /usr/include/yaml-cpp/traits.h \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/yaml-cpp/node/node.h \
/usr/include/yaml-cpp/node/detail/iterator_fwd.h \
/usr/include/yaml-cpp/node/ptr.h /usr/include/yaml-cpp/node/type.h \
/usr/include/yaml-cpp/node/impl.h \
/usr/include/yaml-cpp/node/detail/memory.h \
/usr/include/yaml-cpp/node/detail/node.h \
/usr/include/yaml-cpp/node/detail/node_ref.h \
/usr/include/yaml-cpp/node/detail/node_data.h \
/usr/include/yaml-cpp/node/detail/node_iterator.h \
/usr/include/c++/14/iterator /usr/include/c++/14/bits/stream_iterator.h \
/usr/include/yaml-cpp/node/iterator.h \
/usr/include/yaml-cpp/node/detail/iterator.h \
/usr/include/yaml-cpp/node/convert.h /usr/include/c++/14/valarray \
/usr/include/c++/14/algorithm /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/ranges_algo.h \
/usr/include/c++/14/pstl/glue_algorithm_defs.h \
/usr/include/c++/14/bits/valarray_array.h \
/usr/include/c++/14/bits/valarray_array.tcc \
/usr/include/c++/14/bits/valarray_before.h \
/usr/include/c++/14/bits/slice_array.h \
/usr/include/c++/14/bits/valarray_after.h \
/usr/include/c++/14/bits/gslice.h \
/usr/include/c++/14/bits/gslice_array.h \
/usr/include/c++/14/bits/mask_array.h \
/usr/include/c++/14/bits/indirect_array.h \
/usr/include/yaml-cpp/node/detail/impl.h \
/usr/include/yaml-cpp/node/parse.h /usr/include/yaml-cpp/node/emit.h \
/usr/include/gpiod.h /usr/lib/gcc/aarch64-linux-gnu/14/include/stdbool.h \
/usr/include/c++/14/future /usr/include/c++/14/mutex \
/usr/include/c++/14/bits/unique_lock.h \
/usr/include/c++/14/condition_variable \
/usr/include/c++/14/bits/atomic_futex.h \
/home/group5/RTES_Table_Tennis/src/libs/IMU/Include/i2c_mutex.hpp \
/usr/include/aarch64-linux-gnu/sys/epoll.h \
/usr/include/aarch64-linux-gnu/sys/ioctl.h \
/usr/include/aarch64-linux-gnu/bits/ioctls.h \
/usr/lib/linux/uapi/arm64/asm/ioctls.h /usr/include/asm-generic/ioctls.h \
/usr/include/linux/ioctl.h /usr/lib/linux/uapi/arm64/asm/ioctl.h \
/usr/include/asm-generic/ioctl.h \
/usr/include/aarch64-linux-gnu/bits/ioctl-types.h \
/usr/include/aarch64-linux-gnu/sys/ttydefaults.h \
/usr/include/aarch64-linux-gnu/bits/epoll.h \
/usr/include/aarch64-linux-gnu/sys/eventfd.h \
/usr/include/aarch64-linux-gnu/bits/eventfd.h
```

11.11 src/libs/IMU/Icm20948driver.cpp File Reference

```
#include "Icm20948driver.hpp"
#include <chrono>
#include <iostream>
#include <stdexcept>
#include <string>
#include <thread>
```

```
#include <linux/i2c-dev.h>
#include <linux/i2c.h>
#include <fcntl.h>
#include <sys/ioctl.h>
#include <unistd.h>
#include <cerrno>
#include <cstdio>
#include <cstring>
```

Include dependency graph for Icm20948driver.cpp:



Namespaces

- namespace icm20948

  Settings types.

Macros

- #define G2MSQR 9.80665f
- #define DEG2RAD 0.017453293f
- #define MAGN_SCALE_FACTOR 0.149975574f

Functions

- float icm20948::accel_scale_factor (accel_scale scale)

  Convert accel range enum to raw-count scale factor.
- std::string icm20948::accel_scale_to_str (accel_scale scale)
- std::string icm20948::accel_dlpf_config_to_str (accel_dlpf_config config)

  Stringify accel DLPF enum.
- float icm20948::gyro_scale_factor (gyro_scale scale)

  Convert gyro range enum to raw-count scale factor.
- std::string icm20948::gyro_scale_to_str (gyro_scale scale)

  Stringify gyro range enum.
- std::string icm20948::gyro_dlpf_config_to_str (gyro_dlpf_config config)

  Stringify gyro DLPF enum.
- std::string icm20948::magn_mode_to_str (magn_mode mode)

  Stringify magnetometer mode enum.

11.11.1 Macro Definition Documentation

11.11.1.1 DEG2RAD

```
#define DEG2RAD 0.017453293f
```

Definition at line 19 of file Icm20948driver.cpp.

11.11.1.2 G2MSQR

```
#define G2MSQR 9.80665f
```

Definition at line 18 of file Icm20948driver.cpp.

11.11.1.3 MAGN_SCALE_FACTOR

```
#define MAGN_SCALE_FACTOR 0.149975574f
```

Definition at line 20 of file lcm20948driver.cpp.

11.12 lcm20948driver.cpp

Go to the documentation of this file.

```
#include "Icm20948driver.hpp"

#include <chrono>
#include <iostream>
#include <stdexcept>
#include <string>
#include <thread>

#include <linux/i2c-dev.h>
#include <linux/i2c.h>
#include <fcntl.h>
#include <sys/ioctl.h>
#include <unistd.h>
#include <cerrno>
#include <cstdio>
#include <cstring>

#define G2MSQR            9.80665f
#define DEG2RAD           0.017453293f
#define MAGN_SCALE_FACTOR 0.149975574f

namespace icm20948
{
    float accel_scale_factor(accel_scale scale)
    {
        switch (scale) {
            case ACCEL_2G:  return 1 / 16384.0f;
            case ACCEL_4G:  return 1 / 8192.0f;
            case ACCEL_8G:  return 1 / 4096.0f;
            case ACCEL_16G: return 1 / 2048.0f;
            default:
                throw std::invalid_argument(
                    "Invalid argument in accel_scale_factor(): " + std::to_string(scale));
        }
    }

    // Stringify accel range enum.
    std::string accel_scale_to_str(accel_scale scale)
    {
        switch (scale) {
            case ACCEL_2G:  return "2G";
            case ACCEL_4G:  return "4G";
            case ACCEL_8G:  return "8G";
            case ACCEL_16G: return "16G";
            default:        return "<invalid accelerometer scale>";
        }
    }
    std::string accel_dlpf_config_to_str(accel_dlpf_config config)
    {
        switch (config) {
            case ACCEL_DLPF_246HZ:   return "246Hz";
            case ACCEL_DLPF_246HZ_2: return "246Hz";
            case ACCEL_DLPF_111_4HZ: return "111.4Hz";
            case ACCEL_DLPF_50_4HZ:  return "50.4Hz";
            case ACCEL_DLPF_23_9HZ:  return "23.9Hz";
            case ACCEL_DLPF_11_5HZ:  return "11.5Hz";
            case ACCEL_DLPF_5_7HZ:   return "5.7Hz";
            case ACCEL_DLPF_473HZ:   return "473Hz";
            default:                 return "<invalid accelerometer DLPF config>";
        }
    }
    float gyro_scale_factor(gyro_scale scale)
    {
        switch (scale) {
            case GYRO_250DPS:  return 1 / 131.0f;
            case GYRO_500DPS:  return 1 / 65.5f;
            case GYRO_1000DPS: return 1 / 32.8f;
            case GYRO_2000DPS: return 1 / 16.4f;
            default:
                throw std::invalid_argument(
                    "Invalid argument in gyro_scale_factor(): " + std::to_string(scale));
        }
    }
    std::string gyro_scale_to_str(gyro_scale scale)
    {
```

```
        switch (scale) {
            case GYRO_250DPS:  return "250DPS";
            case GYRO_500DPS:  return "500DPS";
            case GYRO_1000DPS: return "1000DPS";
            case GYRO_2000DPS: return "2000DPS";
            default:           return "<invalid gyroscope scale>";
        }
    }
    std::string gyro_dlpf_config_to_str(gyro_dlpf_config config)
    {
        switch (config) {
            case GYRO_DLPF_196_6HZ: return "196.6Hz";
            case GYRO_DLPF_151_8HZ: return "151.8Hz";
            case GYRO_DLPF_119_5HZ: return "119.5Hz";
            case GYRO_DLPF_51_2HZ:  return "51.2Hz";
            case GYRO_DLPF_23_9HZ:  return "23.9Hz";
            case GYRO_DLPF_11_6HZ:  return "11.6Hz";
            case GYRO_DLPF_5_7HZ:   return "5.7Hz";
            case GYRO_DLPF_361_4HZ: return "361.4Hz";
            default:                return "<invalid gyroscope DLPF config>";
        }
    }
    std::string magn_mode_to_str(magn_mode mode)
    {
        switch (mode) {
            case MAGN_SHUTDOWN:  return "Shutdown";
            case MAGN_SINGLE:    return "Single";
            case MAGN_10HZ:      return "10Hz";
            case MAGN_20HZ:      return "20Hz";
            case MAGN_50HZ:      return "50Hz";
            case MAGN_100HZ:     return "100Hz";
            case MAGN_SELF_TEST: return "Self-test";
            default:             return "<invalid magnetometer mode>";
        }
    }
    settings::settings(YAML::Node node)
    {
        for (auto it = node.begin(); it != node.end(); ++it) {
            const std::string key = it->first.as<std::string>();

            if (key == "accelerometer") {
                for (auto a = it->second.begin(); a != it->second.end(); ++a) {
                    const std::string ak = a->first.as<std::string>();
                    if (ak == "sample_rate_div")
                        accel.sample_rate_div = static_cast<uint16_t>(a->second.as<unsigned>());
                    else if (ak == "scale")
                        accel.scale = static_cast<accel_scale>(a->second.as<unsigned>());
                    else if (ak == "dlpf") {
                        for (auto d = a->second.begin(); d != a->second.end(); ++d) {
                            const std::string dk = d->first.as<std::string>();
                            if (dk == "enable")
                                accel.dlpf_enable = static_cast<bool>(d->second.as<int>());
                            else if (dk == "cutoff")
                                accel.dlpf_config =
static_cast<accel_dlpf_config>(d->second.as<unsigned>());
                        }
                    }
                }
            } else if (key == "gyroscope") {
                for (auto g = it->second.begin(); g != it->second.end(); ++g) {
                    const std::string gk = g->first.as<std::string>();
                    if (gk == "sample_rate_div")
                        gyro.sample_rate_div = static_cast<uint8_t>(g->second.as<unsigned>());
                    else if (gk == "scale")
                        gyro.scale = static_cast<gyro_scale>(g->second.as<unsigned>());
                    else if (gk == "dlpf") {
                        for (auto d = g->second.begin(); d != g->second.end(); ++d) {
                            const std::string dk = d->first.as<std::string>();
                            if (dk == "enable")
                                gyro.dlpf_enable = static_cast<bool>(d->second.as<int>());
                            else if (dk == "cutoff")
                                gyro.dlpf_config =
static_cast<gyro_dlpf_config>(d->second.as<unsigned>());
                        }
                    }
                }
            }
        }
    }

    ICM20948_I2C::ICM20948_I2C(unsigned i2c_bus,
                               unsigned i2c_address,
                               icm20948::settings settings)
        : _i2c_fd(-1),
          _i2c_bus(i2c_bus),
          _i2c_address(i2c_address),
          _current_bank(0),
```

```
        _accel_scale_factor(0.0f),
        _gyro_scale_factor(0.0f),
        _magn_scale_factor(MAGN_SCALE_FACTOR),
        settings(settings)
    {
        char bus_path[32];
        std::snprintf(bus_path, sizeof(bus_path), "/dev/i2c-%u", i2c_bus);

        _i2c_fd = ::open(bus_path, O_RDWR);
        if (_i2c_fd < 0) {
            throw std::runtime_error(std::string("ICM20948: cannot open ") + bus_path +
                                     ": " + std::strerror(errno));
        }
        ::ioctl(_i2c_fd, I2C_TIMEOUT, 20);

        if (::ioctl(_i2c_fd, I2C_SLAVE, i2c_address) < 0) {
            ::close(_i2c_fd);
            _i2c_fd = -1;
            throw std::runtime_error(std::string("ICM20948: I2C_SLAVE failed: ") +
                                     std::strerror(errno));
        }

        accel[0] = accel[1] = accel[2] = 0.0f;
        gyro[0]  = gyro[1]  = gyro[2]  = 0.0f;
        magn[0]  = magn[1]  = magn[2]  = 0.0f;
    }

    ICM20948_I2C::~ICM20948_I2C()
    {
        if (_i2c_fd >= 0) {
            ::close(_i2c_fd);
            _i2c_fd = -1;
        }
    }

    bool ICM20948_I2C::init()
    {
        uint8_t device_id = 0;
        bool success = true;

        success &= _set_bank(0);
        success &= _read_byte(ICM20948_WHO_AM_I_BANK, ICM20948_WHO_AM_I_ADDR, device_id);
        success &= (device_id == ICM20948_BANK0_WHO_AM_I_VALUE);
        success &= reset();
        success &= wake();
        success &= set_settings();
        success &= _enable_data_ready_interrupt();

        // Magnetometer intentionally disabled for stability testing.
        // bool magn_ok = false;
        // for (int i = 0; i < 3; ++i) {
        //     magn_ok = _magnetometer_init();
        //     if (magn_ok) break;
        // }
        // if (!magn_ok) {
        //     std::cerr « "[IMU] Magnetometer init failed, continuing without it\n";
        // }

        return success;
    }

    bool ICM20948_I2C::reset()
    {
        bool success = _write_bit(ICM20948_PWR_MGMT_1_BANK,
                                  ICM20948_PWR_MGMT_1_ADDR, 7, true);
        std::this_thread::sleep_for(std::chrono::milliseconds(5));

        if (success) {
            bool still_resetting = true;
            while (still_resetting && success) {
                std::this_thread::sleep_for(std::chrono::milliseconds(25));
                success &= _read_bit(ICM20948_PWR_MGMT_1_BANK,
                                     ICM20948_PWR_MGMT_1_ADDR, 7, still_resetting);
            }
        }

        if (success) _current_bank = 0;
        return success;
    }

    bool ICM20948_I2C::wake()
    {
        bool success = _write_bit(ICM20948_PWR_MGMT_1_BANK,
                                  ICM20948_PWR_MGMT_1_ADDR, 6, false);
        std::this_thread::sleep_for(std::chrono::milliseconds(5));
        return success;
    }
```

```
00264
00265     bool ICM20948_I2C::set_settings()
00266     {
00268         bool success = true;
00269         success &= _set_accel_sample_rate_div();
00270         success &= _set_accel_range_dlpf();
00271         success &= _set_gyro_sample_rate_div();
00272         success &= _set_gyro_range_dlpf();
00273         return success;
00274     }
00275
00276     bool ICM20948_I2C::_enable_data_ready_interrupt()
00277     {
00279         return _write_byte(ICM20948_INT_ENABLE_1_BANK,
00280                            ICM20948_INT_ENABLE_1_ADDR, 0x01);
00281     }
00282
00283     bool ICM20948_I2C::read_accel_gyro()
00284     {
00285         uint8_t buf[12] = {};
00287         if (!_read_block_bytes(ICM20948_ACCEL_OUT_BANK,
00288                                ICM20948_ACCEL_XOUT_H_ADDR, buf, 12)) {
00289             return false;
00290         }
00291
00292         const int16_t accel_x_raw = static_cast<int16_t>((buf[0] « 8) | buf[1]);
00293         const int16_t accel_y_raw = static_cast<int16_t>((buf[2] « 8) | buf[3]);
00294         const int16_t accel_z_raw = static_cast<int16_t>((buf[4] « 8) | buf[5]);
00295         const int16_t gyro_x_raw  = static_cast<int16_t>((buf[6] « 8) | buf[7]);
00296         const int16_t gyro_y_raw  = static_cast<int16_t>((buf[8] « 8) | buf[9]);
00297         const int16_t gyro_z_raw  = static_cast<int16_t>((buf[10] « 8) | buf[11]);
00298
00300         accel[0] = static_cast<float>(accel_x_raw) * _accel_scale_factor * G2MSQR;
00301         accel[1] = static_cast<float>(accel_y_raw) * _accel_scale_factor * G2MSQR;
00302         accel[2] = static_cast<float>(accel_z_raw) * _accel_scale_factor * G2MSQR;
00303
00304         gyro[0] = static_cast<float>(gyro_x_raw) * _gyro_scale_factor * DEG2RAD;
00305         gyro[1] = static_cast<float>(gyro_y_raw) * _gyro_scale_factor * DEG2RAD;
00306         gyro[2] = static_cast<float>(gyro_z_raw) * _gyro_scale_factor * DEG2RAD;
00307
00308         return true;
00309     }
00310
00311     bool ICM20948_I2C::read_magn()
00312     {
00313         uint8_t buf[6] = {};
00315         if (!_read_block_bytes(ICM20948_EXT_SLV_SENS_DATA_00_BANK,
00316                                ICM20948_EXT_SLV_SENS_DATA_00_ADDR, buf, 6)) {
00317             return false;
00318         }
00319
00320         const int16_t mag_x_raw = static_cast<int16_t>((buf[1] « 8) | buf[0]);
00321         const int16_t mag_y_raw = static_cast<int16_t>((buf[3] « 8) | buf[2]);
00322         const int16_t mag_z_raw = static_cast<int16_t>((buf[5] « 8) | buf[4]);
00323
00325         magn[0] = static_cast<float>(mag_x_raw) * _magn_scale_factor;
00326         magn[1] = static_cast<float>(mag_y_raw) * _magn_scale_factor;
00327         magn[2] = static_cast<float>(mag_z_raw) * _magn_scale_factor;
00328
00329         return true;
00330     }
00331
00332     bool ICM20948_I2C::read_sample(IMUSample& sample)
00333     {
00335         if (!read_accel_gyro()) return false;
00336
00339
00340         magn[0] = 0.0f;
00341         magn[1] = 0.0f;
00342         magn[2] = 0.0f;
00343
00344         sample.ax = accel[0]; sample.ay = accel[1]; sample.az = accel[2];
00345         sample.gx = gyro[0];  sample.gy = gyro[1];  sample.gz = gyro[2];
00346         sample.mx = magn[0];  sample.my = magn[1];  sample.mz = magn[2];
00347
00348         return true;
00349     }
00350
00351     bool ICM20948_I2C::_set_bank(uint8_t bank)
00352     {
00353
00355         if (_current_bank == bank) return true;
00356
00357         uint8_t val = 0;
00358         switch (bank) {
00359             case 0: val = ICM20948_REG_BANK_SEL_BANK0_VALUE; break;
00360             case 1: val = ICM20948_REG_BANK_SEL_BANK1_VALUE; break;
```

```
            case 2: val = ICM20948_REG_BANK_SEL_BANK2_VALUE; break;
            case 3: val = ICM20948_REG_BANK_SEL_BANK3_VALUE; break;
            default:
                throw std::runtime_error(
                    "Invalid bank number in _set_bank(): " + std::to_string(bank));
        }

        uint8_t buf[2] = { ICM20948_REG_BANK_SEL_ADDR, val };
        if (::write(_i2c_fd, buf, 2) == 2) {
            _current_bank = bank;
            return true;
        }
        return false;
    }

    bool ICM20948_I2C::_set_accel_sample_rate_div()
    {
        uint8_t lsb = settings.accel.sample_rate_div & 0xff;
        uint8_t msb = (settings.accel.sample_rate_div » 8) & 0x0f;
        bool ok = true;
        ok &= _write_byte(ICM20948_ACCEL_SMPLRT_DIV_1_BANK,
                          ICM20948_ACCEL_SMPLRT_DIV_1_ADDR, msb);
        ok &= _write_byte(ICM20948_ACCEL_SMPLRT_DIV_2_BANK,
                          ICM20948_ACCEL_SMPLRT_DIV_2_ADDR, lsb);
        return ok;
    }

    bool ICM20948_I2C::_set_accel_range_dlpf()
    {
        uint8_t byte = 0;
        byte |= static_cast<uint8_t>(!!static_cast<uint8_t>(settings.accel.dlpf_enable));
        byte |= static_cast<uint8_t>(static_cast<uint8_t>(settings.accel.scale) « 1);
        byte |= static_cast<uint8_t>(static_cast<uint8_t>(settings.accel.dlpf_config) « 3);

        bool ok = _write_byte(ICM20948_ACCEL_CONFIG_1_BANK,
                              ICM20948_ACCEL_CONFIG_1_ADDR, byte);
        if (ok) _accel_scale_factor = accel_scale_factor(settings.accel.scale);
        return ok;
    }

    bool ICM20948_I2C::_set_gyro_sample_rate_div()
    {
        return _write_byte(ICM20948_GYRO_SMPLRT_DIV_BANK,
                           ICM20948_GYRO_SMPLRT_DIV_ADDR,
                           settings.gyro.sample_rate_div);
    }

    bool ICM20948_I2C::_set_gyro_range_dlpf()
    {
        uint8_t byte = 0;
        byte |= static_cast<uint8_t>(!!static_cast<uint8_t>(settings.gyro.dlpf_enable));
        byte |= static_cast<uint8_t>(static_cast<uint8_t>(settings.gyro.scale) « 1);
        byte |= static_cast<uint8_t>(static_cast<uint8_t>(settings.gyro.dlpf_config) « 3);

        bool ok = _write_byte(ICM20948_GYRO_CONFIG_1_BANK,
                              ICM20948_GYRO_CONFIG_1_ADDR, byte);
        if (ok) _gyro_scale_factor = gyro_scale_factor(settings.gyro.scale);
        return ok;
    }

    bool ICM20948_I2C::_magnetometer_init()
    {
        bool ok = true;
        ok &= _magnetometer_enable();
        if (!ok) { std::cerr « "Failed on _magnetometer_enable()\n"; return false; }
        ok &= _magnetometer_set_mode();
        if (!ok) { std::cerr « "Failed on _magnetometer_set_mode()\n"; return false; }
        ok &= _magnetometer_configured();
        if (!ok) { std::cerr « "Failed on _magnetometer_configured()\n"; return false; }
        ok &= _magnetometer_set_readout();
        if (!ok) { std::cerr « "Failed on _magnetometer_set_readout()\n"; return false; }
        return ok;
    }

    bool ICM20948_I2C::_magnetometer_enable()
    {
        bool ok = _write_bit(ICM20948_INT_PIN_CFG_BANK,
                             ICM20948_INT_PIN_CFG_ADDR, 1, false);
        std::this_thread::sleep_for(std::chrono::milliseconds(10));

        ok &= _write_byte(ICM20948_I2C_MST_CTRL_BANK,
                          ICM20948_I2C_MST_CTRL_ADDR, 0x4D);
        std::this_thread::sleep_for(std::chrono::milliseconds(10));

        ok &= _write_bit(ICM20948_USER_CTRL_BANK,
                         ICM20948_USER_CTRL_ADDR, 5, true);
        std::this_thread::sleep_for(std::chrono::milliseconds(100));
```

```cpp

        return ok;
    }

    bool ICM20948_I2C::_magnetometer_set_mode()
    {
        bool ok = _write_mag_byte(AK09916_CNTL2_ADDR,
                                  static_cast<uint8_t>(MAGN_SHUTDOWN));
        ok &= _write_mag_byte(AK09916_CNTL2_ADDR,
                              static_cast<uint8_t>(settings.magn.mode));
        return ok;
    }

    bool ICM20948_I2C::_magnetometer_configured()
    {
        uint8_t mag_id = 0;
        for (int i = 0; i < 5; ++i) {
            if (_read_mag_byte(0x01, mag_id)) return true;
            _chip_i2c_master_reset();
            std::cerr « "Magnetometer not configured properly, resetting chip I2C master\n";
            std::this_thread::sleep_for(std::chrono::milliseconds(25));
        }
        return false;
    }

    bool ICM20948_I2C::_magnetometer_set_readout()
    {
        bool ok = _write_byte(ICM20948_I2C_SLV0_ADDR_BANK,
                              ICM20948_I2C_SLV0_ADDR_ADDR, 0x8C);
        std::this_thread::sleep_for(std::chrono::milliseconds(5));
        ok &= _write_byte(ICM20948_I2C_SLV0_REG_BANK,
                          ICM20948_I2C_SLV0_REG_ADDR, 0x11);
        std::this_thread::sleep_for(std::chrono::milliseconds(5));
        ok &= _write_byte(ICM20948_I2C_SLV0_CTRL_BANK,
                          ICM20948_I2C_SLV0_CTRL_ADDR, 0x89);
        std::this_thread::sleep_for(std::chrono::milliseconds(5));
        return ok;
    }

    bool ICM20948_I2C::_chip_i2c_master_reset()
    {
        return _write_bit(ICM20948_USER_CTRL_BANK,
                          ICM20948_USER_CTRL_ADDR, 1, true);
    }

    bool ICM20948_I2C::_write_byte(uint8_t bank, uint8_t reg, uint8_t byte)
    {
        bool ok = _set_bank(bank);
        uint8_t buf[2] = { reg, byte };
        ok &= (::write(_i2c_fd, buf, 2) == 2);
        return ok;
    }

    bool ICM20948_I2C::_read_byte(uint8_t bank, uint8_t reg, uint8_t& byte)
    {
        if (!_set_bank(bank)) return false;

        struct i2c_msg msgs[2];
        msgs[0].addr  = static_cast<__u16>(_i2c_address);
        msgs[0].flags = 0;
        msgs[0].len   = 1;
        msgs[0].buf   = &reg;

        msgs[1].addr  = static_cast<__u16>(_i2c_address);
        msgs[1].flags = I2C_M_RD;
        msgs[1].len   = 1;
        msgs[1].buf   = &byte;

        struct i2c_rdwr_ioctl_data txn;
        txn.msgs  = msgs;
        txn.nmsgs = 2;

        bool ok = ::ioctl(_i2c_fd, I2C_RDWR, &txn) == 2;
        if (ok) ::usleep(5000);
        return ok;
    }

    bool ICM20948_I2C::_write_bit(uint8_t bank, uint8_t reg,
                                  uint8_t bit_pos, bool bit)
    {
        uint8_t existing = 0;
        if (!_read_byte(bank, reg, existing)) return false;

        uint8_t updated =
            static_cast<uint8_t>(
                (existing & ~static_cast<uint8_t>(1U « bit_pos)) |
                (static_cast<uint8_t>(!!bit) « bit_pos));
```

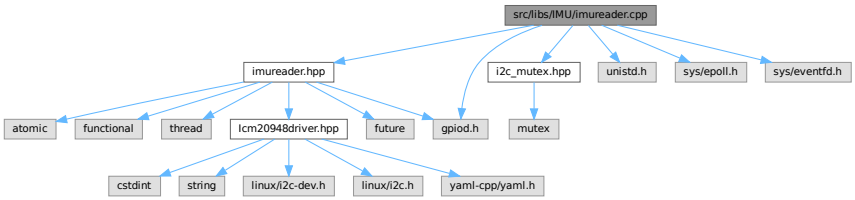
```
    return _write_byte(bank, reg, updated);
}

bool ICM20948_I2C::_read_bit(uint8_t bank, uint8_t reg,
                             uint8_t bit_pos, bool& bit)
{
    uint8_t existing = 0;
    if (!_read_byte(bank, reg, existing)) return false;
    bit = ((existing >> bit_pos) & 1U) != 0U;
    return true;
}

bool ICM20948_I2C::_read_block_bytes(uint8_t bank, uint8_t start_reg,
                                     uint8_t* bytes, int length)
{
    if (!_set_bank(bank)) return false;

    struct i2c_msg msgs[2];
    msgs[0].addr  = static_cast<__u16>(_i2c_address);
    msgs[0].flags = 0;
    msgs[0].len   = 1;
    msgs[0].buf   = &start_reg;

    msgs[1].addr  = static_cast<__u16>(_i2c_address);
    msgs[1].flags = I2C_M_RD;
    msgs[1].len   = static_cast<__u16>(length);
    msgs[1].buf   = bytes;

    struct i2c_rdwr_ioctl_data txn;
    txn.msgs  = msgs;
    txn.nmsgs = 2;

    bool ok = ::ioctl(_i2c_fd, I2C_RDWR, &txn) == 2;
    if (ok) ::usleep(5000);
    return ok;
}

bool ICM20948_I2C::_write_mag_byte(uint8_t mag_reg, uint8_t byte)
{
    bool ok = true;
    ok &= _write_byte(ICM20948_I2C_SLV4_ADDR_BANK,
                      ICM20948_I2C_SLV4_ADDR_ADDR, ICM20948_MAGN_I2C_ADDR);
    ok &= _write_byte(ICM20948_I2C_SLV4_REG_BANK,
                      ICM20948_I2C_SLV4_REG_ADDR, mag_reg);
    ok &= _write_byte(ICM20948_I2C_SLV4_DO_BANK,
                      ICM20948_I2C_SLV4_DO_ADDR, byte);
    ok &= _write_byte(ICM20948_I2C_SLV4_CTRL_BANK,
                      ICM20948_I2C_SLV4_CTRL_ADDR, 0x80);

    bool done = false;
    for (int i = 0; i < 20; ++i) {
        _read_bit(ICM20948_I2C_MST_STATUS_BANK,
                  ICM20948_I2C_MST_STATUS_ADDR, 6, done);
        if (done) break;
        std::this_thread::sleep_for(std::chrono::milliseconds(5));
    }

    if (!done) std::cerr << "Could not write to magnetometer (acknowledgement error)\n";
    return ok && done;
}

bool ICM20948_I2C::_read_mag_byte(uint8_t mag_reg, uint8_t& byte)
{
    bool ok = true;
    ok &= _write_byte(ICM20948_I2C_SLV4_ADDR_BANK,
                      ICM20948_I2C_SLV4_ADDR_ADDR,
                      static_cast<uint8_t>(ICM20948_MAGN_I2C_ADDR | 0x80));
    ok &= _write_byte(ICM20948_I2C_SLV4_REG_BANK,
                      ICM20948_I2C_SLV4_REG_ADDR, mag_reg);
    ok &= _write_byte(ICM20948_I2C_SLV4_CTRL_BANK,
                      ICM20948_I2C_SLV4_CTRL_ADDR, 0x80);

    bool done = false;
    for (int i = 0; i < 20; ++i) {
        _read_bit(ICM20948_I2C_MST_STATUS_BANK,
                  ICM20948_I2C_MST_STATUS_ADDR, 6, done);
        if (done) break;
        std::this_thread::sleep_for(std::chrono::milliseconds(5));
    }

    if (!done) std::cerr << "Could not read from magnetometer (acknowledgement error)\n";
    if (!done) return false;

    return _read_byte(ICM20948_I2C_SLV4_DI_BANK,
                      ICM20948_I2C_SLV4_DI_ADDR, byte);
}
```

```
00639
00640     bool ICM20948_I2C::check_DRDY_INT()
00641     {
00642         uint8_t int_status;
00643
00644         if (!_read_byte(ICM20948_INT_STATUS_1_BANK,
00645                         ICM20948_INT_STATUS_1_ADDR,
00646                         int_status)) {
00647             std::cerr « "Unable to read INT_STATUS" « std::endl;
00648             return false;
00649         }
00650
00651         return (int_status & 0x01) != 0;
00652     }
00653
00654 } // namespace icm20948
```

11.13 src/libs/IMU/imureader.cpp File Reference

```
#include "imureader.hpp"
#include "i2c_mutex.hpp"
#include <unistd.h>
#include <sys/epoll.h>
#include <sys/eventfd.h>
#include <gpiod.h>
```

Include dependency graph for imureader.cpp:



11.14 imureader.cpp

Go to the documentation of this file.

```
00001 #include "imureader.hpp"
00002 #include "i2c_mutex.hpp"
00003
00004 #include <unistd.h>
00005 #include <sys/epoll.h>
00006 #include <sys/eventfd.h>
00007 #include <gpiod.h>
00008
00009 IMUReader::IMUReader(unsigned           i2c_bus,
00010                      unsigned           i2c_address,
00011                      const char*        gpiochip,
00012                      unsigned           rdy_line,
00013                      icm20948::settings settings)
00014     : imu_(i2c_bus, i2c_address, settings),
00015       running_(false),
00016       gpiochip_path_(gpiochip),
00017       rdy_line_(rdy_line),
00018       chip_(nullptr),
00019       request_(nullptr),
00020       gpio_fd_(-1),
00021       stop_fd_(-1)
00022 {
00023 }
00024
00025 IMUReader::~IMUReader()
00026 {
00028     stop();
00029 }
00030
00031 bool IMUReader::init()
00032 {
00034     return imu_.init();
```

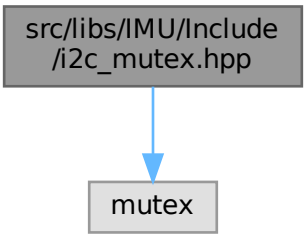
```
}

icm20948::ICM20948_I2C& IMUReader::getIMU()
{
    return imu_;
}

void IMUReader::setCallback(SampleCallback cb)
{
    callback_ = std::move(cb);
}

bool IMUReader::start()
{
    if (running_) return false;

    chip_ = gpiod_chip_open(gpiochip_path_);
    if (!chip_) return false;

    gpiod_line_settings* line_settings = gpiod_line_settings_new();
    gpiod_line_settings_set_direction(line_settings, GPIOD_LINE_DIRECTION_INPUT);
    gpiod_line_settings_set_edge_detection(line_settings, GPIOD_LINE_EDGE_RISING);

    gpiod_line_config* line_cfg = gpiod_line_config_new();
    unsigned int offset = static_cast<unsigned int>(rdy_line_);
    gpiod_line_config_add_line_settings(line_cfg, &offset, 1, line_settings);

    gpiod_request_config* req_cfg = gpiod_request_config_new();
    gpiod_request_config_set_consumer(req_cfg, "imu-rdy");

    request_ = gpiod_chip_request_lines(chip_, req_cfg, line_cfg);

    gpiod_request_config_free(req_cfg);
    gpiod_line_config_free(line_cfg);
    gpiod_line_settings_free(line_settings);

    if (!request_) {
        gpiod_chip_close(chip_);
        return false;
    }

    gpio_fd_ = gpiod_line_request_get_fd(request_);
    stop_fd_ = eventfd(0, EFD_NONBLOCK);

    running_ = true;
    worker_thread_ = std::thread(&IMUReader::worker, this);
    return true;
}

void IMUReader::stop()
{
    running_ = false;

    if (stop_fd_ >= 0) {
        uint64_t val = 1;
        write(stop_fd_, &val, sizeof(val));
    }

    if (worker_thread_.joinable()) worker_thread_.join();

    if (request_) {
        gpiod_line_request_release(request_);
        request_ = nullptr;
    }
    if (chip_) {
        gpiod_chip_close(chip_);
        chip_ = nullptr;
    }

    if (stop_fd_ >= 0) {
        close(stop_fd_);
        stop_fd_ = -1;
    }
}

void IMUReader::worker()
{
    int epoll_fd = epoll_create1(0);

    epoll_event gpio_ev{};
    gpio_ev.events  = EPOLLIN | EPOLLET;
    gpio_ev.data.fd = gpio_fd_;
    epoll_ctl(epoll_fd, EPOLL_CTL_ADD, gpio_fd_, &gpio_ev);

    epoll_event stop_ev{};
    stop_ev.events  = EPOLLIN;
    stop_ev.data.fd = stop_fd_;
```

```
    epoll_ctl(epoll_fd, EPOLL_CTL_ADD, stop_fd_, &stop_ev);

    gpiod_edge_event_buffer* buffer = gpiod_edge_event_buffer_new(1);

    epoll_event fired{};
    while (running_) {
        int n = epoll_wait(epoll_fd, &fired, 1, -1);
        if (n <= 0 || !running_) break;
        if (fired.data.fd == stop_fd_) break;

        if (fired.data.fd == gpio_fd_) {
            int got = gpiod_line_request_read_edge_events(request_, buffer, 1);
            if (got > 0) {
                gpiod_edge_event* event = gpiod_edge_event_buffer_get_event(buffer, 0);
                if (gpiod_edge_event_get_event_type(event) == GPIOD_EDGE_EVENT_RISING_EDGE) {
                    icm20948::IMUSample sample{};

                    bool ok = false;
                    {
                        std::lock_guard<std::mutex> lk(i2c1_mutex());
                        ok = imu_.read_sample(sample);


                        if (ok) {
                        imu_.check_DRDY_INT();
                                }
                    }

                    if (ok && callback_) {
                        callback_(sample.ax, sample.ay, sample.az,
                                  sample.gx, sample.gy, sample.gz,
                                  sample.mx, sample.my);
                    }
                }
            }
        }
    }

    gpiod_edge_event_buffer_free(buffer);
    close(epoll_fd);
}
```

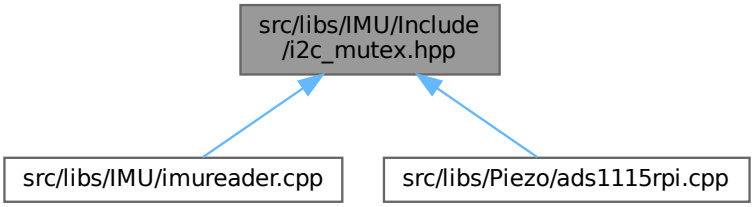
11.15 src/libs/IMU/Include/i2c_mutex.hpp File Reference

`#include <mutex>`

Include dependency graph for i2c_mutex.hpp:



This graph shows which files directly or indirectly include this file:



Functions

- std::mutex & i2c1_mutex ()

11.15.1 Function Documentation

11.15.1.1 i2c1_mutex()

```
std::mutex & i2c1_mutex ( )  [inline]
```

Shared mutex protecting access to the /dev/i2c-1 bus.
SOLID principles: S - This helper provides one shared synchronisation point for I2C bus access. D - Centralising the bus lock in one place avoids duplicating global mutex definitions across multiple modules and makes cross-thread I2C ownership easier to reason about. Shared mutex for the I2C-1 bus. IMUReader (ICM20948) and ADS1115rpi run on separate threads and both access /dev/i2c-1. Without this lock, their ioctl(I2C_SLAVE) and read/write sequences can interleave, causing transactions to target the wrong device and triggering the ADC retry failures seen in logs. Function-local static storage gives one shared mutex instance for the whole process while keeping the global namespace clean.
Definition at line 18 of file i2c_mutex.hpp.

11.16 i2c_mutex.hpp

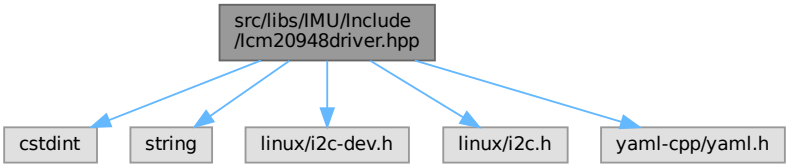
Go to the documentation of this file.

```
00001
00009 #pragma once
00010 #include <mutex>
00011
00018 inline std::mutex& i2c1_mutex()
00019 {
00022     static std::mutex m;
00023     return m;
00024 }
```

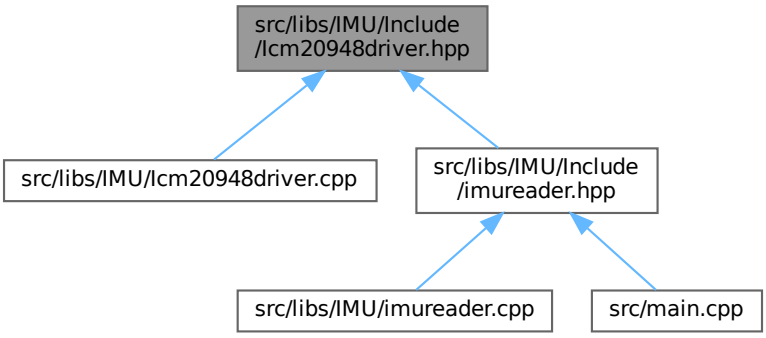
11.17 src/libs/IMU/Include/Icm20948driver.hpp File Reference

```
#include <cstdint>
#include <string>
#include <linux/i2c-dev.h>
#include <linux/i2c.h>
#include "yaml-cpp/yaml.h"
```

Include dependency graph for Icm20948driver.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct icm20948::accel_settings
- struct icm20948::gyro_settings

  Gyroscope runtime configuration.
- struct icm20948::magn_settings

  Magnetometer runtime configuration.
- struct icm20948::settings
- struct icm20948::IMUSample
- class icm20948::ICM20948_I2C

Namespaces

- namespace icm20948

  Settings types.

Macros

- #define ICM20948_I2C_ADDR 0x69
- #define ICM20948_MAGN_I2C_ADDR 0x0C
- #define ICM20948_REG_BANK_SEL_ADDR 0x7F
- #define ICM20948_REG_BANK_SEL_BANK0_VALUE 0x00
- #define ICM20948_REG_BANK_SEL_BANK1_VALUE 0x10
- #define ICM20948_REG_BANK_SEL_BANK2_VALUE 0x20

- #define ICM20948_REG_BANK_SEL_BANK3_VALUE 0x30
- #define ICM20948_WHO_AM_I_ADDR 0x00
- #define ICM20948_USER_CTRL_ADDR 0x03
- #define ICM20948_PWR_MGMT_1_ADDR 0x06
- #define ICM20948_INT_PIN_CFG_ADDR 0x0F
- #define ICM20948_INT_ENABLE_1_ADDR 0x11
- #define ICM20948_I2C_MST_STATUS_ADDR 0x17
- #define ICM20948_ACCEL_XOUT_H_ADDR 0x2D
- #define ICM20948_ACCEL_XOUT_L_ADDR 0x2E
- #define ICM20948_ACCEL_YOUT_H_ADDR 0x2F
- #define ICM20948_ACCEL_YOUT_L_ADDR 0x30
- #define ICM20948_ACCEL_ZOUT_H_ADDR 0x31
- #define ICM20948_ACCEL_ZOUT_L_ADDR 0x32
- #define ICM20948_GYRO_XOUT_H_ADDR 0x33
- #define ICM20948_GYRO_XOUT_L_ADDR 0x34
- #define ICM20948_GYRO_YOUT_H_ADDR 0x35
- #define ICM20948_GYRO_YOUT_L_ADDR 0x36
- #define ICM20948_GYRO_ZOUT_H_ADDR 0x37
- #define ICM20948_GYRO_ZOUT_L_ADDR 0x38
- #define ICM20948_TEMP_OUT_H_ADDR 0x39
- #define ICM20948_TEMP_OUT_L_ADDR 0x3A
- #define ICM20948_EXT_SLV_SENS_DATA_00_ADDR 0x3B
- #define ICM20948_WHO_AM_I_BANK 0
- #define ICM20948_USER_CTRL_BANK 0
- #define ICM20948_PWR_MGMT_1_BANK 0
- #define ICM20948_INT_PIN_CFG_BANK 0
- #define ICM20948_INT_ENABLE_1_BANK 0
- #define ICM20948_I2C_MST_STATUS_BANK 0
- #define ICM20948_ACCEL_OUT_BANK 0
- #define ICM20948_GYRO_OUT_BANK 0
- #define ICM20948_EXT_SLV_SENS_DATA_00_BANK 0
- #define ICM20948_GYRO_SMPLRT_DIV_ADDR 0x00
- #define ICM20948_GYRO_CONFIG_1_ADDR 0x01
- #define ICM20948_ACCEL_SMPLRT_DIV_1_ADDR 0x10
- #define ICM20948_ACCEL_SMPLRT_DIV_2_ADDR 0x11
- #define ICM20948_ACCEL_CONFIG_1_ADDR 0x14
- #define ICM20948_GYRO_SMPLRT_DIV_BANK 2

  Bank IDs for accel/gyro configuration registers.
- #define ICM20948_GYRO_CONFIG_1_BANK 2
- #define ICM20948_ACCEL_SMPLRT_DIV_1_BANK 2
- #define ICM20948_ACCEL_SMPLRT_DIV_2_BANK 2
- #define ICM20948_ACCEL_CONFIG_1_BANK 2
- #define ICM20948_I2C_MST_CTRL_ADDR 0x01
- #define ICM20948_I2C_SLV0_ADDR_ADDR 0x03
- #define ICM20948_I2C_SLV0_REG_ADDR 0x04
- #define ICM20948_I2C_SLV0_CTRL_ADDR 0x05
- #define ICM20948_I2C_SLV4_ADDR_ADDR 0x13
- #define ICM20948_I2C_SLV4_REG_ADDR 0x14
- #define ICM20948_I2C_SLV4_CTRL_ADDR 0x15
- #define ICM20948_I2C_SLV4_DO_ADDR 0x16
- #define ICM20948_I2C_SLV4_DI_ADDR 0x17
- #define ICM20948_I2C_MST_CTRL_BANK 3

  Bank IDs for the internal I2C master interface.
- #define ICM20948_I2C_SLV0_ADDR_BANK 3

- #define ICM20948_I2C_SLV0_REG_BANK 3
- #define ICM20948_I2C_SLV0_CTRL_BANK 3
- #define ICM20948_I2C_SLV4_ADDR_BANK 3
- #define ICM20948_I2C_SLV4_REG_BANK 3
- #define ICM20948_I2C_SLV4_CTRL_BANK 3
- #define ICM20948_I2C_SLV4_DO_BANK 3
- #define ICM20948_I2C_SLV4_DI_BANK 3
- #define AK09916_CNTL2_ADDR 0x31

  AK09916 magnetometer control register and expected WHO_AM_I value.
- #define ICM20948_BANK0_WHO_AM_I_VALUE 0xEA
- #define ICM20948_INT_STATUS_BANK 0
- #define ICM20948_INT_STATUS_1_BANK 0
- #define ICM20948_INT_STATUS_ADDR 0x19
- #define ICM20948_INT_STATUS_1_ADDR 0x1A

Typedefs

- typedef struct icm20948::accel_settings icm20948::accel_settings
- typedef struct icm20948::gyro_settings icm20948::gyro_settings

  Gyroscope runtime configuration.
- typedef struct icm20948::magn_settings icm20948::magn_settings

  Magnetometer runtime configuration.
- typedef struct icm20948::settings icm20948::settings

Enumerations

- enum icm20948::accel_scale { icm20948::ACCEL_2G = 0 , icm20948::ACCEL_4G , icm20948::ACCEL_8G , icm20948::ACCEL_16G }
- enum icm20948::accel_dlpf_config { icm20948::ACCEL_DLPF_246HZ = 0 , icm20948::ACCEL_DLPF_246HZ_2 , icm20948::ACCEL_DLPF_111_4HZ , icm20948::ACCEL_DLPF_50_4HZ , icm20948::ACCEL_DLPF_23_9HZ , icm20948::ACCEL_DLPF_11_5HZ , icm20948::ACCEL_DLPF_5_7HZ , icm20948::ACCEL_DLPF_473HZ }

  Accelerometer digital low-pass filter config.
- enum icm20948::gyro_scale { icm20948::GYRO_250DPS = 0 , icm20948::GYRO_500DPS , icm20948::GYRO_1000DPS , icm20948::GYRO_2000DPS }

  Gyroscope full-scale range.
- enum icm20948::gyro_dlpf_config { icm20948::GYRO_DLPF_196_6HZ = 0 , icm20948::GYRO_DLPF_151_8HZ , icm20948::GYRO_DLPF_119_5HZ , icm20948::GYRO_DLPF_51_2HZ , icm20948::GYRO_DLPF_23_9HZ , icm20948::GYRO_DLPF_11_6HZ , icm20948::GYRO_DLPF_5_7HZ , icm20948::GYRO_DLPF_361_4HZ }

  Gyroscope digital low-pass filter config.
- enum icm20948::magn_mode { icm20948::MAGN_SHUTDOWN = 0 , icm20948::MAGN_SINGLE = 1 , icm20948::MAGN_10HZ = 2 , icm20948::MAGN_20HZ = 4 , icm20948::MAGN_50HZ = 6 , icm20948::MAGN_100HZ = 8 , icm20948::MAGN_SELF_TEST = 16 }

  Magnetometer operating mode.

Functions

- float icm20948::accel_scale_factor (accel_scale scale)

  Convert accel range enum to raw-count scale factor.
- std::string icm20948::accel_scale_to_str (accel_scale scale)
- std::string icm20948::accel_dlpf_config_to_str (accel_dlpf_config config)

Stringify accel DLPF enum.

- float icm20948::gyro_scale_factor (gyro_scale scale)

  Convert gyro range enum to raw-count scale factor.
- std::string icm20948::gyro_scale_to_str (gyro_scale scale)

  Stringify gyro range enum.
- std::string icm20948::gyro_dlpf_config_to_str (gyro_dlpf_config config)

  Stringify gyro DLPF enum.
- std::string icm20948::magn_mode_to_str (magn_mode mode)

  Stringify magnetometer mode enum.

11.17.1 Macro Definition Documentation

11.17.1.1 AK09916_CNTL2_ADDR

```
#define AK09916_CNTL2_ADDR 0x31
```

AK09916 magnetometer control register and expected WHO_AM_I value.
Definition at line 114 of file Icm20948driver.hpp.

11.17.1.2 ICM20948_ACCEL_CONFIG_1_ADDR

```
#define ICM20948_ACCEL_CONFIG_1_ADDR 0x14
```

Definition at line 80 of file Icm20948driver.hpp.

11.17.1.3 ICM20948_ACCEL_CONFIG_1_BANK

```
#define ICM20948_ACCEL_CONFIG_1_BANK 2
```

Definition at line 87 of file Icm20948driver.hpp.

11.17.1.4 ICM20948_ACCEL_OUT_BANK

```
#define ICM20948_ACCEL_OUT_BANK 0
```

Definition at line 69 of file Icm20948driver.hpp.

11.17.1.5 ICM20948_ACCEL_SMPLRT_DIV_1_ADDR

```
#define ICM20948_ACCEL_SMPLRT_DIV_1_ADDR 0x10
```

Definition at line 78 of file Icm20948driver.hpp.

11.17.1.6 ICM20948_ACCEL_SMPLRT_DIV_1_BANK

```
#define ICM20948_ACCEL_SMPLRT_DIV_1_BANK 2
```

Definition at line 85 of file Icm20948driver.hpp.

11.17.1.7 ICM20948_ACCEL_SMPLRT_DIV_2_ADDR

```
#define ICM20948_ACCEL_SMPLRT_DIV_2_ADDR 0x11
```

Definition at line 79 of file Icm20948driver.hpp.

11.17.1.8 ICM20948_ACCEL_SMPLRT_DIV_2_BANK

```
#define ICM20948_ACCEL_SMPLRT_DIV_2_BANK 2
```

Definition at line 86 of file Icm20948driver.hpp.

11.17.1.9 ICM20948_ACCEL_XOUT_H_ADDR

```
#define ICM20948_ACCEL_XOUT_H_ADDR 0x2D
```

Definition at line 44 of file Icm20948driver.hpp.

11.17.1.10 ICM20948_ACCEL_XOUT_L_ADDR

```
#define ICM20948_ACCEL_XOUT_L_ADDR 0x2E
```

Definition at line 45 of file Icm20948driver.hpp.

11.17.1.11 ICM20948_ACCEL_YOUT_H_ADDR

```
#define ICM20948_ACCEL_YOUT_H_ADDR 0x2F
```

Definition at line 46 of file Icm20948driver.hpp.

11.17.1.12 ICM20948_ACCEL_YOUT_L_ADDR

```
#define ICM20948_ACCEL_YOUT_L_ADDR 0x30
```

Definition at line 47 of file Icm20948driver.hpp.

11.17.1.13 ICM20948_ACCEL_ZOUT_H_ADDR

```
#define ICM20948_ACCEL_ZOUT_H_ADDR 0x31
```

Definition at line 48 of file Icm20948driver.hpp.

11.17.1.14 ICM20948_ACCEL_ZOUT_L_ADDR

```
#define ICM20948_ACCEL_ZOUT_L_ADDR 0x32
```

Definition at line 49 of file Icm20948driver.hpp.

11.17.1.15 ICM20948_BANK0_WHO_AM_I_VALUE

```
#define ICM20948_BANK0_WHO_AM_I_VALUE 0xEA
```

Definition at line 115 of file Icm20948driver.hpp.

11.17.1.16 ICM20948_EXT_SLV_SENS_DATA_00_ADDR

```
#define ICM20948_EXT_SLV_SENS_DATA_00_ADDR 0x3B
```

Definition at line 58 of file Icm20948driver.hpp.

11.17.1.17 ICM20948_EXT_SLV_SENS_DATA_00_BANK

```
#define ICM20948_EXT_SLV_SENS_DATA_00_BANK 0
```

Definition at line 71 of file Icm20948driver.hpp.

11.17.1.18 ICM20948_GYRO_CONFIG_1_ADDR

```
#define ICM20948_GYRO_CONFIG_1_ADDR 0x01
```

Definition at line 77 of file Icm20948driver.hpp.

11.17.1.19 ICM20948_GYRO_CONFIG_1_BANK

```
#define ICM20948_GYRO_CONFIG_1_BANK 2
```

Definition at line 84 of file Icm20948driver.hpp.

11.17.1.20 ICM20948_GYRO_OUT_BANK

```
#define ICM20948_GYRO_OUT_BANK 0
```

Definition at line 70 of file Icm20948driver.hpp.

11.17.1.21 ICM20948_GYRO_SMPLRT_DIV_ADDR

```
#define ICM20948_GYRO_SMPLRT_DIV_ADDR 0x00
```

Bank 2 Configuration registers for sample rate, scale, and digital filtering.
Definition at line 76 of file Icm20948driver.hpp.

11.17.1.22 ICM20948_GYRO_SMPLRT_DIV_BANK

```
#define ICM20948_GYRO_SMPLRT_DIV_BANK 2
```

Bank IDs for accel/gyro configuration registers.
Definition at line 83 of file Icm20948driver.hpp.

11.17.1.23 ICM20948_GYRO_XOUT_H_ADDR

```
#define ICM20948_GYRO_XOUT_H_ADDR 0x33
```

Definition at line 50 of file Icm20948driver.hpp.

11.17.1.24 ICM20948_GYRO_XOUT_L_ADDR

```
#define ICM20948_GYRO_XOUT_L_ADDR 0x34
```

Definition at line 51 of file Icm20948driver.hpp.

11.17.1.25 ICM20948_GYRO_YOUT_H_ADDR

```
#define ICM20948_GYRO_YOUT_H_ADDR 0x35
```

Definition at line 52 of file Icm20948driver.hpp.

11.17.1.26 ICM20948_GYRO_YOUT_L_ADDR

```
#define ICM20948_GYRO_YOUT_L_ADDR 0x36
```

Definition at line 53 of file Icm20948driver.hpp.

11.17.1.27 ICM20948_GYRO_ZOUT_H_ADDR

```
#define ICM20948_GYRO_ZOUT_H_ADDR 0x37
```

Definition at line 54 of file Icm20948driver.hpp.

11.17.1.28 ICM20948_GYRO_ZOUT_L_ADDR

```
#define ICM20948_GYRO_ZOUT_L_ADDR 0x38
```

Definition at line 55 of file Icm20948driver.hpp.

11.17.1.29 ICM20948_I2C_ADDR

```
#define ICM20948_I2C_ADDR 0x69
```

This header defines the register map, configuration types, and public driver interface for the ICM-20948 IMU and its onboard magnetometer.
SOLID principles: S - This file centralises device-specific constants, configuration structures, and the public interface for the IMU driver in one place. O - Configuration is represented through settings types, allowing the driver to be reconfigured without changing client code that uses the interface. I - Configuration is grouped into smaller settings types (accelerometer, gyroscope, magnetometer) so callers interact with the parts of the device configuration they actually need. Register addresses and constants for the ICM-20948 IMU and onboard magnetometer. Register map
Definition at line 24 of file Icm20948driver.hpp.

11.17.1.30 ICM20948_I2C_MST_CTRL_ADDR

```
#define ICM20948_I2C_MST_CTRL_ADDR 0x01
```

Bank 3 Registers for the internal I2C master used to communicate with the onboard magnetometer through the main IMU device.
Definition at line 92 of file Icm20948driver.hpp.

11.17.1.31 ICM20948_I2C_MST_CTRL_BANK

```
#define ICM20948_I2C_MST_CTRL_BANK 3
```

Bank IDs for the internal I2C master interface.
Definition at line 103 of file Icm20948driver.hpp.

11.17.1.32 ICM20948_I2C_MST_STATUS_ADDR

```
#define ICM20948_I2C_MST_STATUS_ADDR 0x17
```

Definition at line 43 of file Icm20948driver.hpp.

11.17.1.33 ICM20948_I2C_MST_STATUS_BANK

```
#define ICM20948_I2C_MST_STATUS_BANK 0
```

Definition at line 68 of file Icm20948driver.hpp.

11.17.1.34 ICM20948_I2C_SLV0_ADDR_ADDR

```
#define ICM20948_I2C_SLV0_ADDR_ADDR 0x03
```

Definition at line 93 of file Icm20948driver.hpp.

11.17.1.35 ICM20948_I2C_SLV0_ADDR_BANK

```
#define ICM20948_I2C_SLV0_ADDR_BANK 3
```

Definition at line 104 of file Icm20948driver.hpp.

11.17.1.36 ICM20948_I2C_SLV0_CTRL_ADDR

```
#define ICM20948_I2C_SLV0_CTRL_ADDR 0x05
```

Definition at line 95 of file Icm20948driver.hpp.

11.17.1.37 ICM20948_I2C_SLV0_CTRL_BANK

```
#define ICM20948_I2C_SLV0_CTRL_BANK 3
```

Definition at line 106 of file Icm20948driver.hpp.

11.17.1.38 ICM20948_I2C_SLV0_REG_ADDR

```
#define ICM20948_I2C_SLV0_REG_ADDR 0x04
```

Definition at line 94 of file Icm20948driver.hpp.

11.17.1.39 ICM20948_I2C_SLV0_REG_BANK

```
#define ICM20948_I2C_SLV0_REG_BANK 3
```

Definition at line 105 of file Icm20948driver.hpp.

11.17.1.40 ICM20948_I2C_SLV4_ADDR_ADDR

```
#define ICM20948_I2C_SLV4_ADDR_ADDR 0x13
```

Definition at line 96 of file Icm20948driver.hpp.

11.17.1.41 ICM20948_I2C_SLV4_ADDR_BANK

```
#define ICM20948_I2C_SLV4_ADDR_BANK 3
```

Definition at line 107 of file Icm20948driver.hpp.

11.17.1.42 ICM20948_I2C_SLV4_CTRL_ADDR

```
#define ICM20948_I2C_SLV4_CTRL_ADDR 0x15
```

Definition at line 98 of file Icm20948driver.hpp.

11.17.1.43 ICM20948_I2C_SLV4_CTRL_BANK

```
#define ICM20948_I2C_SLV4_CTRL_BANK 3
```

Definition at line 109 of file Icm20948driver.hpp.

11.17.1.44 ICM20948_I2C_SLV4_DI_ADDR

```
#define ICM20948_I2C_SLV4_DI_ADDR 0x17
```

Definition at line 100 of file Icm20948driver.hpp.

11.17.1.45 ICM20948_I2C_SLV4_DI_BANK

```
#define ICM20948_I2C_SLV4_DI_BANK 3
```

Definition at line 111 of file Icm20948driver.hpp.

11.17.1.46 ICM20948_I2C_SLV4_DO_ADDR

```
#define ICM20948_I2C_SLV4_DO_ADDR 0x16
```

Definition at line 99 of file Icm20948driver.hpp.

11.17.1.47 ICM20948_I2C_SLV4_DO_BANK

```
#define ICM20948_I2C_SLV4_DO_BANK 3
```

Definition at line 110 of file Icm20948driver.hpp.

11.17.1.48 ICM20948_I2C_SLV4_REG_ADDR

```
#define ICM20948_I2C_SLV4_REG_ADDR 0x14
```

Definition at line 97 of file Icm20948driver.hpp.

11.17.1.49 ICM20948_I2C_SLV4_REG_BANK

```
#define ICM20948_I2C_SLV4_REG_BANK 3
```

Definition at line 108 of file Icm20948driver.hpp.

11.17.1.50 ICM20948_INT_ENABLE_1_ADDR

```
#define ICM20948_INT_ENABLE_1_ADDR 0x11
```

Definition at line 42 of file Icm20948driver.hpp.

11.17.1.51 ICM20948_INT_ENABLE_1_BANK

```
#define ICM20948_INT_ENABLE_1_BANK 0
```

Definition at line 67 of file Icm20948driver.hpp.

11.17.1.52 ICM20948_INT_PIN_CFG_ADDR

```
#define ICM20948_INT_PIN_CFG_ADDR 0x0F
```

Definition at line 41 of file Icm20948driver.hpp.

11.17.1.53 ICM20948_INT_PIN_CFG_BANK

```
#define ICM20948_INT_PIN_CFG_BANK 0
```

Definition at line 66 of file Icm20948driver.hpp.

11.17.1.54 ICM20948_INT_STATUS_1_ADDR

```
#define ICM20948_INT_STATUS_1_ADDR 0x1A
```

Definition at line 125 of file Icm20948driver.hpp.

11.17.1.55 ICM20948_INT_STATUS_1_BANK

```
#define ICM20948_INT_STATUS_1_BANK 0
```

Definition at line 122 of file Icm20948driver.hpp.

11.17.1.56 ICM20948_INT_STATUS_ADDR

```
#define ICM20948_INT_STATUS_ADDR 0x19
```

Definition at line 124 of file Icm20948driver.hpp.

11.17.1.57 ICM20948_INT_STATUS_BANK

```
#define ICM20948_INT_STATUS_BANK 0
```

Interrupt status registers. These support event-driven sampling by allowing higher-level code to check whether new data is ready before attempting a read.
Definition at line 121 of file Icm20948driver.hpp.

11.17.1.58 ICM20948_MAGN_I2C_ADDR

```
#define ICM20948_MAGN_I2C_ADDR 0x0C
```

Definition at line 25 of file Icm20948driver.hpp.

11.17.1.59 ICM20948_PWR_MGMT_1_ADDR

```
#define ICM20948_PWR_MGMT_1_ADDR 0x06
```

Definition at line 40 of file Icm20948driver.hpp.

11.17.1.60 ICM20948_PWR_MGMT_1_BANK

```
#define ICM20948_PWR_MGMT_1_BANK 0
```

Definition at line 65 of file Icm20948driver.hpp.

11.17.1.61 ICM20948_REG_BANK_SEL_ADDR

```
#define ICM20948_REG_BANK_SEL_ADDR 0x7F
```

Bank-select register and encoded bank values. These constants isolate device-specific magic numbers from the driver logic.
Definition at line 29 of file Icm20948driver.hpp.

11.17.1.62 ICM20948_REG_BANK_SEL_BANK0_VALUE

```
#define ICM20948_REG_BANK_SEL_BANK0_VALUE 0x00
```

Definition at line 30 of file Icm20948driver.hpp.

11.17.1.63 ICM20948_REG_BANK_SEL_BANK1_VALUE

```
#define ICM20948_REG_BANK_SEL_BANK1_VALUE 0x10
```

Definition at line 31 of file Icm20948driver.hpp.

11.17.1.64 ICM20948_REG_BANK_SEL_BANK2_VALUE

```
#define ICM20948_REG_BANK_SEL_BANK2_VALUE 0x20
```

Definition at line 32 of file Icm20948driver.hpp.

11.17.1.65 ICM20948_REG_BANK_SEL_BANK3_VALUE

```
#define ICM20948_REG_BANK_SEL_BANK3_VALUE 0x30
```

Definition at line 33 of file Icm20948driver.hpp.

11.17.1.66 ICM20948_TEMP_OUT_H_ADDR

```
#define ICM20948_TEMP_OUT_H_ADDR 0x39
```

Definition at line 56 of file Icm20948driver.hpp.

11.17.1.67 ICM20948_TEMP_OUT_L_ADDR

```
#define ICM20948_TEMP_OUT_L_ADDR 0x3A
```

Definition at line 57 of file lcm20948driver.hpp.

11.17.1.68 ICM20948_USER_CTRL_ADDR

```
#define ICM20948_USER_CTRL_ADDR 0x03
```

Definition at line 39 of file lcm20948driver.hpp.

11.17.1.69 ICM20948_USER_CTRL_BANK

```
#define ICM20948_USER_CTRL_BANK 0
```

Definition at line 64 of file lcm20948driver.hpp.

11.17.1.70 ICM20948_WHO_AM_I_ADDR

```
#define ICM20948_WHO_AM_I_ADDR 0x00
```

Bank 0 Output and control registers used for identity, power, interrupts, and reading decoded sensor data.
Definition at line 38 of file lcm20948driver.hpp.

11.17.1.71 ICM20948_WHO_AM_I_BANK

```
#define ICM20948_WHO_AM_I_BANK 0
```

Bank IDs for the corresponding registers above. Keeping bank IDs alongside register definitions reduces the chance of mismatched bank/register access in the implementation.
Definition at line 63 of file lcm20948driver.hpp.

11.18 lcm20948driver.hpp

Go to the documentation of this file.

```
00001
00013 #pragma once
00014
00015 #include <cstdint>
00016 #include <string>
00017 #include <linux/i2c-dev.h>
00018 #include <linux/i2c.h>
00019 #include "yaml-cpp/yaml.h"
00020
00023
00024 #define ICM20948_I2C_ADDR       0x69
00025 #define ICM20948_MAGN_I2C_ADDR  0x0C
00026
00029 #define ICM20948_REG_BANK_SEL_ADDR          0x7F
00030 #define ICM20948_REG_BANK_SEL_BANK0_VALUE   0x00
00031 #define ICM20948_REG_BANK_SEL_BANK1_VALUE   0x10
00032 #define ICM20948_REG_BANK_SEL_BANK2_VALUE   0x20
00033 #define ICM20948_REG_BANK_SEL_BANK3_VALUE   0x30
00034
00038 #define ICM20948_WHO_AM_I_ADDR              0x00
00039 #define ICM20948_USER_CTRL_ADDR             0x03
00040 #define ICM20948_PWR_MGMT_1_ADDR            0x06
00041 #define ICM20948_INT_PIN_CFG_ADDR           0x0F
00042 #define ICM20948_INT_ENABLE_1_ADDR          0x11
00043 #define ICM20948_I2C_MST_STATUS_ADDR        0x17
00044 #define ICM20948_ACCEL_XOUT_H_ADDR          0x2D
00045 #define ICM20948_ACCEL_XOUT_L_ADDR          0x2E
00046 #define ICM20948_ACCEL_YOUT_H_ADDR          0x2F
00047 #define ICM20948_ACCEL_YOUT_L_ADDR          0x30
00048 #define ICM20948_ACCEL_ZOUT_H_ADDR          0x31
00049 #define ICM20948_ACCEL_ZOUT_L_ADDR          0x32
00050 #define ICM20948_GYRO_XOUT_H_ADDR           0x33
00051 #define ICM20948_GYRO_XOUT_L_ADDR           0x34
00052 #define ICM20948_GYRO_YOUT_H_ADDR           0x35
00053 #define ICM20948_GYRO_YOUT_L_ADDR           0x36
00054 #define ICM20948_GYRO_ZOUT_H_ADDR           0x37
00055 #define ICM20948_GYRO_ZOUT_L_ADDR           0x38
00056 #define ICM20948_TEMP_OUT_H_ADDR            0x39
00057 #define ICM20948_TEMP_OUT_L_ADDR            0x3A
00058 #define ICM20948_EXT_SLV_SENS_DATA_00_ADDR  0x3B
00059
00063 #define ICM20948_WHO_AM_I_BANK              0
```

```
#define ICM20948_USER_CTRL_BANK             0
#define ICM20948_PWR_MGMT_1_BANK            0
#define ICM20948_INT_PIN_CFG_BANK           0
#define ICM20948_INT_ENABLE_1_BANK          0
#define ICM20948_I2C_MST_STATUS_BANK        0
#define ICM20948_ACCEL_OUT_BANK             0
#define ICM20948_GYRO_OUT_BANK              0
#define ICM20948_EXT_SLV_SENS_DATA_00_BANK  0


#define ICM20948_GYRO_SMPLRT_DIV_ADDR       0x00
#define ICM20948_GYRO_CONFIG_1_ADDR         0x01
#define ICM20948_ACCEL_SMPLRT_DIV_1_ADDR    0x10
#define ICM20948_ACCEL_SMPLRT_DIV_2_ADDR    0x11
#define ICM20948_ACCEL_CONFIG_1_ADDR        0x14

#define ICM20948_GYRO_SMPLRT_DIV_BANK       2
#define ICM20948_GYRO_CONFIG_1_BANK         2
#define ICM20948_ACCEL_SMPLRT_DIV_1_BANK    2
#define ICM20948_ACCEL_SMPLRT_DIV_2_BANK    2
#define ICM20948_ACCEL_CONFIG_1_BANK        2

#define ICM20948_I2C_MST_CTRL_ADDR          0x01
#define ICM20948_I2C_SLV0_ADDR_ADDR         0x03
#define ICM20948_I2C_SLV0_REG_ADDR          0x04
#define ICM20948_I2C_SLV0_CTRL_ADDR         0x05
#define ICM20948_I2C_SLV4_ADDR_ADDR         0x13
#define ICM20948_I2C_SLV4_REG_ADDR          0x14
#define ICM20948_I2C_SLV4_CTRL_ADDR         0x15
#define ICM20948_I2C_SLV4_DO_ADDR           0x16
#define ICM20948_I2C_SLV4_DI_ADDR           0x17

#define ICM20948_I2C_MST_CTRL_BANK          3
#define ICM20948_I2C_SLV0_ADDR_BANK         3
#define ICM20948_I2C_SLV0_REG_BANK          3
#define ICM20948_I2C_SLV0_CTRL_BANK         3
#define ICM20948_I2C_SLV4_ADDR_BANK         3
#define ICM20948_I2C_SLV4_REG_BANK          3
#define ICM20948_I2C_SLV4_CTRL_BANK         3
#define ICM20948_I2C_SLV4_DO_BANK           3
#define ICM20948_I2C_SLV4_DI_BANK           3

#define AK09916_CNTL2_ADDR                  0x31
#define ICM20948_BANK0_WHO_AM_I_VALUE       0xEA


#define ICM20948_INT_STATUS_BANK            0
#define ICM20948_INT_STATUS_1_BANK          0

#define ICM20948_INT_STATUS_ADDR            0x19
#define ICM20948_INT_STATUS_1_ADDR          0x1A


namespace icm20948
{
    typedef enum {
        ACCEL_2G  = 0,
        ACCEL_4G,
        ACCEL_8G,
        ACCEL_16G
    } accel_scale;

    typedef enum {
        ACCEL_DLPF_246HZ   = 0,
        ACCEL_DLPF_246HZ_2,
        ACCEL_DLPF_111_4HZ,
        ACCEL_DLPF_50_4HZ,
        ACCEL_DLPF_23_9HZ,
        ACCEL_DLPF_11_5HZ,
        ACCEL_DLPF_5_7HZ,
        ACCEL_DLPF_473HZ
    } accel_dlpf_config;

    typedef struct accel_settings {
        uint16_t          sample_rate_div;
        accel_scale       scale;
        bool              dlpf_enable;
        accel_dlpf_config dlpf_config;

        accel_settings(uint16_t          sample_rate_div = 0,
                       accel_scale       scale           = ACCEL_2G,
                       bool              dlpf_enable     = true,
                       accel_dlpf_config dlpf_config     = ACCEL_DLPF_246HZ)
            : sample_rate_div(sample_rate_div), scale(scale),
              dlpf_enable(dlpf_enable), dlpf_config(dlpf_config) {}
    } accel_settings;
```

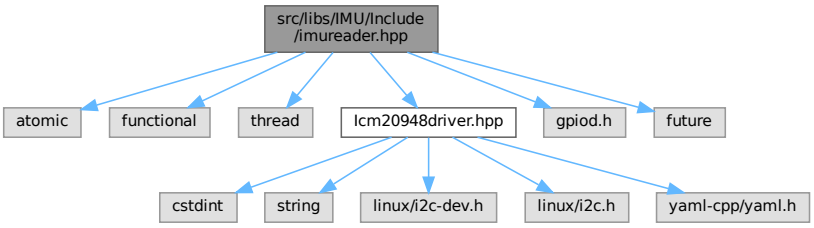
```
    typedef enum {
        GYRO_250DPS  = 0,
        GYRO_500DPS,
        GYRO_1000DPS,
        GYRO_2000DPS
    } gyro_scale;

    typedef enum {
        GYRO_DLPF_196_6HZ = 0,
        GYRO_DLPF_151_8HZ,
        GYRO_DLPF_119_5HZ,
        GYRO_DLPF_51_2HZ,
        GYRO_DLPF_23_9HZ,
        GYRO_DLPF_11_6HZ,
        GYRO_DLPF_5_7HZ,
        GYRO_DLPF_361_4HZ
    } gyro_dlpf_config;

    typedef struct gyro_settings {
        uint8_t          sample_rate_div;
        gyro_scale       scale;
        bool             dlpf_enable;
        gyro_dlpf_config dlpf_config;

        gyro_settings(uint8_t          sample_rate_div = 0,
                      gyro_scale       scale           = GYRO_250DPS,
                      bool             dlpf_enable     = true,
                      gyro_dlpf_config dlpf_config     = GYRO_DLPF_196_6HZ)
            : sample_rate_div(sample_rate_div), scale(scale),
              dlpf_enable(dlpf_enable), dlpf_config(dlpf_config) {}
    } gyro_settings;

    typedef enum {
        MAGN_SHUTDOWN  = 0,
        MAGN_SINGLE    = 1,
        MAGN_10HZ      = 2,
        MAGN_20HZ      = 4,
        MAGN_50HZ      = 6,
        MAGN_100HZ     = 8,
        MAGN_SELF_TEST = 16
    } magn_mode;

    typedef struct magn_settings {
        magn_mode mode;
        explicit magn_settings(magn_mode mode = MAGN_100HZ) : mode(mode) {}
    } magn_settings;

    typedef struct settings {
        accel_settings accel;
        gyro_settings  gyro;
        magn_settings  magn;

        settings(accel_settings accel = accel_settings(),
                 gyro_settings  gyro  = gyro_settings(),
                 magn_settings  magn  = magn_settings())
            : accel(accel), gyro(gyro), magn(magn) {}

        explicit settings(YAML::Node config_file_node);
    } settings;

    float       accel_scale_factor(accel_scale scale);
    std::string accel_scale_to_str(accel_scale scale);
    std::string accel_dlpf_config_to_str(accel_dlpf_config config);

    float       gyro_scale_factor(gyro_scale scale);
    std::string gyro_scale_to_str(gyro_scale scale);
    std::string gyro_dlpf_config_to_str(gyro_dlpf_config config);

    std::string magn_mode_to_str(magn_mode mode);

    struct IMUSample {
        float ax, ay, az;
        float gx, gy, gz;
        float mx, my, mz;
    };

    class ICM20948_I2C
    {
    public:
        float accel[3];
        float gyro[3];
        float magn[3];

        icm20948::settings settings;

        ICM20948_I2C(unsigned           i2c_bus,
```

```
                    unsigned          i2c_address = ICM20948_I2C_ADDR,
                    icm20948::settings settings    = icm20948::settings());
        ~ICM20948_I2C();

        bool init();
        bool reset();
        bool wake();
        bool set_settings();
        bool read_accel_gyro();
        bool read_magn();
        bool read_sample(IMUSample& sample);
        bool check_DRDY_INT();

    private:
        int       _i2c_fd;
        unsigned  _i2c_bus;
        unsigned  _i2c_address;

        uint8_t   _current_bank;

        float     _accel_scale_factor;
        float     _gyro_scale_factor;
        float     _magn_scale_factor;

        bool _enable_data_ready_interrupt();
        bool _set_bank(uint8_t bank);
        bool _set_accel_sample_rate_div();
        bool _set_accel_range_dlpf();
        bool _set_gyro_sample_rate_div();
        bool _set_gyro_range_dlpf();

        bool _magnetometer_init();
        bool _magnetometer_enable();
        bool _magnetometer_set_mode();
        bool _magnetometer_configured();
        bool _magnetometer_set_readout();
        bool _chip_i2c_master_reset();

        bool _write_byte(uint8_t bank, uint8_t reg, uint8_t byte);
        bool _read_byte(uint8_t bank, uint8_t reg, uint8_t& byte);
        bool _write_bit(uint8_t bank, uint8_t reg, uint8_t bit_pos, bool bit);
        bool _read_bit(uint8_t bank, uint8_t reg, uint8_t bit_pos, bool& bit);
        bool _read_block_bytes(uint8_t bank, uint8_t start_reg, uint8_t* bytes, int length);
        bool _write_mag_byte(uint8_t mag_reg, uint8_t byte);
        bool _read_mag_byte(uint8_t mag_reg, uint8_t& byte);


    };

}
```

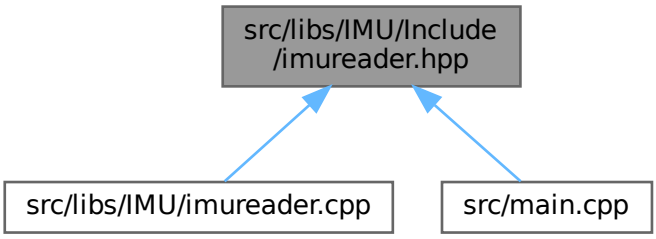
11.19 src/libs/IMU/Include/imureader.hpp File Reference

```
#include <atomic>
#include <functional>
#include <thread>
#include "Icm20948driver.hpp"
#include <gpiod.h>
#include <future>
```

Include dependency graph for imureader.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class IMUReader

11.20 imureader.hpp

Go to the documentation of this file.

```
#pragma once

#include <atomic>
#include <functional>
#include <thread>

#include "Icm20948driver.hpp"
#include <gpiod.h>
#include <future>

class IMUReader
{
public:
    using SampleCallback = std::function<void(float ax, float ay, float az,
                                              float gx, float gy, float gz,
                                              float mx, float my)>;

    IMUReader(unsigned           i2c_bus,
              unsigned           i2c_address = ICM20948_I2C_ADDR,
              const char*        gpiochip    = "/dev/gpiochip0",
              unsigned           rdy_line    = 27,
              icm20948::settings settings    = icm20948::settings());

    ~IMUReader();

    bool init();

    void setCallback(SampleCallback cb);

    bool start();

    void stop();
    icm20948::ICM20948_I2C& getIMU();
private:
    void worker();

    icm20948::ICM20948_I2C imu_;

    SampleCallback         callback_;

    std::thread            worker_thread_;

    std::atomic_bool       running_;

    const char*         gpiochip_path_;
    unsigned            rdy_line_;

    gpiod_chip*         chip_;
    gpiod_line_request* request_;
```

```

    int                 gpio_fd_;
    int                 stop_fd_;
};
```

11.21 README.md File Reference

11.22 src/libs/IMU/README.md File Reference

11.23 src/libs/Motor/README.md File Reference

11.24 src/libs/IMU_math/CMakeFiles/imu_math_↩ lib.dir/positiondetector.cpp.o.d File Reference

11.25 positiondetector.cpp.o.d

Go to the documentation of this file.

```
libs/IMU_math/CMakeFiles/imu_math_lib.dir/positiondetector.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/positiondetector.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/positiondetector.hpp \
 /usr/include/c++/14/cmath /usr/include/c++/14/bits/requires_hosted.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h \
 /usr/include/c++/14/bits/cpp_type_traits.h \
 /usr/include/c++/14/bits/version.h /usr/include/c++/14/ext/type_traits.h \
 /usr/include/math.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
 /usr/include/aarch64-linux-gnu/bits/time64.h \
 /usr/include/aarch64-linux-gnu/bits/math-vector.h \
 /usr/include/aarch64-linux-gnu/bits/libm-simd-decl-stubs.h \
 /usr/include/aarch64-linux-gnu/bits/floatn.h \
 /usr/include/aarch64-linux-gnu/bits/floatn-common.h \
 /usr/include/aarch64-linux-gnu/bits/flt-eval-method.h \
 /usr/include/aarch64-linux-gnu/bits/fp-logb.h \
 /usr/include/aarch64-linux-gnu/bits/fp-fast.h \
 /usr/include/aarch64-linux-gnu/bits/mathcalls-macros.h \
 /usr/include/aarch64-linux-gnu/bits/mathcalls-helper-functions.h \
 /usr/include/aarch64-linux-gnu/bits/mathcalls.h \
 /usr/include/aarch64-linux-gnu/bits/mathcalls-narrow.h \
 /usr/include/aarch64-linux-gnu/bits/iscanonical.h \
 /usr/include/c++/14/bits/std_abs.h /usr/include/stdlib.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/include/aarch64-linux-gnu/bits/waitflags.h \
 /usr/include/aarch64-linux-gnu/bits/waitstatus.h \
 /usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
 /usr/include/aarch64-linux-gnu/sys/types.h \
 /usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/time_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-intn.h /usr/include/endian.h \
 /usr/include/aarch64-linux-gnu/bits/endian.h \
 /usr/include/aarch64-linux-gnu/bits/endianness.h \
 /usr/include/aarch64-linux-gnu/bits/byteswap.h \
 /usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
 /usr/include/aarch64-linux-gnu/sys/select.h \
 /usr/include/aarch64-linux-gnu/bits/select.h \
 /usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
 /usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
 /usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
```

```
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/specfun.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/type_traits \
/usr/include/c++/14/bits/move.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/compare /usr/include/c++/14/concepts \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/stl_iterator.h /usr/include/c++/14/new \
/usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/limits /usr/include/c++/14/tr1/gamma.tcc \
/usr/include/c++/14/tr1/special_function_util.h \
/usr/include/c++/14/tr1/bessel_function.tcc \
/usr/include/c++/14/tr1/beta_function.tcc \
/usr/include/c++/14/tr1/ell_integral.tcc \
/usr/include/c++/14/tr1/exp_integral.tcc \
/usr/include/c++/14/tr1/hypergeometric.tcc \
/usr/include/c++/14/tr1/legendre_function.tcc \
/usr/include/c++/14/tr1/modified_bessel_func.tcc \
/usr/include/c++/14/tr1/poly_hermite.tcc \
/usr/include/c++/14/tr1/poly_laguerre.tcc \
/usr/include/c++/14/tr1/riemann_zeta.tcc /usr/include/c++/14/functional \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h /usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
/usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h \
/usr/include/c++/14/bits/memoryfwd.h \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/erase_if.h \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array
```

11.26 src/libs/IMU_math/CMakeFiles/imu_math_lib.dir/Swing↩ Calibrator.cpp.o.d File Reference

11.27 SwingCalibrator.cpp.o.d

Go to the documentation of this file.

```
libs/IMU_math/CMakeFiles/imu_math_lib.dir/SwingCalibrator.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/SwingCalibrator.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingCalibrator.hpp \
 /usr/include/c++/14/functional \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
```

```
/usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
/usr/include/features.h /usr/include/features-time64.h \
/usr/include/aarch64-linux-gnu/bits/wordsize.h \
/usr/include/aarch64-linux-gnu/bits/timesize.h \
/usr/include/aarch64-linux-gnu/sys/cdefs.h \
/usr/include/aarch64-linux-gnu/bits/long-double.h \
/usr/include/aarch64-linux-gnu/gnu/stubs.h \
/usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
/usr/include/c++/14/pstl/pstl_config.h \
/usr/include/c++/14/bits/stl_function.h /usr/include/c++/14/bits/move.h \
/usr/include/c++/14/type_traits /usr/include/c++/14/bits/version.h \
/usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/compare /usr/include/c++/14/concepts \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/new \
/usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h /usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
/usr/include/c++/14/bits/functexcept.h /usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/requires_hosted.h \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h \
/usr/include/c++/14/bits/memoryfwd.h \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/erase_if.h \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array
```

11.28 src/libs/IMU_math/CMakeFiles/imu_math_lib.dir/Swing↩ Detector.cpp.o.d File Reference

11.29 SwingDetector.cpp.o.d

Go to the documentation of this file.

```
libs/IMU_math/CMakeFiles/imu_math_lib.dir/SwingDetector.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/SwingDetector.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingDetector.hpp \
 /usr/include/c++/14/functional \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
```

```
/usr/include/aarch64-linux-gnu/bits/long-double.h \
/usr/include/aarch64-linux-gnu/gnu/stubs.h \
/usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
/usr/include/c++/14/pstl/pstl_config.h \
/usr/include/c++/14/bits/stl_function.h /usr/include/c++/14/bits/move.h \
/usr/include/c++/14/type_traits /usr/include/c++/14/bits/version.h \
/usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/compare /usr/include/c++/14/concepts \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/new \
/usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h /usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
/usr/include/c++/14/bits/functexcept.h /usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/requires_hosted.h \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h \
/usr/include/c++/14/bits/memoryfwd.h \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/erase_if.h \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array
```

11.30 src/libs/IMU_math/CMakeFiles/imu_math_lib.dir/SwingProcessor.cpp.o.d File Reference

11.31 SwingProcessor.cpp.o.d

Go to the documentation of this file.

```
libs/IMU_math/CMakeFiles/imu_math_lib.dir/SwingProcessor.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/SwingProcessor.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingProcessor.hpp \
 /home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/SwingCalibrator.hpp \
 /usr/include/c++/14/functional \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
```

```
/usr/include/c++/14/pstl/pstl_config.h \
/usr/include/c++/14/bits/stl_function.h /usr/include/c++/14/bits/move.h \
/usr/include/c++/14/type_traits /usr/include/c++/14/bits/version.h \
/usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/compare /usr/include/c++/14/concepts \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/new \
/usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h /usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
/usr/include/c++/14/bits/functexcept.h /usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/requires_hosted.h \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h \
/usr/include/c++/14/bits/memoryfwd.h \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/erase_if.h \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array \
/home/group5/RTES_Table_Tennis/src/libs/IMU_math/Include/positiondetector.hpp \
/usr/include/c++/14/cmath /usr/include/math.h \
/usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
/usr/include/aarch64-linux-gnu/bits/types.h \
/usr/include/aarch64-linux-gnu/bits/typesizes.h \
/usr/include/aarch64-linux-gnu/bits/time64.h \
/usr/include/aarch64-linux-gnu/bits/math-vector.h \
/usr/include/aarch64-linux-gnu/bits/libm-simd-decl-stubs.h \
/usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/include/aarch64-linux-gnu/bits/flt-eval-method.h \
/usr/include/aarch64-linux-gnu/bits/fp-logb.h \
/usr/include/aarch64-linux-gnu/bits/fp-fast.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-macros.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-helper-functions.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls.h \
/usr/include/aarch64-linux-gnu/bits/mathcalls-narrow.h \
/usr/include/aarch64-linux-gnu/bits/iscanonical.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/stdlib.h \
/usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
/usr/include/aarch64-linux-gnu/sys/types.h \
/usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
/usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
/usr/include/aarch64-linux-gnu/bits/types/time_t.h \
/usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
/usr/include/aarch64-linux-gnu/bits/stdint-intn.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/endian.h \
/usr/include/aarch64-linux-gnu/bits/endianness.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
```

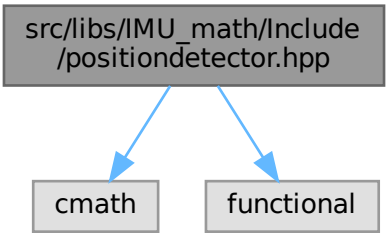
```
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/specfun.h /usr/include/c++/14/limits \
/usr/include/c++/14/tr1/gamma.tcc \
/usr/include/c++/14/tr1/special_function_util.h \
/usr/include/c++/14/tr1/bessel_function.tcc \
/usr/include/c++/14/tr1/beta_function.tcc \
/usr/include/c++/14/tr1/ell_integral.tcc \
/usr/include/c++/14/tr1/exp_integral.tcc \
/usr/include/c++/14/tr1/hypergeometric.tcc \
/usr/include/c++/14/tr1/legendre_function.tcc \
/usr/include/c++/14/tr1/modified_bessel_func.tcc \
/usr/include/c++/14/tr1/poly_hermite.tcc \
/usr/include/c++/14/tr1/poly_laguerre.tcc \
/usr/include/c++/14/tr1/riemann_zeta.tcc /usr/include/c++/14/chrono \
/usr/include/c++/14/bits/chrono.h /usr/include/c++/14/ratio \
/usr/include/c++/14/cstdint \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
/usr/include/aarch64-linux-gnu/bits/wchar.h \
/usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
/usr/include/aarch64-linux-gnu/bits/stdint-least.h \
/usr/include/c++/14/ctime /usr/include/time.h \
/usr/include/aarch64-linux-gnu/bits/time.h \
/usr/include/aarch64-linux-gnu/bits/timex.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
/usr/include/c++/14/bits/parse_numbers.h /usr/include/c++/14/sstream \
/usr/include/c++/14/istream /usr/include/c++/14/ios \
/usr/include/c++/14/iosfwd /usr/include/c++/14/bits/stringfwd.h \
/usr/include/c++/14/bits/postypes.h /usr/include/c++/14/cwchar \
/usr/include/wchar.h /usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/exception /usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h /usr/include/linux/types.h \
/usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
/usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
/usr/include/aarch64-linux-gnu/bits/cpu-set.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/basic_string.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/c++/14/cstdio /usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
```

```
/usr/include/c++/14/cerrno /usr/include/errno.h \
/usr/include/aarch64-linux-gnu/bits/errno.h /usr/include/linux/errno.h \
/usr/lib/linux/uapi/arm64/asm/errno.h /usr/include/asm-generic/errno.h \
/usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc /usr/include/c++/14/ostream \
/usr/include/c++/14/bits/ostream.tcc \
/usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/bits/unique_ptr.h \
/usr/include/c++/14/ext/concurrence.h /usr/include/c++/14/bits/align.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/charconv /usr/include/c++/14/optional \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h
```

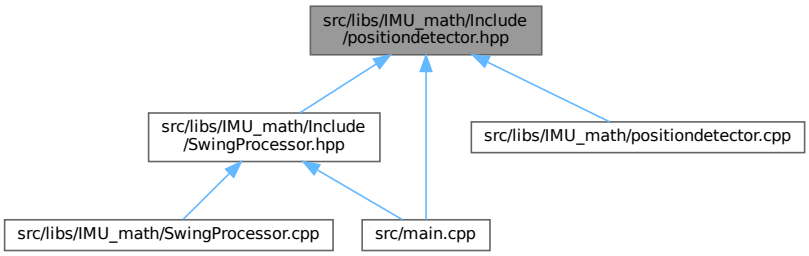
11.32 src/libs/IMU_math/Include/positiondetector.hpp File Reference

```
#include <cmath>
#include <functional>
```

Include dependency graph for positiondetector.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class PositionDetector

  Declares the position detector class, including its callback type, configuration settings, and main functions for processing samples and resetting state.
- struct PositionDetector::Config

11.33 positiondetector.hpp

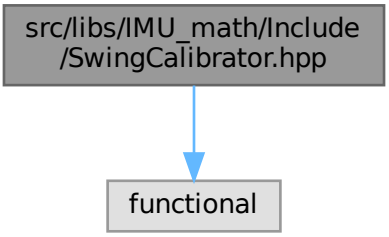
Go to the documentation of this file.

```
#pragma once

#include <cmath>
#include <functional>

class PositionDetector {
public:
    using StateCallback = std::function<void(bool upright)>;

    struct Config {
        float refX =  9.88f;
        float refY = -0.87f;
        float refZ =  0.10f;

        float minG = 7.0f;
        float maxG = 12.5f;
        float minDot = 0.90f;

        float refHeadingDeg = 25.0f;
        float headingToleranceDeg = 180.0f;

        int stabilitySamples = 10;
    };

    PositionDetector();
    explicit PositionDetector(const Config& c);
    virtual ~PositionDetector() = default;

    virtual void onSample(float ax, float ay, float az, float mx, float my);
    void setCallback(StateCallback cb) { callback = std::move(cb); }
    void resetState();

private:
    static float wrapAngleDeg(float a);

    Config cfg;
    float nRefX, nRefY, nRefZ;
    int stableCount;
    bool confirmed;
    StateCallback callback;
};
```

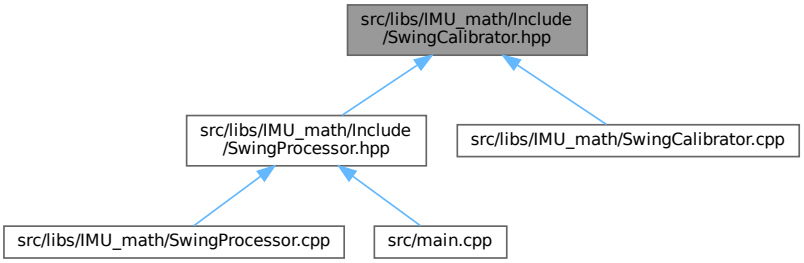
11.34 src/libs/IMU_math/Include/SwingCalibrator.hpp File Reference

`#include <functional>`

Include dependency graph for SwingCalibrator.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class SwingCalibrator

11.35 SwingCalibrator.hpp

Go to the documentation of this file.

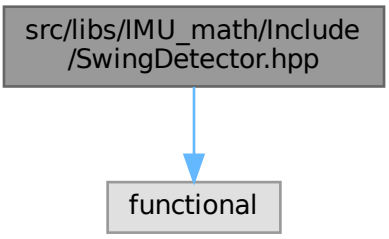
```
#pragma once

#include <functional>

class SwingCalibrator {
public:

    using DoneCallback = std::function<void(float bx, float by, float bz, bool initial)>;

    explicit SwingCalibrator(int initial_samples = 200, int recal_samples = 50);

    void setCallback(DoneCallback cb);

    bool feed(float ax, float ay, float az);

    void triggerRecal();

    bool isReady() const;

    float biasX() const { return bias_x_; }
    float biasY() const { return bias_y_; }
```

```
    float biasZ() const { return bias_z_; }

private:
    enum class State { INITIAL, READY, RECAL };

    int   initial_samples_;
    int   recal_samples_;
    State state_;
    int   count_;

    float sum_x_, sum_y_, sum_z_;

    float bias_x_, bias_y_, bias_z_;

    DoneCallback callback_;

    void accumulate(float ax, float ay, float az, int target, bool initial);
};
```

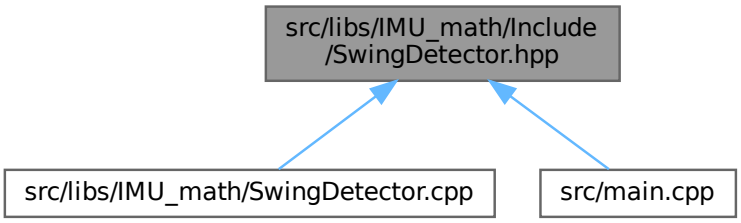
11.36 src/libs/IMU_math/Include/SwingDetector.hpp File Reference

`#include <functional>`

Include dependency graph for SwingDetector.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class SwingDetector

11.37 SwingDetector.hpp

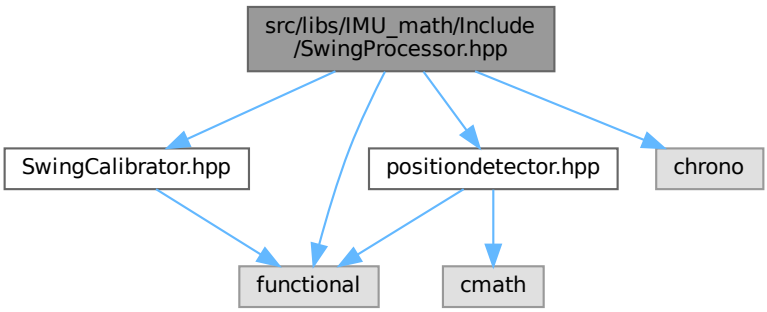
Go to the documentation of this file.

```
#pragma once

#include <functional>

class SwingDetector {
public:
    using EventCallback = std::function<void(const char* level)>;

    SwingDetector();

    void setCallback(EventCallback cb);

    void detect(float accel_mag);

    void reset();

private:
    enum class Level { NONE, LOW, MEDIUM, HIGH };

    static Level    classify(float accel_mag);

    static const char* label(Level level);

    Level         last_level_;

    EventCallback callback_;
};
```

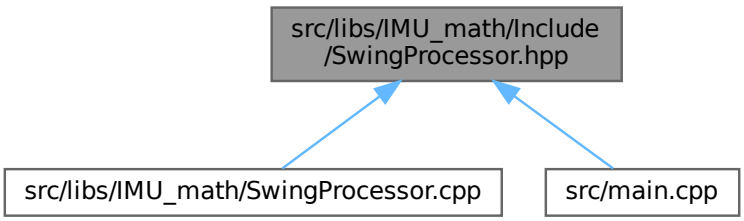
11.38 src/libs/IMU_math/Include/SwingProcessor.hpp File Reference

```
#include "SwingCalibrator.hpp"
#include "positiondetector.hpp"
#include <chrono>
#include <functional>
```

Include dependency graph for SwingProcessor.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class SwingProcessor

  raised to reject motor EMI noise floor

Variables

- static constexpr unsigned I2C_BUS = 1

  Defines the I2C bus and the main constants used for calibration, recalibration, and acceleration deadband control.
- static constexpr int CALIB_SAMPLES = 50
- static constexpr int RECAL_SAMPLES = 20

  ∼0.5s at 100Hz — startup bias
- static constexpr float ACCEL_DEADBAND = 1.0f

  ∼0.2s recal on position confirm

11.38.1 Variable Documentation

11.38.1.1 ACCEL_DEADBAND

`constexpr float ACCEL_DEADBAND = 1.0f [static], [constexpr]`
∼0.2s recal on position confirm
Definition at line 20 of file SwingProcessor.hpp.

11.38.1.2 CALIB_SAMPLES

`constexpr int CALIB_SAMPLES = 50 [static], [constexpr]`
Definition at line 18 of file SwingProcessor.hpp.

11.38.1.3 I2C_BUS

`constexpr unsigned I2C_BUS = 1 [static], [constexpr]`
Defines the I2C bus and the main constants used for calibration, recalibration, and acceleration deadband control.
This header declares the swing processor class. It combines calibration, position detection, force input, and sample processing to control when swing feedback becomes active. SOLID Principles: D - MagnitudeCallback and PositionStateCallback are std::function output abstractions.
Definition at line 17 of file SwingProcessor.hpp.

11.38.1.4 RECAL_SAMPLES

`constexpr int RECAL_SAMPLES = 20 [static], [constexpr]`
∼0.5s at 100Hz — startup bias
Definition at line 19 of file SwingProcessor.hpp.

11.39 SwingProcessor.hpp

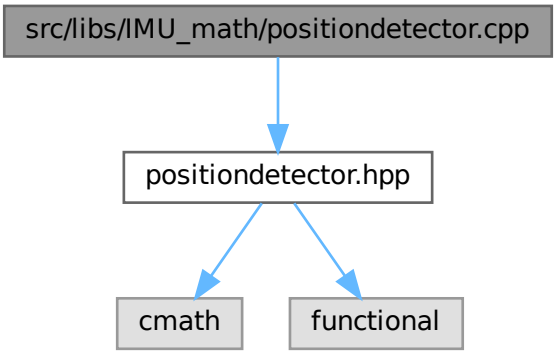
Go to the documentation of this file.

```
#pragma once

#include "SwingCalibrator.hpp"
#include "positiondetector.hpp"

#include <chrono>
#include <functional>

static constexpr unsigned I2C_BUS        = 1;
static constexpr int      CALIB_SAMPLES  = 50;
static constexpr int      RECAL_SAMPLES  = 20;
static constexpr float    ACCEL_DEADBAND = 1.0f;

class SwingProcessor : public PositionDetector {
public:
    using MagnitudeCallback     = std::function<void(float accel_mag)>;
    using PositionStateCallback = std::function<void(bool upright)>;

    explicit SwingProcessor(float accel_deadband = ACCEL_DEADBAND,
                            int   calib_samples  = CALIB_SAMPLES,
                            int   recal_samples  = RECAL_SAMPLES);

    void setMagnitudeCallback(MagnitudeCallback cb);
    void setPositionCallback(PositionStateCallback cb);

    void onForceReady(bool correctForce);
    bool isActive() const { return isActive_; }
    void onPositionReady(bool upright);
    void reset();

    void onSample(float ax, float ay, float az,
                  float mx, float my) override;

    void operator()(float ax, float ay, float az,
                    float gx, float gy, float gz,
                    float mx, float my);

private:
    float                accel_deadband_;
    SwingCalibrator      calibrator_;
    MagnitudeCallback    mag_callback_;
    PositionStateCallback pos_callback_;

    bool isArmed_      = false;
    bool isActive_     = false;
    bool hasLeftStart_ = false;


    static constexpr int COOLDOWN_MS = 500;
    std::chrono::steady_clock::time_point reset_time_{};


};
```

11.40 src/libs/IMU_math/positiondetector.cpp File Reference

`#include "positiondetector.hpp"`

Include dependency graph for positiondetector.cpp:



11.41 positiondetector.cpp

Go to the documentation of this file.

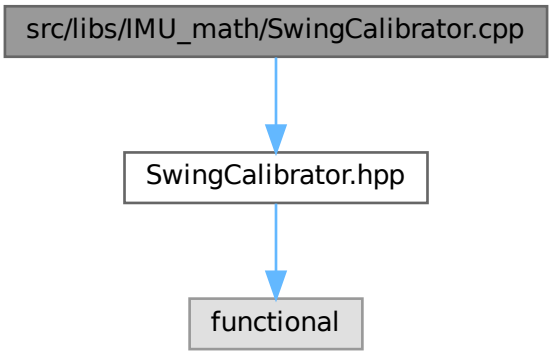
```
#include "positiondetector.hpp"

PositionDetector::PositionDetector()
    : PositionDetector(Config{}) {}

PositionDetector::PositionDetector(const Config& c)
    : cfg(c), stableCount(0), confirmed(false)
{
    float mag = std::sqrt(c.refX*c.refX + c.refY*c.refY + c.refZ*c.refZ);
    nRefX = c.refX / mag;
    nRefY = c.refY / mag;
    nRefZ = c.refZ / mag;
}

float PositionDetector::wrapAngleDeg(float a)
{
    while (a > 180.0f) a -= 360.0f;
    while (a < -180.0f) a += 360.0f;
    return a;
}

void PositionDetector::resetState()
{
    stableCount = 0;
    confirmed   = false;
}

void PositionDetector::onSample(float ax, float ay, float az, float mx, float my)
{
    float g = std::sqrt(ax * ax + ay * ay + az * az);

    bool accelOK = false;
    bool headingOK = false;

    float headingDeg = std::atan2(my, mx) * 180.0f / 3.14159265f;

    float headingError = wrapAngleDeg(headingDeg - cfg.refHeadingDeg);
    headingOK = (std::fabs(headingError) <= cfg.headingToleranceDeg);

    if (g >= cfg.minG && g <= cfg.maxG) {
        float nx = ax / g;
        float ny = ay / g;
        float nz = az / g;

        float dot = nx * nRefX + ny * nRefY + nz * nRefZ;
        accelOK = (dot >= cfg.minDot);
```

```
    }

    bool ok = accelOK && headingOK;

    if (ok) {
        if (stableCount < cfg.stabilitySamples) {
            ++stableCount;
        }
    } else {
        stableCount = 0;
    }

    bool nowConfirmed = (stableCount >= cfg.stabilitySamples);

    if (nowConfirmed != confirmed) {
        confirmed = nowConfirmed;
        if (callback) {
            callback(confirmed);
        }
    }
}
```

11.42 src/libs/IMU_math/SwingCalibrator.cpp File Reference

`#include "SwingCalibrator.hpp"`

Include dependency graph for SwingCalibrator.cpp:



11.43 SwingCalibrator.cpp

Go to the documentation of this file.

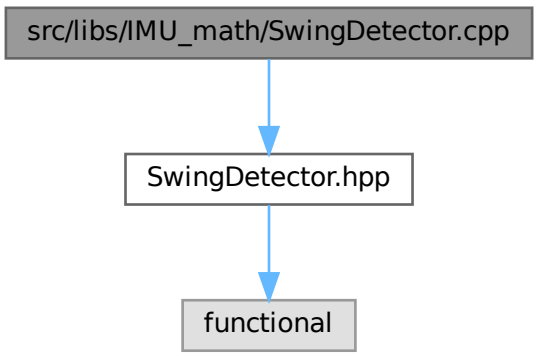
```
#include "SwingCalibrator.hpp"

SwingCalibrator::SwingCalibrator(int initial_samples, int recal_samples)
    : initial_samples_(initial_samples),
      recal_samples_(recal_samples),
      state_(State::INITIAL),
      count_(0),
      sum_x_(0.f), sum_y_(0.f), sum_z_(0.f),
      bias_x_(0.f), bias_y_(0.f), bias_z_(0.f)
{}

void SwingCalibrator::setCallback(DoneCallback cb)
{
    callback_ = std::move(cb);
}

bool SwingCalibrator::feed(float ax, float ay, float az)
{
    switch (state_) {
    case State::INITIAL:
        accumulate(ax, ay, az, initial_samples_, true);
```

```
        return true;

    case State::RECAL:
        accumulate(ax, ay, az, recal_samples_, false);
        return true;

    case State::READY:
        return false;
    }

    return false;
}

void SwingCalibrator::triggerRecal()
{
    if (state_ != State::READY) return;

    state_ = State::RECAL;
    count_ = 0;
    sum_x_ = sum_y_ = sum_z_ = 0.f;
}

bool SwingCalibrator::isReady() const
{
    return state_ == State::READY;
}

void SwingCalibrator::accumulate(float ax, float ay, float az, int target, bool initial)
{
    sum_x_ += ax;
    sum_y_ += ay;
    sum_z_ += az;
    ++count_;

    if (count_ == target) {
        bias_x_ = sum_x_ / target;
        bias_y_ = sum_y_ / target;
        bias_z_ = sum_z_ / target;

        state_  = State::READY;
        count_  = 0;
        sum_x_  = sum_y_ = sum_z_ = 0.f;

        if (callback_) callback_(bias_x_, bias_y_, bias_z_, initial);
    }
}
```

11.44 src/libs/IMU_math/SwingDetector.cpp File Reference

`#include "SwingDetector.hpp"`

Include dependency graph for SwingDetector.cpp:



11.45 SwingDetector.cpp

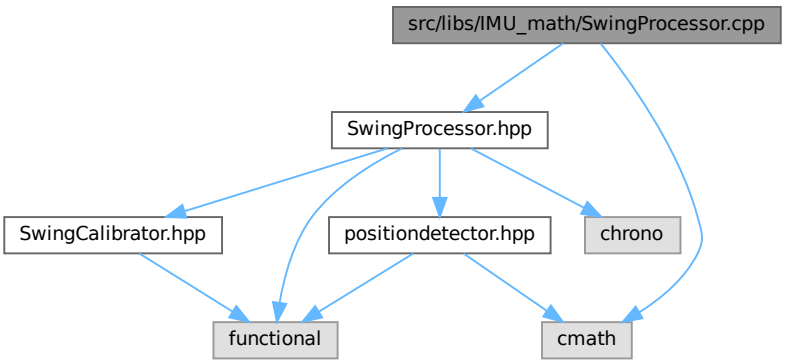
Go to the documentation of this file.

```
#include "SwingDetector.hpp"

SwingDetector::SwingDetector()
    : last_level_(Level::NONE)
{}

void SwingDetector::setCallback(EventCallback cb)
{
    callback_ = std::move(cb);
}

void SwingDetector::reset()
{
    last_level_ = Level::NONE;
}

void SwingDetector::detect(float accel_mag)
{
    Level current = classify(accel_mag);

    if (current != last_level_) {
        last_level_ = current;
        if (callback_) callback_(label(current));
    }
}

SwingDetector::Level SwingDetector::classify(float a)
{
    if (a >= 25.f)  return Level::HIGH;
    if (a >= 15.f)  return Level::MEDIUM;
    if (a >= 10.f)  return Level::LOW;
    return Level::NONE;
}

const char* SwingDetector::label(Level level)
{
    switch (level) {
    case Level::HIGH:   return "Highest Duty Cycle";
    case Level::MEDIUM: return "Medium Duty Cycle";
    case Level::LOW:    return "Low Duty Cycle";
    default:            return "No Duty Cycle";
    }
}
```

11.46 src/libs/IMU_math/SwingProcessor.cpp File Reference

```
#include "SwingProcessor.hpp"
#include <cmath>
```

Include dependency graph for SwingProcessor.cpp:



11.47 SwingProcessor.cpp

Go to the documentation of this file.

```
#include "SwingProcessor.hpp"

#include <cmath>

SwingProcessor::SwingProcessor(float accel_deadband, int calib_samples, int recal_samples)
    : accel_deadband_(accel_deadband),
      calibrator_(calib_samples, recal_samples)
{
    calibrator_.setCallback([](float, float, float, bool) {});

    PositionDetector::setCallback([this](bool upright) {
        if (pos_callback_) pos_callback_(upright);
        onPositionReady(upright);
    });
}
void SwingProcessor::setMagnitudeCallback(MagnitudeCallback cb)
{
    mag_callback_ = std::move(cb);
}

void SwingProcessor::setPositionCallback(PositionStateCallback cb)
{
    pos_callback_ = std::move(cb);
}


void SwingProcessor::onPositionReady(bool upright)
{
    auto elapsed = std::chrono::steady_clock::now() - reset_time_;
    if (elapsed < std::chrono::milliseconds(COOLDOWN_MS)) return;

    if (upright && !isArmed_) {
        isArmed_ = true;
        calibrator_.triggerRecal();
    } else if (!upright && isArmed_) {
        if (!isActive_) {
            isArmed_ = false;
        } else {
            hasLeftStart_ = true;
        }
    }
}

void SwingProcessor::onForceReady(bool correctForce)
{
    if (!correctForce) {
        if (isActive_ && mag_callback_) mag_callback_(0.f);
        return;
    }

    if (!isArmed_) {
        return;
    }

    if (isActive_) return;

    isActive_     = true;
    hasLeftStart_ = false;
}

void SwingProcessor::reset()
{
    isArmed_      = false;
    isActive_     = false;
    hasLeftStart_ = false;
    reset_time_   = std::chrono::steady_clock::now();
}

void SwingProcessor::onSample(float ax, float ay, float az,
                              float mx, float my)
{
    PositionDetector::onSample(ax, ay, az, mx, my);

    if (!isActive_) return;
    if (!hasLeftStart_) return;

    if (calibrator_.feed(ax, ay, az)) return;

    float lin_ax = ax - calibrator_.biasX();
    float lin_ay = ay - calibrator_.biasY();
    float lin_az = az - calibrator_.biasZ();

    float lin_mag = std::sqrt(lin_ax*lin_ax + lin_ay*lin_ay + lin_az*lin_az);
```

```

    if (lin_mag < accel_deadband_) lin_mag = 0.f;

    if (mag_callback_) mag_callback_(lin_mag);
}

void SwingProcessor::operator()(float ax, float ay, float az,
                                float /*gx*/, float /*gy*/, float /*gz*/,
                                float mx, float my)
{
    onSample(ax, ay, az, mx, my);
}
```

11.48 src/libs/LEDs/CMakeFiles/leds_lib.dir/led_controller.cpp.o.d File Reference

11.49 led_controller.cpp.o.d

Go to the documentation of this file.

```
libs/LEDs/CMakeFiles/leds_lib.dir/led_controller.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/LEDs/led_controller.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/led_controller.h \
 /home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/ledcallback.hpp \
 /usr/include/gpiod.hpp /usr/include/gpiodcxx/chip.hpp \
 /usr/include/c++/14/chrono /usr/include/c++/14/bits/requires_hosted.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h /usr/include/c++/14/bits/chrono.h \
 /usr/include/c++/14/ratio /usr/include/c++/14/type_traits \
 /usr/include/c++/14/bits/version.h /usr/include/c++/14/cstdint \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
 /usr/include/aarch64-linux-gnu/bits/time64.h \
 /usr/include/aarch64-linux-gnu/bits/wchar.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-least.h \
 /usr/include/c++/14/limits /usr/include/c++/14/ctime /usr/include/time.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/include/aarch64-linux-gnu/bits/time.h \
 /usr/include/aarch64-linux-gnu/bits/timex.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
 /usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/time_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
 /usr/include/aarch64-linux-gnu/bits/endian.h \
 /usr/include/aarch64-linux-gnu/bits/endianness.h \
 /usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
 /usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
 /usr/include/c++/14/bits/parse_numbers.h \
 /usr/include/c++/14/ext/numeric_traits.h \
 /usr/include/c++/14/bits/cpp_type_traits.h \
 /usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/concepts \
 /usr/include/c++/14/compare /usr/include/c++/14/bit \
 /usr/include/c++/14/sstream /usr/include/c++/14/istream \
 /usr/include/c++/14/ios /usr/include/c++/14/iosfwd \
 /usr/include/c++/14/bits/stringfwd.h \
 /usr/include/c++/14/bits/memoryfwd.h /usr/include/c++/14/bits/postypes.h \
 /usr/include/c++/14/cwchar /usr/include/wchar.h \
 /usr/include/aarch64-linux-gnu/bits/floatn.h \
 /usr/include/aarch64-linux-gnu/bits/floatn-common.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
 /usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
```

```
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/exception /usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/typeinfo /usr/include/c++/14/bits/hash_bytes.h \
/usr/include/c++/14/new /usr/include/c++/14/bits/move.h \
/usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h /usr/include/linux/types.h \
/usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
/usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
/usr/include/aarch64-linux-gnu/bits/cpu-set.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h \
/usr/include/c++/14/bits/refwrap.h /usr/include/c++/14/bits/invoke.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/basic_string.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/stdlib.h /usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/sys/types.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/c++/14/cstdio \
/usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
```

```
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/cerrno /usr/include/errno.h \
/usr/include/aarch64-linux-gnu/bits/errno.h /usr/include/linux/errno.h \
/usr/lib/linux/uapi/arm64/asm/errno.h /usr/include/asm-generic/errno.h \
/usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/uses_allocator_args.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc /usr/include/c++/14/ostream \
/usr/include/c++/14/bits/ostream.tcc \
/usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/c++/14/vector \
/usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/bits/unique_ptr.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/concurrence.h /usr/include/c++/14/bits/align.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/array /usr/include/c++/14/charconv \
/usr/include/c++/14/optional \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h /usr/include/c++/14/iostream \
/usr/include/c++/14/filesystem /usr/include/c++/14/bits/fs_fwd.h \
/usr/include/c++/14/bits/fs_path.h /usr/include/c++/14/codecvt \
/usr/include/c++/14/bits/fs_dir.h /usr/include/c++/14/bits/fs_ops.h \
/usr/include/c++/14/memory \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/bits/atomic_base.h \
/usr/include/c++/14/bits/atomic_lockfree_defines.h \
/usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
```

```
/usr/include/c++/14/bits/std_mutex.h \
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/gpiodcxx/line.hpp \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/gpiodcxx/chip-info.hpp /usr/include/gpiodcxx/edge-event.hpp \
/usr/include/gpiodcxx/timestamp.hpp \
/usr/include/gpiodcxx/edge-event-buffer.hpp \
/usr/include/gpiodcxx/exception.hpp /usr/include/gpiodcxx/info-event.hpp \
/usr/include/gpiodcxx/line.hpp /usr/include/gpiodcxx/line-config.hpp \
/usr/include/c++/14/map /usr/include/c++/14/bits/stl_tree.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/c++/14/bits/erase_if.h /usr/include/gpiodcxx/line-info.hpp \
/usr/include/gpiodcxx/line-request.hpp /usr/include/gpiodcxx/misc.hpp \
/usr/include/gpiodcxx/line-settings.hpp \
/usr/include/gpiodcxx/request-builder.hpp \
/usr/include/gpiodcxx/request-config.hpp
```

11.50 src/libs/LEDs/CMakeFiles/leds_lib.dir/ledcallback.cpp.o.d File Reference

11.51 ledcallback.cpp.o.d

Go to the documentation of this file.

```
libs/LEDs/CMakeFiles/leds_lib.dir/ledcallback.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/LEDs/ledcallback.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/ledcallback.hpp \
 /usr/include/gpiod.hpp /usr/include/gpiodcxx/chip.hpp \
 /usr/include/c++/14/chrono /usr/include/c++/14/bits/requires_hosted.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h /usr/include/c++/14/bits/chrono.h \
 /usr/include/c++/14/ratio /usr/include/c++/14/type_traits \
 /usr/include/c++/14/bits/version.h /usr/include/c++/14/cstdint \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
 /usr/include/aarch64-linux-gnu/bits/time64.h \
 /usr/include/aarch64-linux-gnu/bits/wchar.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-least.h \
 /usr/include/c++/14/limits /usr/include/c++/14/ctime /usr/include/time.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/include/aarch64-linux-gnu/bits/time.h \
 /usr/include/aarch64-linux-gnu/bits/timex.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
 /usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/time_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
 /usr/include/aarch64-linux-gnu/bits/endian.h \
 /usr/include/aarch64-linux-gnu/bits/endianness.h \
 /usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
 /usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
 /usr/include/c++/14/bits/parse_numbers.h \
 /usr/include/c++/14/ext/numeric_traits.h \
 /usr/include/c++/14/bits/cpp_type_traits.h \
 /usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/concepts \
 /usr/include/c++/14/compare /usr/include/c++/14/bit \
 /usr/include/c++/14/sstream /usr/include/c++/14/istream \
 /usr/include/c++/14/ios /usr/include/c++/14/iosfwd \
 /usr/include/c++/14/bits/stringfwd.h \
 /usr/include/c++/14/bits/memoryfwd.h /usr/include/c++/14/bits/postypes.h \
 /usr/include/c++/14/cwchar /usr/include/wchar.h \
```

```
/usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/exception /usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/typeinfo /usr/include/c++/14/bits/hash_bytes.h \
/usr/include/c++/14/new /usr/include/c++/14/bits/move.h \
/usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h /usr/include/linux/types.h \
/usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
/usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
/usr/include/aarch64-linux-gnu/bits/cpu-set.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h \
/usr/include/c++/14/bits/refwrap.h /usr/include/c++/14/bits/invoke.h \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/basic_string.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/stdlib.h /usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/sys/types.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
```

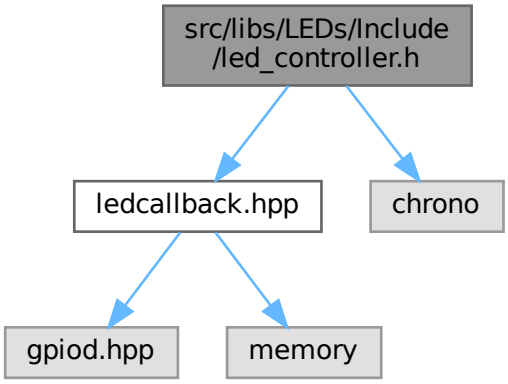
```
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/c++/14/cstdio \
/usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/cerrno /usr/include/errno.h \
/usr/include/aarch64-linux-gnu/bits/errno.h /usr/include/linux/errno.h \
/usr/lib/linux/uapi/arm64/asm/errno.h /usr/include/asm-generic/errno.h \
/usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/uses_allocator_args.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc /usr/include/c++/14/ostream \
/usr/include/c++/14/bits/ostream.tcc \
/usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/c++/14/vector \
/usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/bits/unique_ptr.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/concurrence.h /usr/include/c++/14/bits/align.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/array /usr/include/c++/14/charconv \
/usr/include/c++/14/optional \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h /usr/include/c++/14/iostream \
/usr/include/c++/14/filesystem /usr/include/c++/14/bits/fs_fwd.h \
/usr/include/c++/14/bits/fs_path.h /usr/include/c++/14/codecvt \
/usr/include/c++/14/bits/fs_dir.h /usr/include/c++/14/bits/fs_ops.h \
/usr/include/c++/14/memory \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/bits/atomic_base.h \
/usr/include/c++/14/bits/atomic_lockfree_defines.h \
/usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
```

```
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/std_mutex.h \
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/gpiodcxx/line.hpp \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/gpiodcxx/chip-info.hpp /usr/include/gpiodcxx/edge-event.hpp \
/usr/include/gpiodcxx/timestamp.hpp \
/usr/include/gpiodcxx/edge-event-buffer.hpp \
/usr/include/gpiodcxx/exception.hpp /usr/include/gpiodcxx/info-event.hpp \
/usr/include/gpiodcxx/line.hpp /usr/include/gpiodcxx/line-config.hpp \
/usr/include/c++/14/map /usr/include/c++/14/bits/stl_tree.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/c++/14/bits/erase_if.h /usr/include/gpiodcxx/line-info.hpp \
/usr/include/gpiodcxx/line-request.hpp /usr/include/gpiodcxx/misc.hpp \
/usr/include/gpiodcxx/line-settings.hpp \
/usr/include/gpiodcxx/request-builder.hpp \
/usr/include/gpiodcxx/request-config.hpp
```

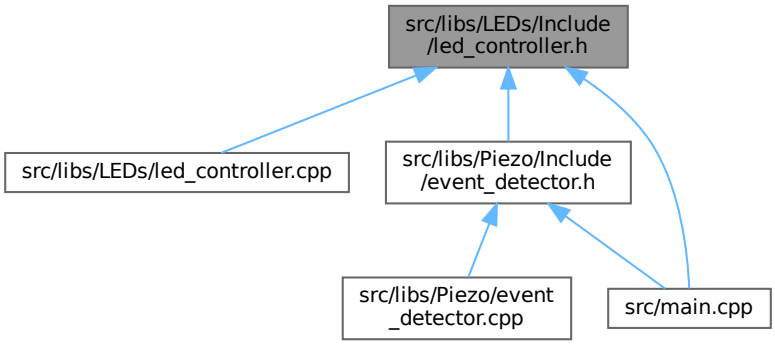
11.52 src/libs/LEDs/Include/led_controller.h File Reference

```
#include "ledcallback.hpp"
#include <chrono>
```

Include dependency graph for led_controller.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct LEDControllerSettings

  Declares the LED controller settings, including the chip number and green LED GPIO pin.
- class LEDController

  Declares the LED controller class and its main functions for setting, switching, flashing, and updating the LED output.

11.53 led_controller.h

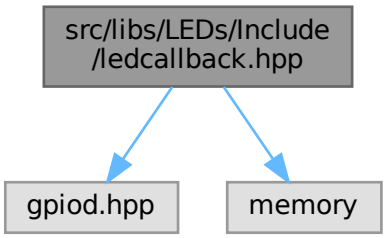
Go to the documentation of this file.

```
#ifndef LED_CONTROLLER_H
#define LED_CONTROLLER_H

#include "ledcallback.hpp"
#include <chrono>

struct LEDControllerSettings {
    int          chipNumber = 0;
    unsigned int greenGpio  = 25;
};

class LEDController : public SimpleLEDController
{
public:
    LEDController(LEDControllerSettings settings = LEDControllerSettings());
    ~LEDController() override;

    void set(bool on) override;

    void greenOn();
    void greenOff();
    void allOff();

    void flashGreen(int flashMs);
    void service();

private:
    void updateOutputs();

    bool greenActive_ = false;
    std::chrono::steady_clock::time_point greenOffTime_{};
};

#endif
```

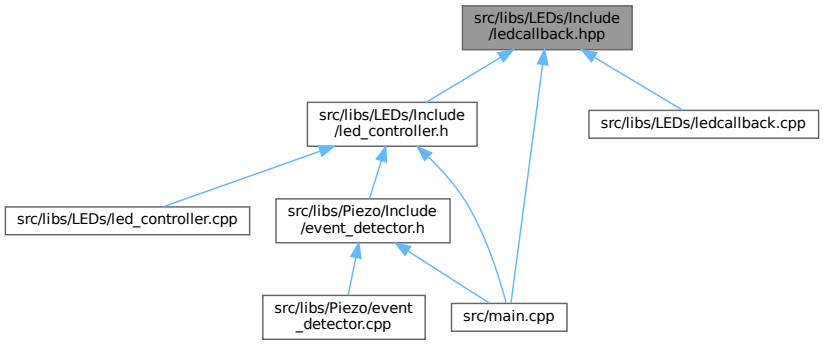
11.54 src/libs/LEDs/Include/ledcallback.hpp File Reference

```
#include <gpiod.hpp>
#include <memory>
```

Include dependency graph for ledcallback.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class SimpleLEDController

11.55 ledcallback.hpp

Go to the documentation of this file.

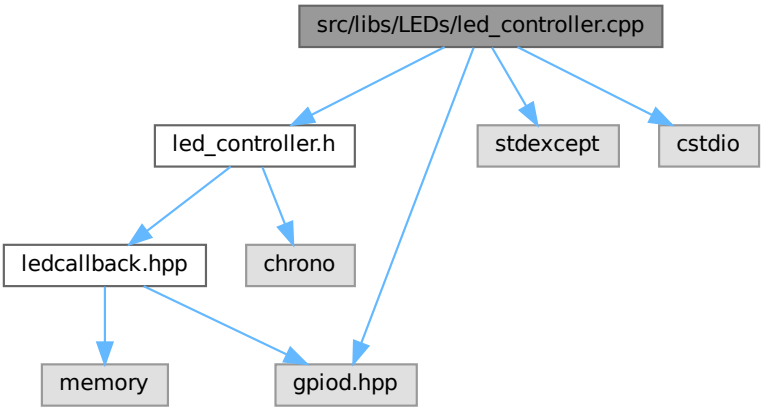
```
00014 #ifndef LEDCALLBACK_HPP
00015 #define LEDCALLBACK_HPP
00016
00017 #include <gpiod.hpp>
00018 #include <memory>
00019
00023 class SimpleLEDController
00024 {
00025 public:
00029     SimpleLEDController(int pinNo, int chipNo = 0);
00030
00033     virtual ~SimpleLEDController();
00034
00037     virtual void set(bool on);
00038
00042     virtual void flashGreen(int /*flashMs*/) {}
```

```
    virtual void service()                    {}

protected:
    SimpleLEDController() = default;

    int pin        = -1;
    int chipNumber = -1;

    std::shared_ptr<gpiod::chip>         chip;
    std::shared_ptr<gpiod::line_request> request;
};

#endif
```

11.56 src/libs/LEDs/led_controller.cpp File Reference

```
#include "led_controller.h"
#include <gpiod.hpp>
#include <stdexcept>
#include <cstdio>
```

Include dependency graph for led_controller.cpp:



11.57 led_controller.cpp

Go to the documentation of this file.

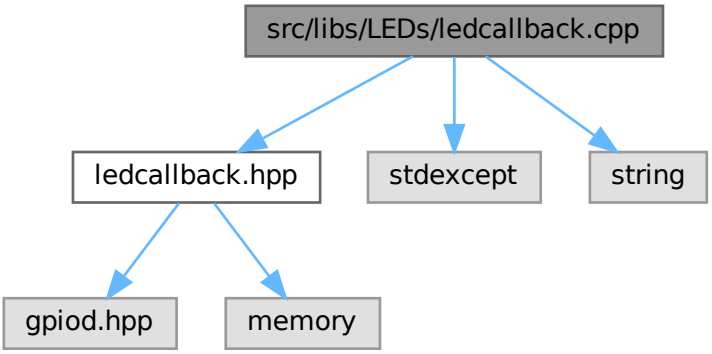
```

#include "led_controller.h"

#include <gpiod.hpp>
#include <stdexcept>
#include <cstdio>

namespace {
gpiod::chip& getChip(int chipNumber)
{
    static std::unique_ptr<gpiod::chip> chip0;
    if (!chip0) {
        chip0 = std::make_unique<gpiod::chip>("gpiochip" + std::to_string(chipNumber));
    }
    return *chip0;
}
}

LEDController::LEDController(LEDControllerSettings settings)
    : SimpleLEDController(settings.greenGpio, settings.chipNumber)
{
    greenActive_ = false;
    greenOffTime_ = std::chrono::steady_clock::time_point{};
```

```
    updateOutputs();
}

LEDController::~LEDController()
{
    allOff();
}

void LEDController::set(bool on)
{
    std::printf("[led] set(%s)\n", on ? "true" : "false");

    greenActive_ = on;
    greenOffTime_ = std::chrono::steady_clock::time_point{};
    updateOutputs();
}

void LEDController::greenOn()
{
    greenActive_ = true;
    greenOffTime_ = std::chrono::steady_clock::time_point{};
    updateOutputs();
}

void LEDController::greenOff()
{
    greenActive_ = false;
    greenOffTime_ = std::chrono::steady_clock::time_point{};
    updateOutputs();
}

void LEDController::allOff()
{
    greenActive_ = false;
    greenOffTime_ = std::chrono::steady_clock::time_point{};
    updateOutputs();
}
void LEDController::flashGreen(int flashMs)
{
    greenActive_ = true;
    greenOffTime_ = std::chrono::steady_clock::now() + std::chrono::milliseconds(flashMs);
    updateOutputs();
}

void LEDController::service()
{
    if (greenActive_ &&
        greenOffTime_ != std::chrono::steady_clock::time_point{} &&
        std::chrono::steady_clock::now() >= greenOffTime_) {
        greenActive_ = false;
        greenOffTime_ = std::chrono::steady_clock::time_point{};
        updateOutputs();
    }
}

void LEDController::updateOutputs()
{
    SimpleLEDController::set(greenActive_);
}
```

11.58 src/libs/LEDs/ledcallback.cpp File Reference

```
#include "ledcallback.hpp"
#include <stdexcept>
#include <string>
```

Include dependency graph for ledcallback.cpp:



11.59 ledcallback.cpp

Go to the documentation of this file.

```
#include "ledcallback.hpp"

#include <stdexcept>
#include <string>

SimpleLEDController::SimpleLEDController(int pinNo, int chipNo)
    : pin(pinNo), chipNumber(chipNo)
{
    const std::string chipPath =
        "/dev/gpiochip" + std::to_string(chipNumber);

    const std::string consumerName =
        "ledcontroller_" + std::to_string(chipNumber) + "_" + std::to_string(pin);

    chip = std::make_shared<gpiod::chip>(chipPath);

    gpiod::line_config line_cfg;
    line_cfg.add_line_settings(
        pin,
        gpiod::line_settings()
            .set_direction(gpiod::line::direction::OUTPUT)
            .set_output_value(gpiod::line::value::INACTIVE));

    auto builder = chip->prepare_request();
    builder.set_consumer(consumerName);
    builder.set_line_config(line_cfg);

    request = std::make_shared<gpiod::line_request>(builder.do_request());
}

SimpleLEDController::~SimpleLEDController()
{
    try {
        if (request)
            request->set_value(pin, gpiod::line::value::INACTIVE);
    } catch (...) {}

    if (request) {
        request->release();
    }

    if (chip) {
        chip->close();
    }
}

void SimpleLEDController::set(bool on)
{
    if (!request) {
```

```
        throw std::runtime_error("LED GPIO line is not initialized");
    }

    request->set_value(
        pin,
        on ? gpiod::line::value::ACTIVE
           : gpiod::line::value::INACTIVE);
}
```

11.60 src/libs/Motor/CMakeFiles/motor_lib.dir/swingfeedback.cpp.o.d File Reference

11.61 swingfeedback.cpp.o.d

Go to the documentation of this file.

```
libs/Motor/CMakeFiles/motor_lib.dir/swingfeedback.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/Motor/swingfeedback.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/Motor/Include/swingfeedback.h \
 /home/group5/RTES_Table_Tennis/src/libs/Motor/Include/rpi_pwm.h \
 /usr/include/c++/14/atomic /usr/include/c++/14/bits/version.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h \
 /usr/include/c++/14/bits/atomic_base.h /usr/include/c++/14/new \
 /usr/include/c++/14/bits/exception.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
 /usr/include/aarch64-linux-gnu/bits/time64.h \
 /usr/include/aarch64-linux-gnu/bits/wchar.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
 /usr/include/aarch64-linux-gnu/bits/stdint-least.h \
 /usr/include/c++/14/bits/atomic_lockfree_defines.h \
 /usr/include/c++/14/bits/move.h /usr/include/c++/14/type_traits \
 /usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/cstdint \
 /usr/include/c++/14/bits/functional_hash.h \
 /usr/include/c++/14/bits/hash_bytes.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
 /usr/include/pthread.h /usr/include/sched.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/include/aarch64-linux-gnu/bits/types/time_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
 /usr/include/aarch64-linux-gnu/bits/endian.h \
 /usr/include/aarch64-linux-gnu/bits/endianness.h \
 /usr/include/aarch64-linux-gnu/bits/sched.h \
 /usr/include/linux/sched/types.h /usr/include/linux/types.h \
 /usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
 /usr/include/asm-generic/int-ll64.h \
 /usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
 /usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
 /usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
 /usr/include/asm-generic/posix_types.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
 /usr/include/aarch64-linux-gnu/bits/cpu-set.h /usr/include/time.h \
 /usr/include/aarch64-linux-gnu/bits/time.h \
 /usr/include/aarch64-linux-gnu/bits/timex.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
 /usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
 /usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
 /usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
 /usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
 /usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
 /usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
 /usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
 /usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
```

```
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
/usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/cerrno \
/usr/include/errno.h /usr/include/aarch64-linux-gnu/bits/errno.h \
/usr/include/linux/errno.h /usr/lib/linux/uapi/arm64/asm/errno.h \
/usr/include/asm-generic/errno.h /usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/std_mutex.h /usr/include/c++/14/chrono \
/usr/include/c++/14/bits/requires_hosted.h \
/usr/include/c++/14/bits/chrono.h /usr/include/c++/14/ratio \
/usr/include/c++/14/limits /usr/include/c++/14/ctime \
/usr/include/c++/14/bits/parse_numbers.h /usr/include/c++/14/concepts \
/usr/include/c++/14/compare /usr/include/c++/14/bit \
/usr/include/c++/14/sstream /usr/include/c++/14/istream \
/usr/include/c++/14/ios /usr/include/c++/14/iosfwd \
/usr/include/c++/14/bits/stringfwd.h \
/usr/include/c++/14/bits/memoryfwd.h /usr/include/c++/14/bits/postypes.h \
/usr/include/c++/14/cwchar /usr/include/wchar.h \
/usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/exception /usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/typeinfo /usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h \
/usr/include/c++/14/bits/refwrap.h /usr/include/c++/14/bits/invoke.h \
```

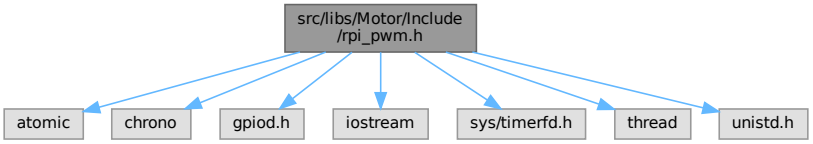
```
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/basic_string.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/stdlib.h /usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/sys/types.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/c++/14/cstdio \
/usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/uses_allocator_args.h /usr/include/c++/14/tuple \
/usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc /usr/include/c++/14/ostream \
/usr/include/c++/14/bits/ostream.tcc \
/usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/c++/14/vector \
/usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/bits/unique_ptr.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/concurrence.h /usr/include/c++/14/bits/align.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/array /usr/include/c++/14/charconv \
/usr/include/c++/14/optional \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h /usr/include/gpiod.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdbool.h \
/usr/include/c++/14/iostream \
/usr/include/aarch64-linux-gnu/sys/timerfd.h \
/usr/include/aarch64-linux-gnu/bits/timerfd.h /usr/include/c++/14/thread \
/usr/include/c++/14/stop_token /usr/include/c++/14/bits/std_thread.h \
/usr/include/c++/14/semaphore /usr/include/c++/14/bits/semaphore_base.h \
/usr/include/c++/14/bits/atomic_timed_wait.h \
/usr/include/c++/14/bits/this_thread_sleep.h \
/usr/include/aarch64-linux-gnu/sys/time.h /usr/include/semaphore.h \
/usr/include/aarch64-linux-gnu/bits/semaphore.h \
```

```
/usr/include/c++/14/functional /usr/include/c++/14/bits/std_function.h \
/usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/erase_if.h /usr/include/c++/14/utility \
/usr/include/c++/14/bits/stl_relops.h
```

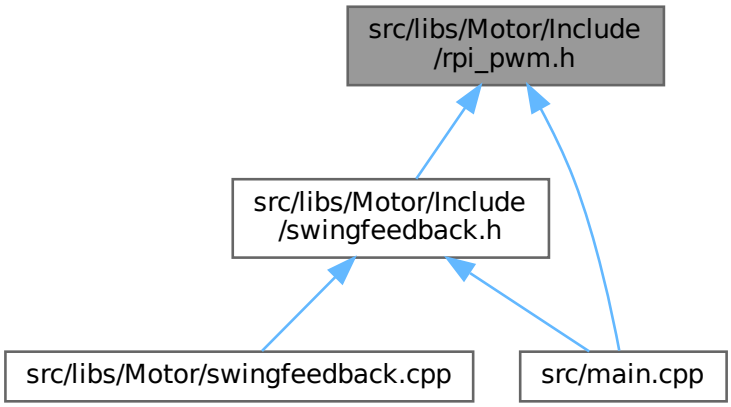
11.62 src/libs/Motor/Include/rpi_pwm.h File Reference

```
#include <atomic>
#include <chrono>
#include <gpiod.h>
#include <iostream>
#include <sys/timerfd.h>
#include <thread>
#include <unistd.h>
```

Include dependency graph for rpi_pwm.h:



This graph shows which files directly or indirectly include this file:



Classes

- class RPI_PWM

Variables

- static constexpr unsigned int PWM_GPIO_LINE = 18
- static constexpr int PWM_FREQ_HZ = 30

11.62.1 Variable Documentation

11.62.1.1 PWM_FREQ_HZ

```
constexpr int PWM_FREQ_HZ = 30  [static], [constexpr]
```

Definition at line 29 of file rpi_pwm.h.

11.62.1.2 PWM_GPIO_LINE

```
constexpr unsigned int PWM_GPIO_LINE = 18  [static], [constexpr]
```

This class uses a dedicated worker thread and libgpiod to provide a software PWM output on a Raspberry Pi GPIO pin. Its main responsibility is to provide a safe interface for starting and stopping PWM generation, allow client code to update the duty cycle without needing to manage any GPIO or timing details, and internally handle the GPIO line request and timing loop.
SOLID principles: S - this class his only responsible for generating PWM on one GPIO line. Higher level classification decides when feedback should happen and why the duty cycle changes. O - Client code can reuse this class for different duty cycle values without the internal logic being modified itself. This behaviour is extended by calling its public interface rather than rewriting the timing logic. Fixed hardware/software config forPWM output Keeping these constant avoids mageic numbers and makes the class behaviour easier to maintain and reason with.
Definition at line 28 of file rpi_pwm.h.

11.63 rpi_pwm.h

Go to the documentation of this file.

```
00001
00016 #pragma once
00017
00018 #include <atomic>
00019 #include <chrono>
00020 #include <gpiod.h>
00021 #include <iostream>
00022 #include <sys/timerfd.h>
00023 #include <thread>
00024 #include <unistd.h>
00025
00028 static constexpr unsigned int PWM_GPIO_LINE = 18;
00029 static constexpr int PWM_FREQ_HZ = 30;
00030
00031 // Declares the PWM class and its main functions for starting, stopping, and updating the duty cycle.
00032 class RPI_PWM {
00033 public:
00034
00038     RPI_PWM() : duty_(0.0f), running_(false), chip_(nullptr), request_(nullptr) {}
00039
00044     bool start() {
00046         chip_ = gpiod_chip_open("/dev/gpiochip0");
00047         if (!chip_) return false;
00048
00050         gpiod_line_settings* settings = gpiod_line_settings_new();
00051         gpiod_line_settings_set_direction(settings, GPIOD_LINE_DIRECTION_OUTPUT);
00052         gpiod_line_settings_set_output_value(settings, GPIOD_LINE_VALUE_INACTIVE);
00053
00055         gpiod_line_config* line_cfg = gpiod_line_config_new();
00056         gpiod_line_config_add_line_settings(line_cfg, &PWM_GPIO_LINE, 1, settings);
00057
00059         gpiod_request_config* req_cfg = gpiod_request_config_new();
00060         gpiod_request_config_set_consumer(req_cfg, "soft_pwm");
00061
00063         request_ = gpiod_chip_request_lines(chip_, req_cfg, line_cfg);
00064
00066         gpiod_line_settings_free(settings);
00067         gpiod_line_config_free(line_cfg);
00068         gpiod_request_config_free(req_cfg);
00069
00071         if (!request_) {
00072             gpiod_chip_close(chip_);
00073             return false;
00074         }
00075
00077         running_ = true;
00078         thread_ = std::thread(&RPI_PWM::loop, this);
00079
00081         std::cout « "Software PWM started on GPIO " « PWM_GPIO_LINE « "\n";
00082         return true;
00083     }
00084
```

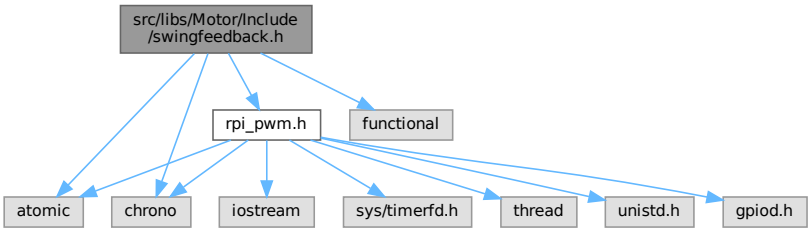
```
    void stop() {
        running_ = false;

        if (thread_.joinable()) thread_.join();

        if (request_) {
            gpiod_line_request_set_value(request_, PWM_GPIO_LINE, GPIOD_LINE_VALUE_INACTIVE);
            gpiod_line_request_release(request_);
            request_ = nullptr;
        }

        if (chip_) {
            gpiod_chip_close(chip_);
            chip_ = nullptr;
        }
    }


    void setDutyCycle(float percent) {
        if (percent < 0.0f)   percent = 0.0f;
        if (percent > 100.0f) percent = 100.0f;

        duty_ = percent;
    }

    float getDutyCycle() const { return duty_.load(); }

    ~RPI_PWM() { stop(); }

private:
    static itimerspec usToTimerspec(long us) {
        itimerspec ts{};

        ts.it_value.tv_sec     = us / 1000000;

        ts.it_value.tv_nsec    = (us % 1000000) * 1000;

        ts.it_interval.tv_sec  = 0;
        ts.it_interval.tv_nsec = 0;
        return ts;
    }


    void waitUs(int tfd, int us) {
        itimerspec ts = usToTimerspec(us);

        timerfd_settime(tfd, 0, &ts, nullptr);

        uint64_t exp;
        ::read(tfd, &exp, sizeof(exp));
    }

    void loop() {
        const int period_us = 1000000 / PWM_FREQ_HZ;

        int tfd = timerfd_create(CLOCK_MONOTONIC, 0);
        if (tfd < 0) return;

        while (running_) {
            // Read the current duty cycle.
            float d    = duty_.load();

            int on_us  = (int)(period_us * d / 100.0f);
            int off_us = period_us - on_us;

            if (d >= 100.0f) {
                gpiod_line_request_set_value(request_, PWM_GPIO_LINE, GPIOD_LINE_VALUE_ACTIVE);
                waitUs(tfd, period_us);

            } else if (d <= 0.0f) {
                gpiod_line_request_set_value(request_, PWM_GPIO_LINE, GPIOD_LINE_VALUE_INACTIVE);
                waitUs(tfd, period_us);

            } else {
                gpiod_line_request_set_value(request_, PWM_GPIO_LINE, GPIOD_LINE_VALUE_ACTIVE);
                waitUs(tfd, on_us);
                gpiod_line_request_set_value(request_, PWM_GPIO_LINE, GPIOD_LINE_VALUE_INACTIVE);
                waitUs(tfd, off_us);
            }
        }

        ::close(tfd);
    }

    std::atomic<float>  duty_;
    std::atomic<bool>   running_;
```

```
    std::thread         thread_;
    gpiod_chip*         chip_;
    gpiod_line_request* request_;
};
```

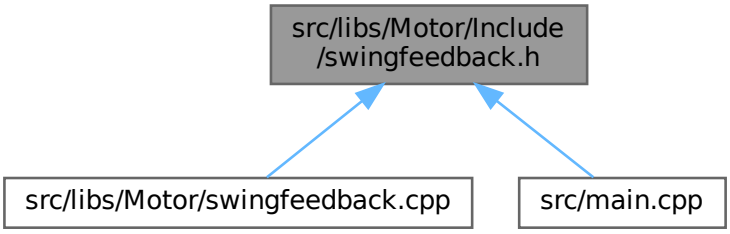
11.64 src/libs/Motor/Include/swingfeedback.h File Reference

```
#include "rpi_pwm.h"
#include <atomic>
#include <chrono>
#include <functional>
```

Include dependency graph for swingfeedback.h:



This graph shows which files directly or indirectly include this file:



Classes

- class SwingFeedback

  Drives PWM feedback from detected swing levels.

11.65 swingfeedback.h

Go to the documentation of this file.

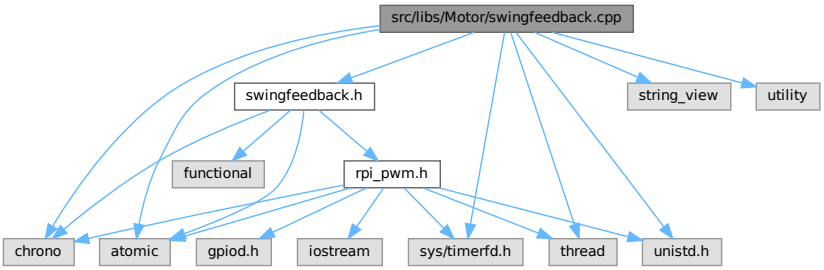
```
/*
    This header declares the swing feedback class.
    It manages PWM feedback output, handles level changes,
    checks for time-outs, and supports a reset callback.
*/

#pragma once

#include "rpi_pwm.h"

```

```
#include <atomic>
#include <chrono>
#include <functional>


class SwingFeedback {
public:
    using ResetCallback = std::function<void()>;

    explicit SwingFeedback(RPI_PWM& pwm);

    void setResetCallback(ResetCallback cb);
    void checkTimeout();
    void onLevel(const char* level);
    void forceOff();

private:
    void worker();

    RPI_PWM&           pwm_;

    ResetCallback      reset_cb_;

    std::atomic<bool>  busy_{false};

    std::atomic<int>   pending_level_{0};

    std::atomic<int64_t> busy_since_ms_{0};

    static constexpr int64_t TIMEOUT_MS = 3000;
};
```

11.66 src/libs/Motor/swingfeedback.cpp File Reference

```
#include "swingfeedback.h"
#include <atomic>
#include <chrono>
#include <string_view>
#include <sys/timerfd.h>
#include <thread>
#include <unistd.h>
#include <utility>
```

Include dependency graph for swingfeedback.cpp:



Functions

- static int64_t nowMs ()

  Returns the current time in milliseconds for timeout tracking.

11.66.1 Function Documentation

11.66.1.1 nowMs()

```
static int64_t nowMs ( )  [static]
```

Returns the current time in milliseconds for timeout tracking.
Definition at line 95 of file swingfeedback.cpp.

11.67 swingfeedback.cpp

Go to the documentation of this file.

```
#include "swingfeedback.h"

#include <atomic>
#include <chrono>
#include <string_view>
#include <sys/timerfd.h>
#include <thread>
#include <unistd.h>
#include <utility>

namespace {

static constexpr float MAX_DUTY = 80.0f;

int levelRank(std::string_view level)
{
    if (level == "Low Duty Cycle")      return 1;
    if (level == "Medium Duty Cycle")   return 2;
    if (level == "Highest Duty Cycle")  return 3;
    return 0;
}

void waitMs(int tfd, int ms)
{
    itimerspec ts{};
    ts.it_value.tv_sec  = ms / 1000;
    ts.it_value.tv_nsec = (ms % 1000) * 1000000L;
    timerfd_settime(tfd, 0, &ts, nullptr);
    uint64_t exp;
    ::read(tfd, &exp, sizeof(exp));
}


void rampUp(RPI_PWM& pwm, int tfd)
{
    pwm.setDutyCycle(MAX_DUTY * 0.33f); waitMs(tfd, 30);
    pwm.setDutyCycle(MAX_DUTY * 0.66f); waitMs(tfd, 30);
    pwm.setDutyCycle(MAX_DUTY);         waitMs(tfd, 30);
}


void rampDown(RPI_PWM& pwm, int tfd)
{
    pwm.setDutyCycle(MAX_DUTY * 0.66f); waitMs(tfd, 30);
    pwm.setDutyCycle(MAX_DUTY * 0.33f); waitMs(tfd, 30);
    pwm.setDutyCycle(0.0f);
}

void playRank(RPI_PWM& pwm, int rank)
{
    int tfd = timerfd_create(CLOCK_MONOTONIC, 0);
    if (tfd < 0) return;

    if (rank == 1) {
        rampUp(pwm, tfd);
        waitMs(tfd, 180);
        rampDown(pwm, tfd);
    } else if (rank == 2) {
        rampUp(pwm, tfd);
        waitMs(tfd, 180);
        rampDown(pwm, tfd);
        waitMs(tfd, 100);
        rampUp(pwm, tfd);
        waitMs(tfd, 180);
        rampDown(pwm, tfd);
    } else if (rank == 3) {
        rampUp(pwm, tfd);
        waitMs(tfd, 450);
        rampDown(pwm, tfd);
    }

    pwm.setDutyCycle(0.0f);
    ::close(tfd);
}

}

```

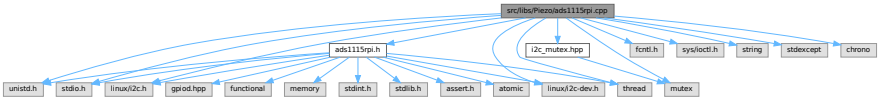
```
static int64_t nowMs()
{
    return std::chrono::duration_cast<std::chrono::milliseconds>(
        std::chrono::steady_clock::now().time_since_epoch()).count();
}

SwingFeedback::SwingFeedback(RPI_PWM& pwm)
    : pwm_(pwm)
{
    pwm_.setDutyCycle(0.0f);
}

void SwingFeedback::setResetCallback(ResetCallback cb)
{
    reset_cb_ = std::move(cb);
}
void SwingFeedback::checkTimeout()
{
    int64_t start = busy_since_ms_.load();
    if (start == 0) return;

    if (nowMs() - start > TIMEOUT_MS) {
        forceOff();
    }
}

void SwingFeedback::worker()
{
    if (busy_.exchange(true)) return;

    busy_since_ms_.store(nowMs());

    int rank = pending_level_.exchange(0);
    if (rank > 0) {
        playRank(pwm_, rank);
    }

    busy_since_ms_.store(0);
    busy_ = false;

    if (reset_cb_) {
        reset_cb_();
    }

    pending_level_.store(0);
}

void SwingFeedback::onLevel(const char* level)
{
    if (!level) return;

    const int rank = levelRank(level);

    if (rank <= 0) {
        if (!busy_.load() && pending_level_.load() > 0) {
            std::thread(&SwingFeedback::worker, this).detach();
        }
        return;
    }

    int current = pending_level_.load();
    while (rank > current && !pending_level_.compare_exchange_weak(current, rank)) {
    }
}

void SwingFeedback::forceOff()
{
    busy_since_ms_.store(0);
    busy_.store(false);
    pending_level_.store(0);
    pwm_.setDutyCycle(0.0f);
    if (reset_cb_) reset_cb_();
}
```

11.68 src/libs/Piezo/ads1115rpi.cpp File Reference

```
#include "ads1115rpi.h"
#include "i2c_mutex.hpp"
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/ioctl.h>
```

```
#include <linux/i2c-dev.h>
#include <linux/i2c.h>
#include <string>
#include <stdexcept>
#include <thread>
#include <chrono>
#include <mutex>
```

Include dependency graph for ads1115rpi.cpp:



Functions

- ADS1115settings makeDefaultADS1115Settings ()

  Creates and returns the default ADS1115 settings.

11.68.1 Function Documentation

11.68.1.1 makeDefaultADS1115Settings()

```
ADS1115settings makeDefaultADS1115Settings ( )
```

Creates and returns the default ADS1115 settings.

This file sets up and manages the ADS1115 ADC on a Raspberry Pi. It initializes I2C and GPIO interrupt handling, reads conversion data when the device signals it is ready, and passes the converted voltage to a callback.

Definition at line 23 of file ads1115rpi.cpp.

11.69 ads1115rpi.cpp

Go to the documentation of this file.

```
00001
00006 #include "ads1115rpi.h"
00007 #include "i2c_mutex.hpp"
00008
00009 #include <stdio.h>
00010 #include <fcntl.h>
00011 #include <unistd.h>
00012 #include <sys/ioctl.h>
00013 #include <linux/i2c-dev.h>
00014 #include <linux/i2c.h>
00015
00016 #include <string>
00017 #include <stdexcept>
00018 #include <thread>
00019 #include <chrono>
00020 #include <mutex>
00021
00023 ADS1115settings makeDefaultADS1115Settings()
00024 {
00025     ADS1115settings s{};
00026     s.i2c_bus      = 1;
00027     s.address      = ADS1115settings::DEFAULT_ADS1115_ADDRESS;
00028     s.samplingRate = ADS1115settings::FS128HZ;
00029     s.pgaGain      = ADS1115settings::FSR2_048;
00030     s.channel      = ADS1115settings::AIN0;
00031     s.drdy_chip    = 0;
00032     s.drdy_gpio    = ADS1115settings::DEFAULT_ALERT_RDY_TO_GPIO;
00033     return s;
00034 }
00035
00037 void ADS1115rpi::start(ADS1115settings settings)
00038 {
00039     if (running) {
00040         throw std::runtime_error("ADS1115: start() called while already running.");
00041     }
00042
```

```
    ads1115settings = settings;

    char gpioFilename[20];
    snprintf(gpioFilename, sizeof(gpioFilename), "/dev/i2c-%d", settings.i2c_bus);

    fd_i2c = open(gpioFilename, O_RDWR);
    if (fd_i2c < 0) {
        throw std::invalid_argument("Could not open I2C.");
    }

    try {
        {
            std::lock_guard<std::mutex> lk(i2c1_mutex());

            if (ioctl(fd_i2c, I2C_SLAVE, settings.address) < 0) {
                throw std::invalid_argument("Could not access I2C address.");
            }
        }

        i2c_writeWord(reg_lo_thres, 0x0000);
        i2c_writeWord(reg_hi_thres, 0x8000);

        unsigned r = (0b10000000 << 8);
        r = r | (1 << 2) | (1 << 3);
        r = r | (settings.samplingRate << 5);
        r = r | (settings.pgaGain << 9);
        r = r | (settings.channel << 12) | (1 << 14);
        i2c_writeWord(reg_config, r);

        const std::string chipPath =
            "/dev/gpiochip" + std::to_string(settings.drdy_chip);
        const std::string consumername =
            "gpioconsumer_" + std::to_string(settings.drdy_chip)
            + "_" + std::to_string(settings.drdy_gpio);

        gpiod::line_config line_cfg;
        line_cfg.add_line_settings(
            settings.drdy_gpio,
            gpiod::line_settings()
                .set_direction(gpiod::line::direction::INPUT)
                .set_edge_detection(gpiod::line::edge::RISING)
                .set_bias(gpiod::line::bias::PULL_UP));

        chip = std::make_shared<gpiod::chip>(chipPath);
        auto builder = chip->prepare_request();
        builder.set_consumer(consumername);
        builder.set_line_config(line_cfg);
        request = std::make_shared<gpiod::line_request>(builder.do_request());

        running = true;
        thr = std::thread(&ADS1115rpi::worker, this);
    } catch (...) {
        running = false;

        if (request) {
            request->release();
            request.reset();
        }

        if (chip) {
            chip->close();
            chip.reset();
        }

        if (fd_i2c >= 0) {
            close(fd_i2c);
            fd_i2c = -1;
        }

        throw;
    }
}

void ADS1115rpi::setChannel(ADS1115settings::Input channel)
{
    unsigned r = i2c_readWord(reg_config);
    r = r & (~(3 << 12));
    r = r | (channel << 12);
    i2c_writeWord(reg_config, r);
    ads1115settings.channel = channel;
}

void ADS1115rpi::dataReady()
{
    int ret = i2c_readConversion();
    if (ret < 0) {
        return;
```

```
    }

    float v = static_cast<float>(ret) / static_cast<float>(0x7fff)
              * fullScaleVoltage();

    if (adsCallbackInterface) {
        adsCallbackInterface(v);
    }
}

void ADS1115rpi::worker()
{
    static constexpr int WARN_EVERY = 10;
    int misses = 0;

    while (running)
    {
        bool edgeSeen = request->wait_edge_events(
            std::chrono::milliseconds(ISR_TIMEOUT_MS));

        if (!running) break;

        if (edgeSeen) {
            misses = 0;

            gpiod::edge_event_buffer buffer;
            request->read_edge_events(buffer, 1);

            dataReady();
        } else {
            ++misses;
            if (misses % WARN_EVERY == 0) {
                fprintf(stderr,
                        "ADS1115: no ALERT/RDY edge for ~%d s (misses=%d) "
                        "-- check DRDY wiring\n",
                        static_cast<int>(misses * ISR_TIMEOUT_MS / 1000),
                        misses);
            }
        }
    }
}

void ADS1115rpi::stop()
{
    if (!running) return;

    running = false;

    if (thr.joinable()) {
        thr.join();
    }

    if (request) {
        request->release();
        request.reset();
    }

    if (chip) {
        chip->close();
        chip.reset();
    }

    if (fd_i2c >= 0) {
        close(fd_i2c);
        fd_i2c = -1;
    }
}

void ADS1115rpi::i2c_selectDevice()
{
    if (ioctl(fd_i2c, I2C_SLAVE, ads1115settings.address) < 0) {
        throw std::invalid_argument("ADS1115: could not select I2C device.");
    }
}

void ADS1115rpi::i2c_writeWord(uint8_t reg, unsigned data)
{
    std::lock_guard<std::mutex> lk(i2c1_mutex());

    i2c_selectDevice();

    uint8_t tmp[3];
    tmp[0] = reg;
    tmp[1] = static_cast<uint8_t>((data & 0xff00) » 8);
    tmp[2] = static_cast<uint8_t>(data & 0x00ff);

    if (write(fd_i2c, tmp, 3) < 0) {
```

```
        throw std::invalid_argument("ADS1115: could not write to i2c.");
    }
}

unsigned ADS1115rpi::i2c_readWord(uint8_t reg)
{
    std::lock_guard<std::mutex> lk(i2c1_mutex());

    i2c_selectDevice();

    uint8_t tmp[2];
    tmp[0] = reg;

    if (write(fd_i2c, tmp, 1) < 0) {
        throw std::invalid_argument("ADS1115: could not select register on i2c.");
    }

    if (read(fd_i2c, tmp, 2) < 0) {
        throw std::invalid_argument("ADS1115: could not read from i2c.");
    }

    return (static_cast<unsigned>(tmp[0]) « 8)
           | static_cast<unsigned>(tmp[1]);
}

int ADS1115rpi::i2c_readConversion()
{
    static constexpr int MAX_RETRIES = 5;

    std::lock_guard<std::mutex> lk(i2c1_mutex());

    uint8_t reg = 0x00;
    uint8_t buf[2] = {};

    struct i2c_msg msgs[2];
    msgs[0].addr  = ads1115settings.address;
    msgs[0].flags = 0;
    msgs[0].len   = 1;
    msgs[0].buf   = &reg;

    msgs[1].addr  = ads1115settings.address;
    msgs[1].flags = I2C_M_RD;
    msgs[1].len   = 2;
    msgs[1].buf   = buf;

    struct i2c_rdwr_ioctl_data txn;
    txn.msgs  = msgs;
    txn.nmsgs = 2;

    for (int attempt = 0; attempt < MAX_RETRIES; ++attempt) {
        int ret = ioctl(fd_i2c, I2C_RDWR, &txn);
        if (ret >= 0) {
            return (static_cast<int>(buf[0]) « 8)
                   | static_cast<int>(buf[1]);
        }
    }

    return -1;
}
```

11.70 src/libs/Piezo/CMakeFiles/piezo_lib.dir/ads1115rpi.cpp.o.d File Reference

11.71 ads1115rpi.cpp.o.d

Go to the documentation of this file.

```
libs/Piezo/CMakeFiles/piezo_lib.dir/ads1115rpi.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/Piezo/ads1115rpi.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/Piezo/Include/ads1115rpi.h \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
 /usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/bits/types.h \
 /usr/include/aarch64-linux-gnu/bits/typesizes.h \
```

```
/usr/include/aarch64-linux-gnu/bits/time64.h \
/usr/include/aarch64-linux-gnu/bits/wchar.h \
/usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
/usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
/usr/include/aarch64-linux-gnu/bits/stdint-least.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/stdio.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/include/c++/14/stdlib.h /usr/include/c++/14/cstdlib \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
/usr/include/c++/14/pstl/pstl_config.h /usr/include/stdlib.h \
/usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
/usr/include/aarch64-linux-gnu/sys/types.h \
/usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
/usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
/usr/include/aarch64-linux-gnu/bits/types/time_t.h \
/usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
/usr/include/endian.h /usr/include/aarch64-linux-gnu/bits/endian.h \
/usr/include/aarch64-linux-gnu/bits/endianness.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/assert.h \
/usr/include/linux/i2c.h /usr/include/linux/types.h \
/usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
/usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h /usr/include/linux/i2c-dev.h \
/usr/include/c++/14/atomic /usr/include/c++/14/bits/version.h \
/usr/include/c++/14/bits/atomic_base.h /usr/include/c++/14/new \
/usr/include/c++/14/bits/exception.h \
/usr/include/c++/14/bits/atomic_lockfree_defines.h \
/usr/include/c++/14/bits/move.h /usr/include/c++/14/type_traits \
/usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/cstdint \
/usr/include/c++/14/bits/functional_hash.h \
/usr/include/c++/14/bits/hash_bytes.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
/usr/include/aarch64-linux-gnu/bits/cpu-set.h /usr/include/time.h \
/usr/include/aarch64-linux-gnu/bits/time.h \
/usr/include/aarch64-linux-gnu/bits/timex.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/c++/14/ext/numeric_traits.h \
/usr/include/c++/14/bits/cpp_type_traits.h \
```

```
/usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/cerrno \
/usr/include/errno.h /usr/include/aarch64-linux-gnu/bits/errno.h \
/usr/include/linux/errno.h /usr/lib/linux/uapi/arm64/asm/errno.h \
/usr/include/asm-generic/errno.h /usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/functexcept.h \
/usr/include/c++/14/bits/exception_defines.h \
/usr/include/c++/14/bits/std_mutex.h /usr/include/c++/14/thread \
/usr/include/c++/14/bits/requires_hosted.h /usr/include/c++/14/compare \
/usr/include/c++/14/concepts /usr/include/c++/14/stop_token \
/usr/include/c++/14/bits/std_thread.h /usr/include/c++/14/iosfwd \
/usr/include/c++/14/bits/stringfwd.h \
/usr/include/c++/14/bits/memoryfwd.h /usr/include/c++/14/bits/postypes.h \
/usr/include/c++/14/cwchar /usr/include/wchar.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/c++/14/tuple /usr/include/c++/14/bits/stl_pair.h \
/usr/include/c++/14/bits/utility.h \
/usr/include/c++/14/bits/uses_allocator.h \
/usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
/usr/include/c++/14/bits/ranges_base.h \
/usr/include/c++/14/initializer_list \
/usr/include/c++/14/bits/stl_iterator.h \
/usr/include/c++/14/bits/stl_iterator_base_types.h \
/usr/include/c++/14/bits/iterator_concepts.h \
/usr/include/c++/14/bits/ptr_traits.h \
/usr/include/c++/14/bits/ranges_cmp.h \
/usr/include/c++/14/bits/stl_construct.h \
/usr/include/c++/14/bits/stl_iterator_base_funcs.h \
/usr/include/c++/14/bits/concept_check.h \
/usr/include/c++/14/debug/assertions.h \
/usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
/usr/include/c++/14/bits/refwrap.h \
/usr/include/c++/14/bits/stl_function.h \
/usr/include/c++/14/backward/binders.h \
/usr/include/c++/14/bits/unique_ptr.h /usr/include/c++/14/ostream \
/usr/include/c++/14/ios /usr/include/c++/14/exception \
/usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/typeinfo /usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/allocator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
/usr/include/c++/14/bits/new_allocator.h \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/stl_algobase.h \
/usr/include/c++/14/debug/debug.h \
/usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
/usr/include/c++/14/bits/range_access.h \
/usr/include/c++/14/bits/basic_string.h \
/usr/include/c++/14/ext/alloc_traits.h \
/usr/include/c++/14/bits/alloc_traits.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdio \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
/usr/include/c++/14/bits/uses_allocator_args.h \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
```

```
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc \
/usr/include/c++/14/bits/ostream.tcc /usr/include/c++/14/semaphore \
/usr/include/c++/14/bits/semaphore_base.h \
/usr/include/c++/14/bits/chrono.h /usr/include/c++/14/ratio \
/usr/include/c++/14/limits /usr/include/c++/14/ctime \
/usr/include/c++/14/bits/parse_numbers.h \
/usr/include/c++/14/bits/atomic_timed_wait.h \
/usr/include/c++/14/bits/this_thread_sleep.h \
/usr/include/aarch64-linux-gnu/sys/time.h /usr/include/semaphore.h \
/usr/include/aarch64-linux-gnu/bits/semaphore.h /usr/include/gpiod.hpp \
/usr/include/gpiodcxx/chip.hpp /usr/include/c++/14/chrono \
/usr/include/c++/14/sstream /usr/include/c++/14/istream \
/usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/c++/14/vector \
/usr/include/c++/14/bits/stl_uninitialized.h \
/usr/include/c++/14/bits/stl_vector.h \
/usr/include/c++/14/bits/stl_bvector.h \
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/ext/aligned_buffer.h \
/usr/include/c++/14/ext/concurrence.h /usr/include/c++/14/bits/align.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/array /usr/include/c++/14/charconv \
/usr/include/c++/14/optional \
/usr/include/c++/14/bits/enable_special_members.h \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h /usr/include/c++/14/iostream \
/usr/include/c++/14/filesystem /usr/include/c++/14/bits/fs_fwd.h \
/usr/include/c++/14/bits/fs_path.h /usr/include/c++/14/codecvt \
/usr/include/c++/14/bits/fs_dir.h /usr/include/c++/14/bits/fs_ops.h \
/usr/include/c++/14/memory \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/gpiodcxx/line.hpp \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/gpiodcxx/chip-info.hpp /usr/include/gpiodcxx/edge-event.hpp \
/usr/include/gpiodcxx/timestamp.hpp \
/usr/include/gpiodcxx/edge-event-buffer.hpp \
/usr/include/gpiodcxx/exception.hpp /usr/include/gpiodcxx/info-event.hpp \
/usr/include/gpiodcxx/line.hpp /usr/include/gpiodcxx/line-config.hpp \
/usr/include/c++/14/map /usr/include/c++/14/bits/stl_tree.h \
/usr/include/c++/14/bits/node_handle.h \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/c++/14/bits/erase_if.h /usr/include/gpiodcxx/line-info.hpp \
/usr/include/gpiodcxx/line-request.hpp /usr/include/gpiodcxx/misc.hpp \
/usr/include/gpiodcxx/line-settings.hpp \
/usr/include/gpiodcxx/request-builder.hpp \
/usr/include/gpiodcxx/request-config.hpp /usr/include/c++/14/functional \
/usr/include/c++/14/bits/std_function.h \
/usr/include/c++/14/unordered_map \
/usr/include/c++/14/bits/unordered_map.h \
/usr/include/c++/14/bits/hashtable.h \
/usr/include/c++/14/bits/hashtable_policy.h \
/home/group5/RTES_Table_Tennis/src/libs/IMU/Include/i2c_mutex.hpp \
/usr/include/c++/14/mutex /usr/include/c++/14/bits/unique_lock.h \
/usr/include/fcntl.h /usr/include/aarch64-linux-gnu/bits/fcntl.h \
/usr/include/aarch64-linux-gnu/bits/fcntl-linux.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_iovec.h \
/usr/include/linux/falloc.h /usr/include/aarch64-linux-gnu/bits/stat.h \
/usr/include/aarch64-linux-gnu/bits/struct_stat.h \
/usr/include/aarch64-linux-gnu/sys/ioctl.h \
```

```
/usr/include/aarch64-linux-gnu/bits/ioctls.h \
/usr/lib/linux/uapi/arm64/asm/ioctls.h /usr/include/asm-generic/ioctls.h \
/usr/include/linux/ioctl.h /usr/lib/linux/uapi/arm64/asm/ioctl.h \
/usr/include/asm-generic/ioctl.h \
/usr/include/aarch64-linux-gnu/bits/ioctl-types.h \
/usr/include/aarch64-linux-gnu/sys/ttydefaults.h
```

11.72 src/libs/Piezo/CMakeFiles/piezo_lib.dir/event_detector.cpp.o.d File Reference

11.73 event_detector.cpp.o.d

Go to the documentation of this file.

```
libs/Piezo/CMakeFiles/piezo_lib.dir/event_detector.cpp.o: \
 /home/group5/RTES_Table_Tennis/src/libs/Piezo/event_detector.cpp \
 /usr/include/stdc-predef.h \
 /home/group5/RTES_Table_Tennis/src/libs/Piezo/Include/event_detector.h \
 /usr/include/c++/14/functional \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++config.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/os_defines.h \
 /usr/include/features.h /usr/include/features-time64.h \
 /usr/include/aarch64-linux-gnu/bits/wordsize.h \
 /usr/include/aarch64-linux-gnu/bits/timesize.h \
 /usr/include/aarch64-linux-gnu/sys/cdefs.h \
 /usr/include/aarch64-linux-gnu/bits/long-double.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs.h \
 /usr/include/aarch64-linux-gnu/gnu/stubs-lp64.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/cpu_defines.h \
 /usr/include/c++/14/pstl/pstl_config.h \
 /usr/include/c++/14/bits/stl_function.h /usr/include/c++/14/bits/move.h \
 /usr/include/c++/14/type_traits /usr/include/c++/14/bits/version.h \
 /usr/include/c++/14/backward/binders.h /usr/include/c++/14/tuple \
 /usr/include/c++/14/bits/stl_pair.h /usr/include/c++/14/bits/utility.h \
 /usr/include/c++/14/compare /usr/include/c++/14/concepts \
 /usr/include/c++/14/bits/uses_allocator.h \
 /usr/include/c++/14/bits/invoke.h /usr/include/c++/14/bits/ranges_util.h \
 /usr/include/c++/14/bits/ranges_base.h \
 /usr/include/c++/14/initializer_list \
 /usr/include/c++/14/bits/stl_iterator.h \
 /usr/include/c++/14/bits/cpp_type_traits.h \
 /usr/include/c++/14/bits/stl_iterator_base_types.h \
 /usr/include/c++/14/bits/iterator_concepts.h \
 /usr/include/c++/14/bits/ptr_traits.h \
 /usr/include/c++/14/bits/ranges_cmp.h \
 /usr/include/c++/14/ext/type_traits.h /usr/include/c++/14/new \
 /usr/include/c++/14/bits/exception.h \
 /usr/include/c++/14/bits/exception_defines.h \
 /usr/include/c++/14/bits/stl_construct.h \
 /usr/include/c++/14/bits/stl_iterator_base_funcs.h \
 /usr/include/c++/14/bits/concept_check.h \
 /usr/include/c++/14/debug/assertions.h \
 /usr/include/c++/14/ext/numeric_traits.h \
 /usr/include/c++/14/bits/max_size_type.h /usr/include/c++/14/numbers \
 /usr/include/c++/14/bits/functional_hash.h \
 /usr/include/c++/14/bits/hash_bytes.h /usr/include/c++/14/bits/refwrap.h \
 /usr/include/c++/14/bits/std_function.h /usr/include/c++/14/typeinfo \
 /usr/include/c++/14/bits/functexcept.h /usr/include/c++/14/unordered_map \
 /usr/include/c++/14/bits/requires_hosted.h \
 /usr/include/c++/14/bits/unordered_map.h \
 /usr/include/c++/14/bits/hashtable.h \
 /usr/include/c++/14/bits/hashtable_policy.h \
 /usr/include/c++/14/bits/stl_algobase.h \
 /usr/include/c++/14/debug/debug.h \
 /usr/include/c++/14/bits/predefined_ops.h /usr/include/c++/14/bit \
 /usr/include/c++/14/ext/aligned_buffer.h \
 /usr/include/c++/14/ext/alloc_traits.h \
 /usr/include/c++/14/bits/alloc_traits.h \
 /usr/include/c++/14/bits/memoryfwd.h \
 /usr/include/c++/14/bits/allocator.h \
 /usr/include/aarch64-linux-gnu/c++/14/bits/c++allocator.h \
 /usr/include/c++/14/bits/new_allocator.h \
 /usr/include/c++/14/bits/enable_special_members.h \
 /usr/include/c++/14/bits/node_handle.h \
 /usr/include/c++/14/bits/range_access.h \
 /usr/include/c++/14/bits/erase_if.h \
 /usr/include/c++/14/bits/memory_resource.h /usr/include/c++/14/cstddef \
 /usr/lib/gcc/aarch64-linux-gnu/14/include/stddef.h \
 /usr/include/c++/14/bits/uses_allocator_args.h \
 /usr/include/c++/14/vector /usr/include/c++/14/bits/stl_uninitialized.h \
 /usr/include/c++/14/bits/stl_vector.h \
 /usr/include/c++/14/bits/stl_bvector.h \
```

```
/usr/include/c++/14/bits/vector.tcc /usr/include/c++/14/array \
/home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/led_controller.h \
/home/group5/RTES_Table_Tennis/src/libs/LEDs/Include/ledcallback.hpp \
/usr/include/gpiod.hpp /usr/include/gpiodcxx/chip.hpp \
/usr/include/c++/14/chrono /usr/include/c++/14/bits/chrono.h \
/usr/include/c++/14/ratio /usr/include/c++/14/cstdint \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdint.h /usr/include/stdint.h \
/usr/include/aarch64-linux-gnu/bits/libc-header-start.h \
/usr/include/aarch64-linux-gnu/bits/types.h \
/usr/include/aarch64-linux-gnu/bits/typesizes.h \
/usr/include/aarch64-linux-gnu/bits/time64.h \
/usr/include/aarch64-linux-gnu/bits/wchar.h \
/usr/include/aarch64-linux-gnu/bits/stdint-intn.h \
/usr/include/aarch64-linux-gnu/bits/stdint-uintn.h \
/usr/include/aarch64-linux-gnu/bits/stdint-least.h \
/usr/include/c++/14/limits /usr/include/c++/14/ctime /usr/include/time.h \
/usr/include/aarch64-linux-gnu/bits/time.h \
/usr/include/aarch64-linux-gnu/bits/timex.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timeval.h \
/usr/include/aarch64-linux-gnu/bits/types/clock_t.h \
/usr/include/aarch64-linux-gnu/bits/types/time_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_tm.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_timespec.h \
/usr/include/aarch64-linux-gnu/bits/endian.h \
/usr/include/aarch64-linux-gnu/bits/endianness.h \
/usr/include/aarch64-linux-gnu/bits/types/clockid_t.h \
/usr/include/aarch64-linux-gnu/bits/types/timer_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_itimerspec.h \
/usr/include/aarch64-linux-gnu/bits/types/locale_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__locale_t.h \
/usr/include/c++/14/bits/parse_numbers.h /usr/include/c++/14/sstream \
/usr/include/c++/14/istream /usr/include/c++/14/ios \
/usr/include/c++/14/iosfwd /usr/include/c++/14/bits/stringfwd.h \
/usr/include/c++/14/bits/postypes.h /usr/include/c++/14/cwchar \
/usr/include/wchar.h /usr/include/aarch64-linux-gnu/bits/floatn.h \
/usr/include/aarch64-linux-gnu/bits/floatn-common.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/stdarg.h \
/usr/include/aarch64-linux-gnu/bits/types/wint_t.h \
/usr/include/aarch64-linux-gnu/bits/types/mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__mbstate_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/FILE.h \
/usr/include/c++/14/exception /usr/include/c++/14/bits/exception_ptr.h \
/usr/include/c++/14/bits/cxxabi_init_exception.h \
/usr/include/c++/14/bits/nested_exception.h \
/usr/include/c++/14/bits/char_traits.h \
/usr/include/c++/14/bits/localefwd.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/c++locale.h \
/usr/include/c++/14/clocale /usr/include/locale.h \
/usr/include/aarch64-linux-gnu/bits/locale.h /usr/include/c++/14/cctype \
/usr/include/ctype.h /usr/include/c++/14/bits/ios_base.h \
/usr/include/c++/14/ext/atomicity.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/gthr-default.h \
/usr/include/pthread.h /usr/include/sched.h \
/usr/include/aarch64-linux-gnu/bits/sched.h \
/usr/include/linux/sched/types.h /usr/include/linux/types.h \
/usr/lib/linux/uapi/arm64/asm/types.h /usr/include/asm-generic/types.h \
/usr/include/asm-generic/int-ll64.h \
/usr/lib/linux/uapi/arm64/asm/bitsperlong.h \
/usr/include/asm-generic/bitsperlong.h /usr/include/linux/posix_types.h \
/usr/include/linux/stddef.h /usr/lib/linux/uapi/arm64/asm/posix_types.h \
/usr/include/asm-generic/posix_types.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_sched_param.h \
/usr/include/aarch64-linux-gnu/bits/cpu-set.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes.h \
/usr/include/aarch64-linux-gnu/bits/thread-shared-types.h \
/usr/include/aarch64-linux-gnu/bits/pthreadtypes-arch.h \
/usr/include/aarch64-linux-gnu/bits/atomic_wide_counter.h \
/usr/include/aarch64-linux-gnu/bits/struct_mutex.h \
/usr/include/aarch64-linux-gnu/bits/struct_rwlock.h \
/usr/include/aarch64-linux-gnu/bits/setjmp.h \
/usr/include/aarch64-linux-gnu/bits/types/__sigset_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct___jmp_buf_tag.h \
/usr/include/aarch64-linux-gnu/bits/pthread_stack_min-dynamic.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/atomic_word.h \
/usr/include/aarch64-linux-gnu/sys/single_threaded.h \
/usr/include/c++/14/bits/locale_classes.h /usr/include/c++/14/string \
/usr/include/c++/14/bits/ostream_insert.h \
/usr/include/c++/14/bits/cxxabi_forced.h \
/usr/include/c++/14/bits/basic_string.h /usr/include/c++/14/string_view \
/usr/include/c++/14/bits/string_view.tcc \
/usr/include/c++/14/ext/string_conversions.h /usr/include/c++/14/cstdlib \
/usr/include/stdlib.h /usr/include/aarch64-linux-gnu/bits/waitflags.h \
/usr/include/aarch64-linux-gnu/bits/waitstatus.h \
/usr/include/aarch64-linux-gnu/sys/types.h /usr/include/endian.h \
/usr/include/aarch64-linux-gnu/bits/byteswap.h \
```

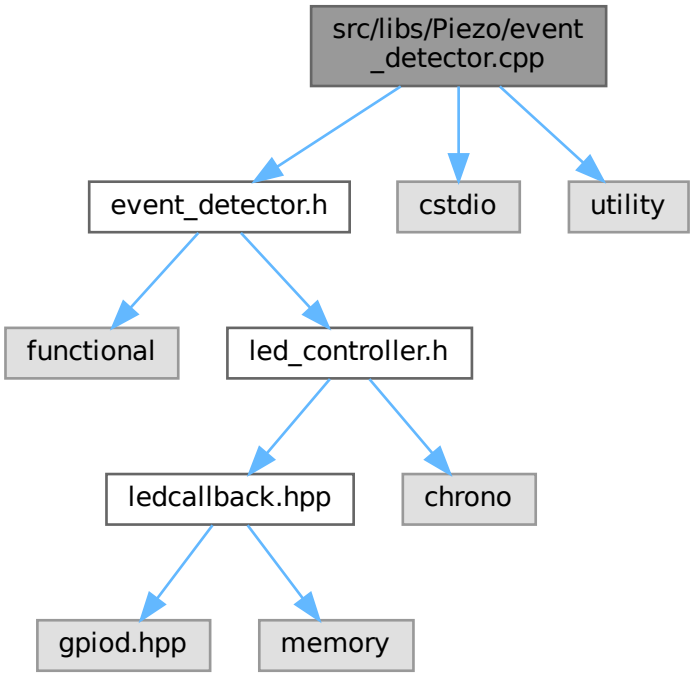
```
/usr/include/aarch64-linux-gnu/bits/uintn-identity.h \
/usr/include/aarch64-linux-gnu/sys/select.h \
/usr/include/aarch64-linux-gnu/bits/select.h \
/usr/include/aarch64-linux-gnu/bits/types/sigset_t.h \
/usr/include/alloca.h /usr/include/aarch64-linux-gnu/bits/stdlib-float.h \
/usr/include/c++/14/bits/std_abs.h /usr/include/c++/14/cstdio \
/usr/include/stdio.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos_t.h \
/usr/include/aarch64-linux-gnu/bits/types/__fpos64_t.h \
/usr/include/aarch64-linux-gnu/bits/types/struct_FILE.h \
/usr/include/aarch64-linux-gnu/bits/types/cookie_io_functions_t.h \
/usr/include/aarch64-linux-gnu/bits/stdio_lim.h \
/usr/include/c++/14/cerrno /usr/include/errno.h \
/usr/include/aarch64-linux-gnu/bits/errno.h /usr/include/linux/errno.h \
/usr/lib/linux/uapi/arm64/asm/errno.h /usr/include/asm-generic/errno.h \
/usr/include/asm-generic/errno-base.h \
/usr/include/aarch64-linux-gnu/bits/types/error_t.h \
/usr/include/c++/14/bits/charconv.h \
/usr/include/c++/14/bits/basic_string.tcc \
/usr/include/c++/14/bits/locale_classes.tcc \
/usr/include/c++/14/system_error \
/usr/include/aarch64-linux-gnu/c++/14/bits/error_constants.h \
/usr/include/c++/14/stdexcept /usr/include/c++/14/streambuf \
/usr/include/c++/14/bits/streambuf.tcc \
/usr/include/c++/14/bits/basic_ios.h \
/usr/include/c++/14/bits/locale_facets.h /usr/include/c++/14/cwctype \
/usr/include/wctype.h /usr/include/aarch64-linux-gnu/bits/wctype-wchar.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_base.h \
/usr/include/c++/14/bits/streambuf_iterator.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/ctype_inline.h \
/usr/include/c++/14/bits/locale_facets.tcc \
/usr/include/c++/14/bits/basic_ios.tcc /usr/include/c++/14/ostream \
/usr/include/c++/14/bits/ostream.tcc \
/usr/include/c++/14/bits/istream.tcc \
/usr/include/c++/14/bits/sstream.tcc /usr/include/c++/14/bits/stl_algo.h \
/usr/include/c++/14/bits/algorithmfwd.h \
/usr/include/c++/14/bits/stl_heap.h \
/usr/include/c++/14/bits/uniform_int_dist.h \
/usr/include/c++/14/bits/stl_tempbuf.h \
/usr/include/c++/14/bits/shared_ptr.h \
/usr/include/c++/14/bits/shared_ptr_base.h \
/usr/include/c++/14/bits/allocated_ptr.h \
/usr/include/c++/14/bits/unique_ptr.h \
/usr/include/c++/14/ext/concurrence.h /usr/include/c++/14/bits/align.h \
/usr/include/c++/14/bits/chrono_io.h /usr/include/c++/14/iomanip \
/usr/include/c++/14/locale \
/usr/include/c++/14/bits/locale_facets_nonio.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/time_members.h \
/usr/include/aarch64-linux-gnu/c++/14/bits/messages_members.h \
/usr/include/libintl.h /usr/include/c++/14/bits/codecvt.h \
/usr/include/c++/14/bits/locale_facets_nonio.tcc \
/usr/include/c++/14/bits/locale_conv.h \
/usr/include/c++/14/bits/quoted_string.h /usr/include/c++/14/format \
/usr/include/c++/14/charconv /usr/include/c++/14/optional \
/usr/include/c++/14/span /usr/include/c++/14/variant \
/usr/include/c++/14/bits/ranges_algobase.h \
/usr/include/c++/14/bits/unicode.h \
/usr/include/c++/14/bits/unicode-data.h /usr/include/c++/14/iostream \
/usr/include/c++/14/filesystem /usr/include/c++/14/bits/fs_fwd.h \
/usr/include/c++/14/bits/fs_path.h /usr/include/c++/14/codecvt \
/usr/include/c++/14/bits/fs_dir.h /usr/include/c++/14/bits/fs_ops.h \
/usr/include/c++/14/memory \
/usr/include/c++/14/bits/stl_raw_storage_iter.h \
/usr/include/c++/14/bits/shared_ptr_atomic.h \
/usr/include/c++/14/bits/atomic_base.h \
/usr/include/c++/14/bits/atomic_lockfree_defines.h \
/usr/include/c++/14/bits/atomic_wait.h /usr/include/c++/14/climits \
/usr/lib/gcc/aarch64-linux-gnu/14/include/limits.h \
/usr/lib/gcc/aarch64-linux-gnu/14/include/syslimits.h \
/usr/include/limits.h /usr/include/aarch64-linux-gnu/bits/posix1_lim.h \
/usr/include/aarch64-linux-gnu/bits/local_lim.h \
/usr/include/linux/limits.h \
/usr/include/aarch64-linux-gnu/bits/posix2_lim.h \
/usr/include/aarch64-linux-gnu/bits/xopen_lim.h \
/usr/include/aarch64-linux-gnu/bits/uio_lim.h /usr/include/unistd.h \
/usr/include/aarch64-linux-gnu/bits/posix_opt.h \
/usr/include/aarch64-linux-gnu/bits/environments.h \
/usr/include/aarch64-linux-gnu/bits/confname.h \
/usr/include/aarch64-linux-gnu/bits/getopt_posix.h \
/usr/include/aarch64-linux-gnu/bits/getopt_core.h \
/usr/include/aarch64-linux-gnu/bits/unistd_ext.h \
/usr/include/linux/close_range.h /usr/include/syscall.h \
/usr/include/aarch64-linux-gnu/sys/syscall.h \
/usr/lib/linux/uapi/arm64/asm/unistd.h \
/usr/lib/linux/uapi/arm64/asm/unistd_64.h \
/usr/include/aarch64-linux-gnu/bits/syscall.h \
/usr/include/c++/14/bits/std_mutex.h \
```

```
/usr/include/c++/14/backward/auto_ptr.h \
/usr/include/c++/14/bits/ranges_uninitialized.h \
/usr/include/c++/14/pstl/glue_memory_defs.h \
/usr/include/c++/14/pstl/execution_defs.h /usr/include/gpiodcxx/line.hpp \
/usr/include/c++/14/utility /usr/include/c++/14/bits/stl_relops.h \
/usr/include/gpiodcxx/chip-info.hpp /usr/include/gpiodcxx/edge-event.hpp \
/usr/include/gpiodcxx/timestamp.hpp \
/usr/include/gpiodcxx/edge-event-buffer.hpp \
/usr/include/gpiodcxx/exception.hpp /usr/include/gpiodcxx/info-event.hpp \
/usr/include/gpiodcxx/line.hpp /usr/include/gpiodcxx/line-config.hpp \
/usr/include/c++/14/map /usr/include/c++/14/bits/stl_tree.h \
/usr/include/c++/14/bits/stl_map.h \
/usr/include/c++/14/bits/stl_multimap.h \
/usr/include/gpiodcxx/line-info.hpp \
/usr/include/gpiodcxx/line-request.hpp /usr/include/gpiodcxx/misc.hpp \
/usr/include/gpiodcxx/line-settings.hpp \
/usr/include/gpiodcxx/request-builder.hpp \
/usr/include/gpiodcxx/request-config.hpp
```

11.74 src/libs/Piezo/event_detector.cpp File Reference

```
#include "event_detector.h"
#include <cstdio>
#include <utility>
```

Include dependency graph for event_detector.cpp:



11.75 event_detector.cpp

Go to the documentation of this file.

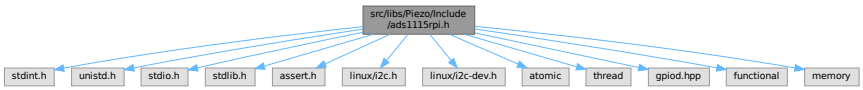
```
00001
00005 #include "event_detector.h"
00006
00007 #include <cstdio>
00008 #include <utility>
00009
```

```
PiezoEventDetector::PiezoEventDetector(SimpleLEDController& ledRef,
                                       PiezoEventDetectorSettings settings)
    : led_(ledRef), settings_(settings)
{
}

void PiezoEventDetector::setForceCallback(ForceCallback cb)
{
    forceCallback_ = std::move(cb);
}

void PiezoEventDetector::reset()
{
    pressed_     = false;
    filter_init_ = false;
    filtered_v_  = 0.0f;
    led_.set(false);
}

void PiezoEventDetector::processSample(float v)
{
    if (!filter_init_) {
        filtered_v_  = v;
        filter_init_ = true;
    } else {
        filtered_v_ = 0.2f * filtered_v_ + 0.8f * v;
    }
    v = filtered_v_;

    if (settings_.enableDebugPrints) {
        std::printf("[piezo] v=%.3f pressed=%s\n",
                    v,
                    pressed_ ? "yes" : "no");
    }

    if (!pressed_ && v >= settings_.pressThreshold) {
        pressed_ = true;
        led_.set(true);

        if (settings_.enableDebugPrints) {
            std::printf("[piezo] pressed -> LED ON\n");
        }

        if (forceCallback_) {
            forceCallback_(true);
        }
        return;
    }

    if (pressed_ && v <= settings_.releaseThreshold) {
        pressed_ = false;
        led_.set(false);

        if (settings_.enableDebugPrints) {
            std::printf("[piezo] released -> LED OFF\n");
        }

        if (forceCallback_) {
            forceCallback_(false);
        }
    }
}
```

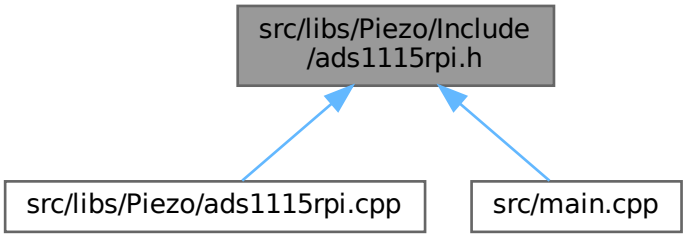
11.76 src/libs/Piezo/Include/ads1115rpi.h File Reference

```
#include <stdint.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>
#include <assert.h>
#include <linux/i2c.h>
#include <linux/i2c-dev.h>
#include <atomic>
#include <thread>
#include <gpiod.hpp>
#include <functional>
#include <memory>
```

Include dependency graph for ads1115rpi.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct ADS1115settings

  Stores the ADS1115 configuration values, including I2C settings, sampling rate, gain, input channel, and GPIO settings.
- class ADS1115rpi

  Declares the ADS1115 Raspberry Pi interface, including callback support and the main functions for starting, stopping, configuring, and reading the device.

Macros

- #define DEBUG

Functions

- ADS1115settings makeDefaultADS1115Settings ()

  Creates and returns the default ADS1115 settings.

11.76.1 Macro Definition Documentation

11.76.1.1 DEBUG

`#define DEBUG`

This header defines the ADS1115 configuration data and the Raspberry Pi interface used to communicate with the ADS1115 device. It provides the available setup options, callback registration, channel control, and the functions used to start, stop, and manage ADC readings.

SOLID principles: S - The settings struct is responsible only for configuration data, while the delivery. O - Client behaviour can be extended through the callback interface without changing the ADC reading logic itself. I/D - Client code interacts with this class through a small public API and a narrow callback interface rather than depending on low-level I2C or GPIO operations directly.

Definition at line 33 of file ads1115rpi.h.

11.76.2 Function Documentation

11.76.2.1 makeDefaultADS1115Settings()

```
ADS1115settings makeDefaultADS1115Settings ( )
```

Creates and returns the default ADS1115 settings.

This file sets up and manages the ADS1115 ADC on a Raspberry Pi. It initializes I2C and GPIO interrupt handling, reads conversion data when the device signals it is ready, and passes the converted voltage to a callback.

Definition at line 23 of file ads1115rpi.cpp.

11.77 ads1115rpi.h

Go to the documentation of this file.

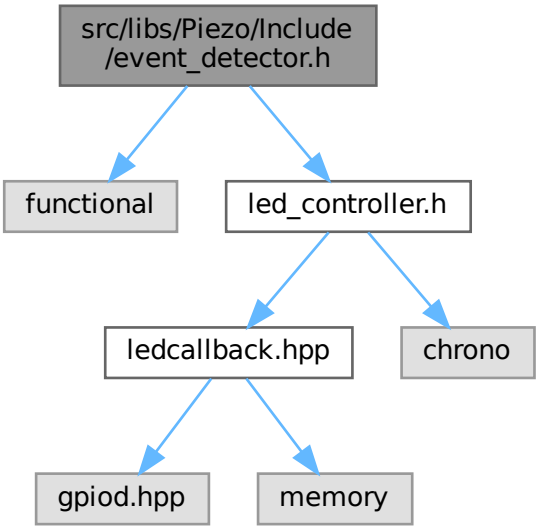
```
00001
00016 #ifndef __ADS1115RPI_H
00017 #define __ADS1115RPI_H
00018
00019 #include <stdint.h>
00020 #include <unistd.h>
00021 #include <stdio.h>
00022 #include <stdlib.h>
00023 #include <assert.h>
00024 #include <linux/i2c.h>
00025 #include <linux/i2c-dev.h>
00026 #include <atomic>
00027 #include <thread>
00028 #include <gpiod.hpp>
00029 #include <functional>
00030 #include <memory>
00031
00032 #ifndef NDEBUG
00033 #define DEBUG
00034 #endif
00035
00037 struct ADS1115settings
00038 {
00039     int i2c_bus = 1;
00040
00041     static constexpr uint8_t DEFAULT_ADS1115_ADDRESS = 0x48;
00042     uint8_t address = DEFAULT_ADS1115_ADDRESS;
00043
00044     enum SamplingRates
00045     {
00046         FS8HZ = 0,
00047         FS16HZ = 1,
00048         FS32HZ = 2,
00049         FS64HZ = 3,
00050         FS128HZ = 4,
00051         FS250HZ = 5,
00052         FS475HZ = 6,
00053         FS860HZ = 7
00054     };
00056     inline unsigned getSamplingRate() const
00057     {
00058         const unsigned SamplingRateEnum2Value[8] =
00059             {8, 16, 32, 64, 128, 250, 475, 860};
00060         return SamplingRateEnum2Value[samplingRate];
00061     }
00062
00063     SamplingRates samplingRate = FS128HZ;
00064
00065     enum PGA
00066     {
00067         FSR2_048 = 2,
00068         FSR1_024 = 3,
00069         FSR0_512 = 4,
00070         FSR0_256 = 5
00071     };
00072
00073     PGA pgaGain = FSR2_048;
00074
00075     enum Input
00076     {
00077         AIN0 = 0,
00078         AIN1 = 1,
00079         AIN2 = 2,
00080         AIN3 = 3
00081     };
00082
00083     Input channel = AIN0;
00084
```

```
    int drdy_chip = 0;

    static constexpr int DEFAULT_ALERT_RDY_TO_GPIO = 26;
    int drdy_gpio = DEFAULT_ALERT_RDY_TO_GPIO;
};

ADS1115settings makeDefaultADS1115Settings();

class ADS1115rpi
{
public:
    ~ADS1115rpi()
    {
        stop();
    }

    using ADSCallbackInterface = std::function<void(float)>;

    void registerCallback(ADSCallbackInterface ci)
    {
        adsCallbackInterface = ci;
    }

    void setChannel(ADS1115settings::Input channel);
    void start(ADS1115settings settings = ADS1115settings());
    ADS1115settings getADS1115settings() const
    {
        return ads1115settings;
    }
    void stop();

    void setRetryEnabled(bool enabled) { retry_enabled_.store(enabled); }

private:
    ADS1115settings ads1115settings;

    void dataReady();
    void worker();

    void i2c_selectDevice();
    void i2c_writeWord(uint8_t reg, unsigned data);
    unsigned i2c_readWord(uint8_t reg);
    int i2c_readConversion();

    const uint8_t reg_config = 1;
    const uint8_t reg_lo_thres = 2;
    const uint8_t reg_hi_thres = 3;

    float fullScaleVoltage()
    {
        switch (ads1115settings.pgaGain)
        {
        case ADS1115settings::FSR2_048:
            return 2.048f;
        case ADS1115settings::FSR1_024:
            return 1.024f;
        case ADS1115settings::FSR0_512:
            return 0.512f;
        case ADS1115settings::FSR0_256:
            return 0.256f;
        }
        assert(1 == 0);
        return 0.0f;
    }

    std::shared_ptr<gpiod::chip> chip;
    std::shared_ptr<gpiod::line_request> request;
    std::thread thr;
    int fd_i2c = -1;
    bool running = false;
    ADSCallbackInterface adsCallbackInterface;
    std::atomic<bool> retry_enabled_{false};

    static constexpr int64_t ISR_TIMEOUT_MS = 500;
};

#endif
```

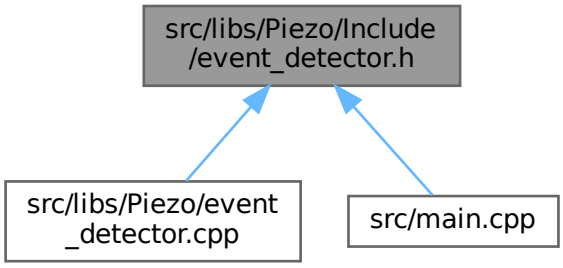
11.78 src/libs/Piezo/Include/event_detector.h File Reference

```
#include <functional>
#include "led_controller.h"
```

Include dependency graph for event_detector.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct PiezoEventDetectorSettings

  Stores the threshold and debug settings for the detector.
- class PiezoEventDetector

  Declares the detector class and its public interface.

11.79 event_detector.h

Go to the documentation of this file.

```
00001
00015 #pragma once
```

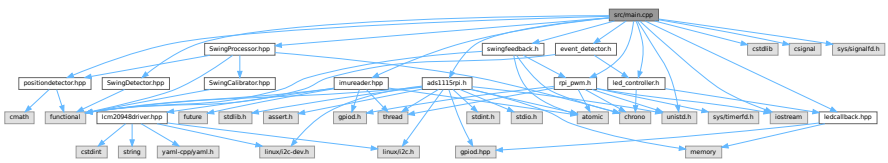
```

#include <functional>

#include "led_controller.h"

struct PiezoEventDetectorSettings
{
    float pressThreshold = 0.95f;
    float releaseThreshold = 0.70f;
    bool enableDebugPrints = false;
};

class PiezoEventDetector
{
public:
    using ForceCallback = std::function<void(bool)>;

    PiezoEventDetector(SimpleLEDController& ledRef,
                       PiezoEventDetectorSettings settings);

    void setForceCallback(ForceCallback cb);
    void processSample(float v);
    void reset();

private:
    SimpleLEDController& led_;
    PiezoEventDetectorSettings settings_;
    ForceCallback forceCallback_;

    bool  pressed_     = false;
    bool  filter_init_ = false;
    float filtered_v_  = 0.0f;
};
```

11.80 src/libs/Piezo/ReadMe.md File Reference

11.81 src/main.cpp File Reference

```
#include "positiondetector.hpp"
#include "imureader.hpp"
#include "SwingProcessor.hpp"
#include "SwingDetector.hpp"
#include "swingfeedback.h"
#include "rpi_pwm.h"
#include <iostream>
#include <cstdlib>
#include <csignal>
#include <sys/signalfd.h>
#include <unistd.h>
#include "ads1115rpi.h"
#include "event_detector.h"
#include "led_controller.h"
#include "ledcallback.hpp"
```

Include dependency graph for main.cpp:



Functions

- int main ()

11.81.1 Function Documentation

11.81.1.1 main()

```
int main ( )
```

Latency test for IMU
change this to test for latency
12-bit divider
8-bit divider
Latency test for Piezo
Definition at line 19 of file main.cpp.

11.82 main.cpp

Go to the documentation of this file.

```
00001 #include "positiondetector.hpp"
00002 #include "imureader.hpp"
00003 #include "SwingProcessor.hpp"
00004 #include "SwingDetector.hpp"
00005 #include "swingfeedback.h"
00006 #include "rpi_pwm.h"
00007
00008 #include <iostream>
00009 #include <cstdlib>
00010 #include <csignal>
00011 #include <sys/signalfd.h>
00012 #include <unistd.h>
00013
00014 #include "ads1115rpi.h"
00015 #include "event_detector.h"
00016 #include "led_controller.h"
00017 #include "ledcallback.hpp"
00018
00019 int main()
00020 {
00021     SimpleLEDController imuLed(22, 0);
00022     LEDController piezoLed(LEDControllerSettings{.chipNumber = 0, .greenGpio = 25});
00023
00024     RPI_PWM pwm;
00025     if (!pwm.start()) {
00026         return EXIT_FAILURE;
00027     }
00028     pwm.setDutyCycle(0.0f);
00029
00030     SwingDetector  swing_detector;
00031     SwingProcessor processor;
00032     SwingFeedback  feedback(pwm);
00033
00034     swing_detector.setCallback([&feedback](const char* level) {
00035         feedback.onLevel(level);
00036     });
00037
00038     processor.setMagnitudeCallback([&swing_detector](float mag) {
00039         swing_detector.detect(mag);
00040     });
00041
00042     processor.setPositionCallback([&imuLed](bool upright) {
00043         imuLed.set(upright);
00044     });
00045
00047     int16_t sample_rate_div = 59;
00048
00049     icm20948::settings imuSettings(
00050         icm20948::accel_settings(sample_rate_div),
00051         icm20948::gyro_settings(static_cast<uint8_t>(sample_rate_div))
00052     );
00053
00054     IMUReader reader(I2C_BUS, ICM20948_I2C_ADDR, "/dev/gpiochip0", 27, imuSettings);
00055
00056     if (!reader.init()) {
00057         pwm.setDutyCycle(0.0f);
00058         return EXIT_FAILURE;
00059     }
00060
00061     reader.setCallback([&](float ax, float ay, float az,
00062                            float gx, float gy, float gz,
00063                            float mx, float my) {
00064         feedback.checkTimeout();
00065         processor(ax, ay, az, gx, gy, gz, mx, my);
00066     });
```

```

    reader.start();

    PiezoEventDetector piezoDetector(piezoLed, PiezoEventDetectorSettings{});

    feedback.setResetCallback([&processor, &swing_detector, &piezoDetector]() {
        swing_detector.reset();
        processor.reset();
        piezoDetector.reset();
    });

    piezoDetector.setForceCallback([&](bool pressed) {
        if (!pressed) return;
        if (processor.isActive()) return;
        const bool wasActive = processor.isActive();
        processor.onForceReady(true);
        (void)wasActive;
    });

    ADS1115rpi ads1115rpi;
    ads1115rpi.registerCallback([&](float v) {
        if (processor.isActive()) return;
        piezoDetector.processSample(v);
    });

    ADS1115settings piezoSettings;
    piezoSettings.samplingRate = ADS1115settings::FS128HZ;
    // available rates: 8, 16, 32, 64, 128 (default), 250, 475, 860

    try {
        ads1115rpi.start(piezoSettings);
    } catch (const std::exception&) {}

    sigset_t mask;
    sigemptyset(&mask);
    sigaddset(&mask, SIGINT);
    sigaddset(&mask, SIGHUP);

    if (sigprocmask(SIG_BLOCK, &mask, nullptr) == -1) {
        reader.stop();
        ads1115rpi.stop();
        feedback.forceOff();
        pwm.setDutyCycle(0.0f);
        return EXIT_FAILURE;
    }

    int sfd = signalfd(-1, &mask, 0);
    if (sfd == -1) {
        reader.stop();
        ads1115rpi.stop();
        feedback.forceOff();
        pwm.setDutyCycle(0.0f);
        return EXIT_FAILURE;
    }

    bool running = true;
    while (running) {
        signalfd_siginfo fdsi;
        ssize_t s = ::read(sfd, &fdsi, sizeof(fdsi));
        if (s != static_cast<ssize_t>(sizeof(fdsi))) break;

        if (fdsi.ssi_signo == SIGINT || fdsi.ssi_signo == SIGHUP) {
            running = false;
        }
    }

    ::close(sfd);
    reader.stop();
    ads1115rpi.stop();
    feedback.forceOff();
    pwm.setDutyCycle(0.0f);

    return EXIT_SUCCESS;
}
```

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